

General Industries Hub

Enterprise Inventory Management

Product Manual & User Catalogue

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GI Hub - Multi-Site Inventory, Procurement & Asset Management

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Version 3.0 - Multi-Site Warehouse Inventory ERP + Procurement Chain Document Scope: Complete operational reference for every role, page, tab, and element built into the system. Companion documents: the Standard Operating Procedure, for day-to-day procedure organised by task rather than by screen; and the architecture reference, for whoever administers the server.

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no running total can drift out of step with reality.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper confirms it arrived. Every handover is recorded.
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Principle	What it means
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock record	Permanent - stock figures move now
4	System	Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then	Warehouse - Receiving

Step	Role	Action	Lands in
		records the physical arrival from the vendor	
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
1	Quality Control (QC)	One site or one warehouse - never both
2	Head of Department	One site
2	Head of Qualities (QC-HOD)	All sites, Surface Shield only, read + escalate
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

The Head of Qualities is level 2 but reads across every site, and that is not a contradiction. Cross-site reach normally comes from being level 3 - and level 3 would also have handed this role every endpoint the rank admits, which is most of the system. Instead it is level 2 with a named cross-site exemption, and the rank grants it nothing at all: a level check refuses it outright, so it reaches only what names the role explicitly. Its whole surface is the Quality Oversight page. See §23.

The QC role is scoped on one of two axes, and exactly one. Material is inspected either where it arrives (a warehouse) or where it lands (a site), and which one depends on how your operation is organised. A QC account is therefore created against a site or against a warehouse, and an account carrying neither - or both - sees nothing at all rather than everything. That is deliberate: an inspector with an unclear remit is a bigger problem than an inspector with none.

2.2 Page access matrix - which role can open which page

[!] A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Read that caveat together with the table below it, not instead of it: the caveat says a page is locked to "its own role", and the table is what names that role. A Head of Department can open Man-Hours and the Material Estimator; those two are locked to the HOD, not away from them. Admin is the single exception and reaches every workspace deliberately, for support.

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Live Dashboard	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Stock	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[X]	[OK]	[OK]
Rack Locator	[OK]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Assets	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Entry Log (all tabs)	[OK]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
HOD Portal -> Approvals	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
HOD Portal -> Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Lining Coverage	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Bulk Excel Import	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
WBS & Work	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Types									
Material Estimator (SME)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Man-Hours / Manpower Tracking	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Execution Entries	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[X]	[OK]
Reports	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK] (read + download)	[OK]
Logistics Portal (Procurement)	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Warehouse Portal (Receiving & DN)	[X]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Quality -> Inspections	[OK] (read)	[OK] (read)	[X]	[OK] (decides)	[OK] (read)	[OK] (read)	[X]	[OK] (read)	[OK]
Quality -> QC Accounts	[X]	[OK]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]
Quality Oversight (\$23)	[X]	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[OK]
Safety &	[OK]	[X]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
People -> PPE Forecast									
Safety & People -> PPE Usable Time	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Safety & People -> Employees	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Records (Inventory, ledgers, POs, PRs)	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[OK] (read)	[OK]
Valuation & 30-Day Burn (\$24.3)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Admin Portal	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Documents - Security - Training - Feedback	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]

Where each role lands after signing in: Store Keeper -> Entry Log -> Issue Stock - Warehouse User -> Warehouse Portal - Supervisor -> Supervisor Portal - QC -> Quality -> Inspections - HOD -> HOD Portal -> Approvals - Logistics -> Logistics Portal - Head of Qualities -> Quality Oversight - Auditor ->

Dashboard - Admin -> Admin Portal -> Console.

2.3 What each role can do

One list per role, in plain words. If you only want to know whether your account can do a thing, read your own entry - each is complete on its own.

2.3.1 Store Keeper - what a Store Keeper can and cannot do

Can: open the Entry Log and post every stock movement - receive, issue, return, adjust, count, returnable items, OCR import; see Incoming Deliveries and Supervisor Requests; open the Dashboard, Stock, Rack Locator, Assets, PPE Forecast, PPE Usable Time, Employees and Execution Entries; read quality inspections; download documents.

Cannot: approve anything (an HOD approves what a Store Keeper stages); open the HOD, Supervisor, Logistics or Warehouse portals; open Reports, Records, the Material Estimator or Man-Hours; see another site's stock.

Scope: one site.

2.3.2 Supervisor - what a Supervisor can and cannot do

Can: raise material requests from the Supervisor Portal; post Execution Entries; open the Dashboard, Stock, Assets and Employees.

Cannot: post stock movements (the Entry Log is the Store Keeper's); approve requests; open Reports, Records, the Material Estimator, Man-Hours or any other role's portal.

Scope: one site.

2.3.3 Quality Control (QC) - what an inspector can and cannot do

Can: decide inspections - pass, fail or hold - from Quality -> Inspections; open the Stock page to see what is waiting and what their decision released.

Cannot: open the Dashboard, post any stock movement, approve a purchase, create QC accounts, or move themselves between sites (that is an Admin decision). See §22.

Scope: exactly one site or exactly one warehouse - never both, never neither.

2.3.4 Head of Department (HOD) - what an HOD can and cannot do

Can - and this is the question people ask most: open the Man-Hours / Manpower Tracking page and the Material Estimator. Both are exact-locked to the HOD (and Admin for support); the lock is what admits an HOD, not what excludes them. That includes all thirteen Man-Hours tabs - roster, timesheets, estimator, variance, scorecard, employee-wise, the four execution views, the Manpower Planner and the SME Session Plan (§19).

Also: approve or reject everything the site stages (receipts, issues, returns, adjustments, counts); create and submit Purchase Requisitions to Logistics; raise Cross-Site Requests; run Reports; open the

Executive Summary, Burn Rate, Low Stock, Lining Coverage, Document Library and Bulk Excel Import; create QC accounts for their site; post Execution Entries; open PPE Forecast, PPE Usable Time and Employees.

Cannot: open the Logistics Portal, the Warehouse Portal, the Supervisor Portal, the Entry Log, Quality Oversight or the Admin Portal; raise a Purchase Order (Logistics does that); see another site's stock - though they may request material from one.

Scope: one site.

2.3.5 Head of Qualities (QC-HOD) - what this role can and cannot do

Can: open the Quality Oversight page and nothing else. Across every site and warehouse, but only for Surface Shield material: the controlled category is filtered in the database on every query, so it is the boundary of the role rather than a filter on a page. Seven views - overview, Surface Shield purchase orders, the MTC register, where it is used, stagnation and expiry, escalations, and the 90/60-day thresholds. They may raise and close escalations and retune those two thresholds.

Cannot: approve or reject a quality inspection; move, issue or receive stock; raise a PR or PO; open any other portal or page; see PPE, tools, consumables or any other category. See §23.

Scope: all sites and all warehouses; Surface Shield only.

2.3.6 Logistics Coordinator - what Logistics can and cannot do

Can: run the whole procurement chain from the Logistics Portal - receive submitted PRs, raise Purchase Orders, assign POs to warehouses, manage vendors, track DN status; open Records (inventory, ledgers, POs, PRs), Reports, the Warehouse Portal, Lining Coverage, the Rack Locator, PPE Forecast, and quality inspections; create QC accounts.

Cannot: open the HOD Portal, the Entry Log, the Supervisor Portal, the Material Estimator, Man-Hours or the Admin Portal - outranking a role does not open its workspace.

Scope: all sites, no site lock.

2.3.7 Warehouse Operator - what a Warehouse User can and cannot do

Can: receive against POs, prepare and dispatch Delivery Notes from the Warehouse Portal; open the Dashboard, Stock, Rack Locator and Assets; read quality inspections; create QC accounts for their warehouse.

Cannot: open the Logistics Portal, the HOD Portal, the Entry Log, Reports, Records, the Material Estimator or Man-Hours.

Scope: one warehouse.

2.3.8 Auditor - what a view-only Auditor can and cannot do

Can: read across every site - Dashboard, Stock, Assets, Records, Reports (including downloads), Lining Coverage, the Material Estimator, the HOD Portal's read-only views (Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site Requests), Employees and quality inspections.

Cannot: change anything at all, anywhere. Every write route is refused by the server, not merely hidden. Not Man-Hours, not the Entry Log, not any approval. See §20.

Scope: all sites, read-only.

2.3.9 Administrator - what an Admin can do

Everything, including every other role's workspace as a support shadow, plus the Admin Portal: users, access requests, overdue actions, inventory master, the audit log and the console.

2.4 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
Quality Control (QC)	Whichever single place they were created against: one site, or one warehouse. A site QC sees inspections raised at that site and has no warehouse business at all; a warehouse QC is pinned to its warehouse exactly as a Warehouse User is. An account with no binding, or with both, sees an empty list - it never falls back to seeing everything. Moving a QC to another site is a request, decided by an Admin, not something the QC or their HOD can do alone.
HOD	Their own site only - but they can REQUEST material from other sites (Cross-Site tab) and submit PRs to Logistics.
Logistics	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Head of Qualities	All sites globally, but ONLY Surface Shield material - every query on their page is filtered to the controlled category in the database, so the category is the boundary of the role rather than a filter on a page. They read, and they send escalations; they cannot approve an inspection, move stock, raise a PR or open any other portal. See §23.
Auditor	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It can open the Dashboard, Stock, Records, Reports and Lining Coverage and change nothing anywhere. See §20.

Role	What they see
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar when shadowing the Warehouse Portal.

2.5 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics, warehouse_user, qc, qc_hod or auditor accounts are seeded. All of them are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if

Element	Purpose	Notes
		clear)
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
Standard header	Live Dashboard, Entry Log	Gold subtitle accent + current date
HOD header	HOD Portal	Purple subtitle accent + current date
Admin header	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

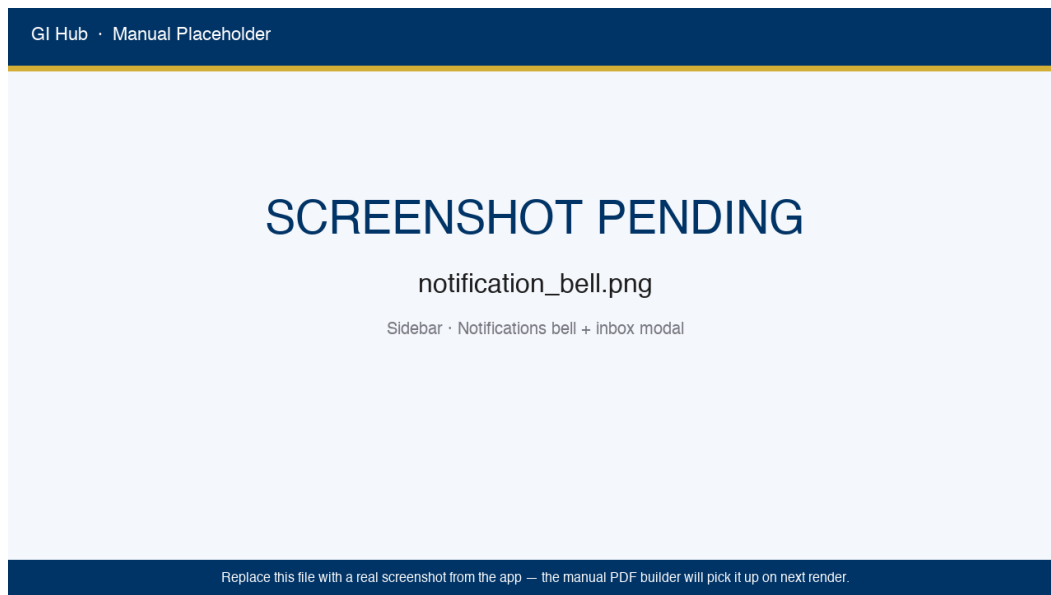
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

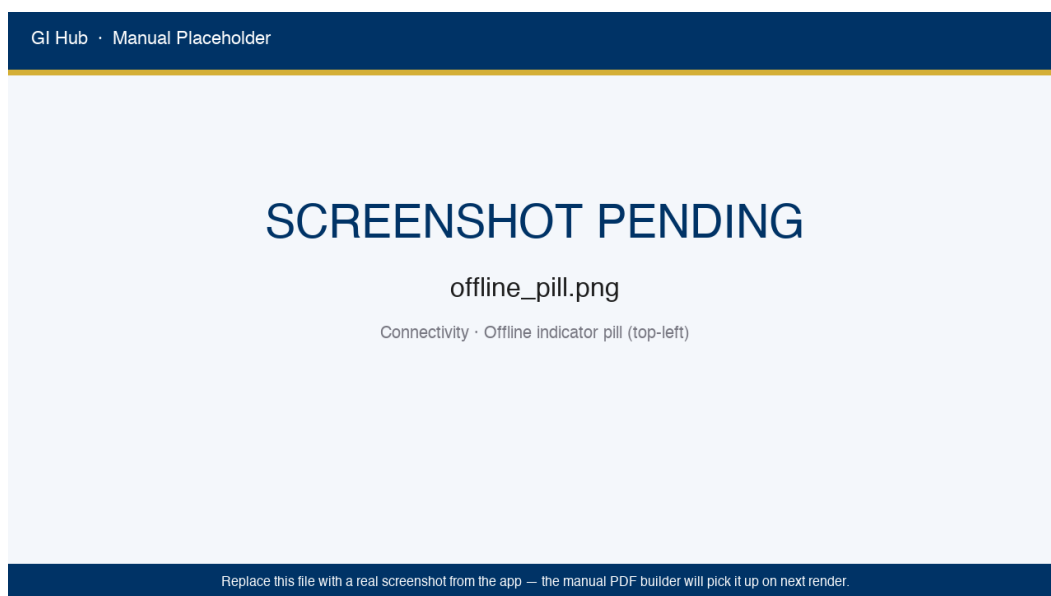
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
- Critical (red) - force-closures, T-0 delivery reminders
- Warning (amber) - T-1/T-2 reminders, reschedule requests
- Success (green) - DN approved / received successfully
- Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification

- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if the notification lookup fails, the badge silently shows 0 instead of breaking the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by the per-event WhatsApp toggles in the system settings (see §9).

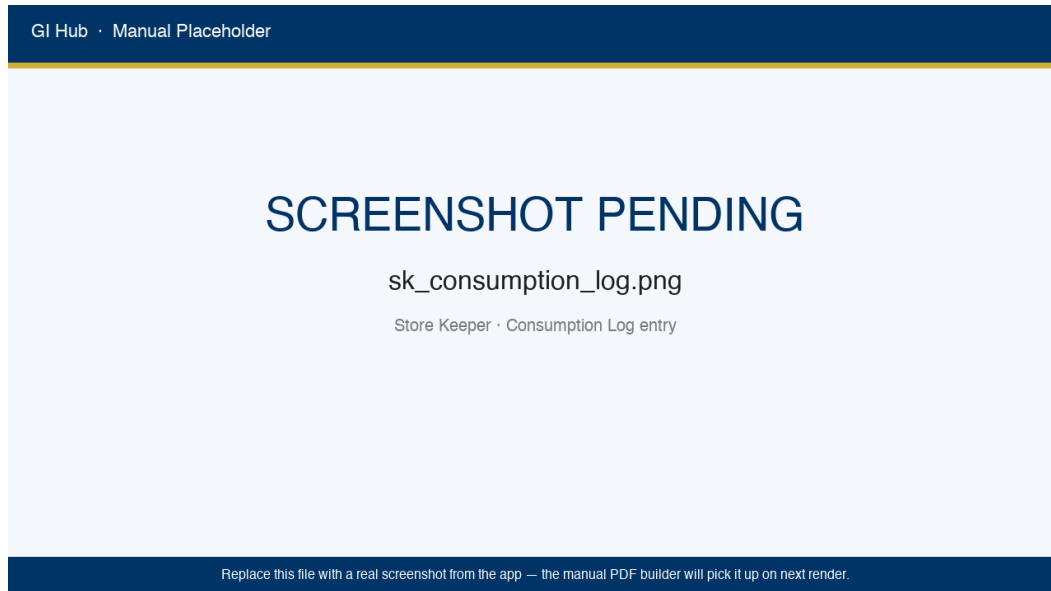
3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

4. Store Keeper Manual

The Store Keeper is the warehouse-floor operator. They see only the Entry Log page and the sidebar shell.



Store Keeper Consumption Log entry - main daily workflow

4.1 Pages visible

- Entry Log
- Locator - find which rack a material is on, or scan a rack to see what should be on it (see §21.8)
- Assets - the serialised tools and equipment register, with its own record and location history for each physical item (see §21.7)

The Locator and Assets pages are available to every role, Store Keeper included. They are deliberately placed at the top of the menu rather than buried inside a portal, because the people who need them are usually standing in the warehouse holding a scanner.

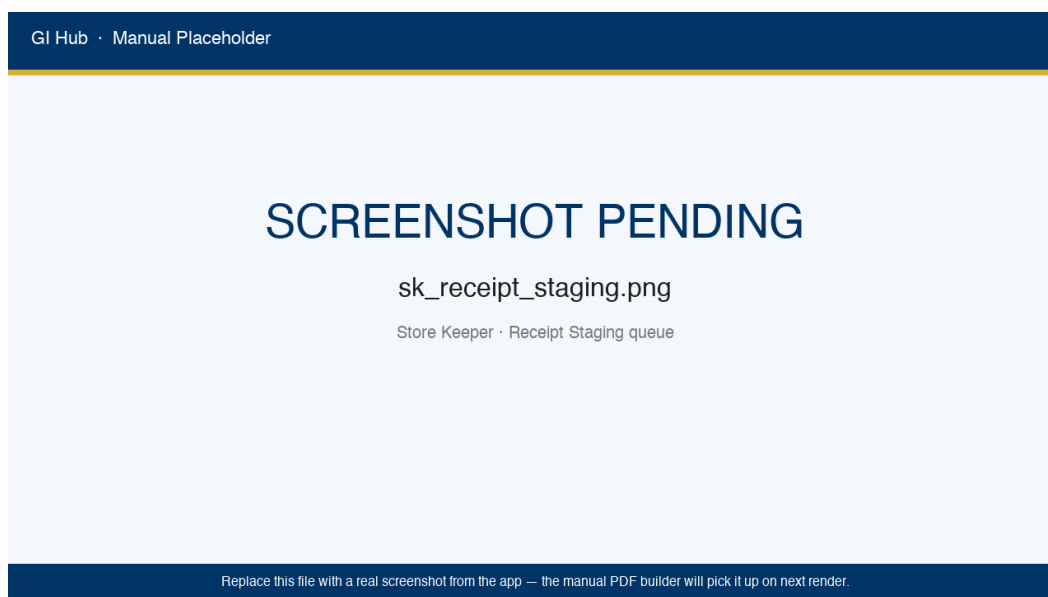
4.2 Entry Log - Tab structure

The Entry Log has four tabs:

1. Consumption Log - record material issued out
2. Receipt Staging - record material arriving in
3. Returnable Items - track tools/items temporarily issued
4. Stock Count - submit physical-count reconciliations

4.3 Entry Log -> Consumption Log

This is where every material consumed by site operations is recorded.



Receipt Staging queue - review incoming materials before submitting to HOD



Supervisor Requests tab - approve / reject worker material requests

4.3.1 Top section: Bulk OCR upload (expander)

Upload Handwritten Consumption List (OCR) - for when you have a handwritten list to bulk-stage instead of typing each row.

Element	Purpose
Input method radio	Switch between "Image upload (vision AI)" and "Paste text"
File uploader	Upload PNG/JPG/JPEG/WEBP of handwritten list (vision model reads it)
Extract rows from image button	Triggers vision AI extraction -> preview rows

Element	Purpose
Paste text area	Alternative - paste tab/comma/pipe-separated rows
Parse pasted rows button	Parse the textarea
OCR Review grid	Editable preview of extracted rows with ambiguous-match pickers
Confirm All & Stage button	Pushes all reviewed rows into pending_issues

4.3.2 Mobile camera barcode/QR scanner (expander)

Barcode / QR Scanner - opens the browser camera and reads SAP codes off labels.

Element	Purpose
Camera feed	Live preview (requires camera permission)
Manual SAP code entry text box	Fallback if camera unavailable - also accepts pasted scanner output
Detected code display	Shows the last successfully scanned code in green

When a code is scanned, the material selectbox auto-populates so you don't have to search.

4.3.3 Scan / Add New Item to Queue (main expander, expanded by default)

This is the workhorse panel. Reading top to bottom:

A. Recently-scanned ring buffer

Element	Purpose
Recent: pills	Up to 5 quick-tap buttons showing your last 5 items (SAP code + truncated description). Tapping one auto-fills the material selectbox.

B. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Type-ahead search across the inventory master. Format: [SAP_Code] Description.

When you pick a material, two cards appear automatically:

C. Item Snapshot card

A compact dark card showing:

- SAP code (gold monospace)
- Material description (bold)
- 3-stat strip:
- SITE STOCK (color-coded: red=empty, amber=below min, green=ok)
- 30-DAY BURN (units consumed last 30 days)

- DAILY RATE
- Inline 70x22 SVG sparkline - last 30 days consumption trend
- Status badge bottom-right (OK / Low / Below Min / Empty)

D. FEFO panel (lot suggestion)

If lots exist for this item at your site:

Element	Purpose
FEFO - First Expiry, First Out header (amber)	Banner
Per-lot row	Lot number (monospace) - xqty (dim) - "Exp <date>" (amber if <90d, red if expired) - USE FIRST badge on the first row

The system will automatically attach the top-FEFO lot number to your consumption row when you submit.

E. Pull from a different lot (FEFO override) - expander

Only renders when 2 or more open lots exist for this item at your site.

Element	Purpose
Lot to pull from instead selectbox	Default: "Keep FEFO suggestion". Other options: all other open lots with expiry + remaining qty shown
Reason for override text box (200 char max)	Mandatory. Min 5 characters to activate the override
Amber confirmation banner	Appears once override is active: "FEFO override active: pulling from <chosen> instead of <suggested>"

When activated:

- Your consumption row gets Lot_Number = chosen + FEFO_Override = reason
- Audit log entry FEFO_OVERRIDE is written with full context
- A WhatsApp alert is queued to the site HOD in real time

F. 2. Fill Entry Details (dynamic form)

The form shows a field for every detail an issue entry carries. Every required field is marked with an asterisk.

Field	Type	Purpose
Date	Date picker (default: today)	When the consumption happened
Quantity	Number input (min 0.1)	How much you're issuing. Above this field is a stock badge showing current site stock; a red [!] warning appears if you type more than available.
Work_Type	Selectbox (sourced from system_settings.Work_Type)	Classification: PM Work, Breakdown, Project, Shutdown, Inspection, etc.

Field	Type	Purpose
Issued_By	Text (auto-filled with your username)	Who handed it over
Issued_To	Text (smart-defaults from your last entry)	Recipient personnel/department
Tank_No	Text	Equipment tag if applicable
Serial_No	Text (optional)	Item serial number if tracked
PR_Number	Text (smart-default)	Linked purchase request if applicable
Remarks	Text (optional)	Free-form notes

Hidden side-effects:

- Site_ID is auto-set to your site
- Lot_Number is auto-set to FEFO top suggestion (or your override)
- status is set to 'draft' until you submit

G. [!] Override Expiry Warning (amber card with checkbox)

If the system detects a short-dated batch at your site for this item, it will hard-block the Add to Grid button until you check this box, confirming you've physically pulled from the expiring batch first.

H. Add to Grid button

Validates that:

- A material is selected
- No required fields are empty
- Quantity <= current site stock (over-issue guard)
- Expiry override is checked when needed

On success: the entry is added to your staging queue as a draft, a "[OK] Added to staging queue" confirmation appears, and the page refreshes.

4.3.4 Staging Queue section

Element	Purpose
Staging Queue header + count badge	Shows how many draft rows you've accumulated
Data editor table	Edit any row inline before submitting
Save Draft Edits button	Persists edits in-place (rows stay as drafts)
Submit Grid to HOD button	Moves every draft row to awaiting-HOD-approval and queues a WhatsApp alert to the site HOD with the full item list

After submission, the rows are locked from your view - they appear in HOD Portal -> EOD Commit tab for approval.

4.4 Entry Log -> Receipt Staging

For when materials arrive at your site.

4.4.1 Upload Delivery Note (OCR) - expander

Identical lane structure to consumption OCR - upload an image of the delivery note OR paste text. Vision AI extracts rows for bulk staging.

4.4.2 Add Receipt to Queue - expander (expanded by default)

A. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Same type-ahead search as consumption

When picked, a blue-tinted info card shows:

- Mat Code
- UOM

B. 2. Fill Receipt Details (dynamic form)

Field	Type	Purpose
Date	Date picker (today)	Delivery date
Quantity	Number input	Units received
Supplier	Text (optional)	Vendor name
Lot_Number	Text (optional)	If blank + Expiry_Date is set, system auto-generates LOT-YYYYMMDD-SAP
Expiry_Date	Date picker (optional)	Lot expiry - when set, triggers lot master entry
PR_Number	Text (optional)	Link to an open Purchase Request
Serial_No / Vehicle_No / DN_No / Pallet_No / Mob_From / Mob_To / Prepared_by / Driver_Name	Text (optional)	Logistics tracking fields
Remarks	Text	Free notes

C. Add to Receipt Queue button

Checks that a material is picked and every required field is filled, then adds the line to your receipt draft queue.

4.4.0 Incoming Delivery Notes from Warehouse (NEW in v3.0)

A new expander appears at the TOP of the Receipt Staging tab. It's only populated when the Warehouse has prepared a DN bound for your site AND your HOD has approved it. If empty, the expander shows:

"Nothing inbound. Logistics -> HOD-approved DNs will appear here for you to confirm physical receipt."

When DNs are inbound, each appears as its own container:

Element	Purpose
DN header line	DN <number> - PO <number> - Warehouse <id> - DN Date <date>
Line count + total qty	At-a-glance: how many SKUs, how many units total
View lines expander	Material_Code, Description, Qty, UOM, Lot_Number, Expiry_Date, Remarks - read-only preview of every line
[OK] Mark as Received button	Confirms physical receipt. One receipt is recorded per delivery line, each keeping its delivery note, warehouse and originating order so the trail is complete; the delivery closes and leaves your list. Stock figures update immediately - the Live Dashboard shows the new quantity the same minute.

When to use:

- The Warehouse delivered the materials physically to your site
- HOD already approved the DN content (you'll see it appear without any action)
- You've inspected the goods and they match the DN qty + lot

When NOT to use:

- Direct deliveries from supplier to your site (those go through the existing Add Receipt to Queue -> Submit to HOD flow below; the email-driven PR/PO path remains supported)
- Partial receipts (the current flow assumes you confirm the full DN qty - for partials, raise a Vendor Return on the diff after confirming and ask your HOD)

This is the FINAL step in the procurement chain. After you click Mark as Received:

- Logistics sees a "DN received at site" entry in their notifications
- Warehouse sees the same
- The PO and PR move toward closure if this DN completes the order

4.4.3 Receipt Draft Queue section

Element	Purpose
Receipt Draft Queue header + count badge	Number of drafts you've accumulated
Data editor table	Inline edit before submit
Save Draft Edits button	Persists
Submit to HOD for Approval button	Moves your drafts to awaiting-HOD-approval and sends the HOD a WhatsApp alert with the item list

4.5 Entry Log -> Returnable Items

For tools, gauges, fittings temporarily handed to personnel - items that should come back.

4.5.0 Smart Scan workflow (new - 2026-06)

A two-step camera flow sits at the top of the tab. Use it to skip typing borrower + tool details by hand; the existing manual form still works exactly as before and stays visible underneath.

Step 1 - Scan the employee badge

1. Click into the Smart Scan (Beta) expander.
2. Hold the borrower's printed badge in front of the camera. (Badges are generated in Admin Portal -> Employees -> Roster + Badges - see §7.6.)
3. The system decodes the QR (which carries only the ID_Number, no PII), looks it up in the Employees master, and shows:
 - [OK] Green success card - EMP-1042 - Ahmed Ali (+9665...) - the borrower's name + phone are now staged.
 - [X] Red error card - Badge couldn't be read OR the employee is inactive/suspended. Re-snap or use the manual form.

Step 2 - Scan the tool

1. After a successful badge scan, a second camera input appears. Point it at the tool.
2. The active image-recognition model (managed in Admin Portal -> Tool Catalogue, which is also where a new model is trained and promoted) classifies the image. Three outcomes:
 - Confidence ≥ 0.75 -> [OK] Auto-fill. The detected tool name + borrower details flow into the manual form below; review and click "Issue Item".
 - Confidence 0.30-0.74 -> [!] A "Top candidates" radio appears. Pick the right one and click "Use this tool". The form pre-fills.
 - No active CV model OR confidence < 0.30 -> An info banner explains the fallback. Borrower fields stay pre-filled; type the tool name into the manual form yourself.

Click "Start over (clear scan)" any time to reset and scan a different borrower.

What happens if the AI is wrong

You're never locked in. Every Smart Scan write-through goes into the manual form's fields - Name, Phone, Material name. Edit anything that's wrong before clicking "Issue Item". The submitted record records that the camera made the identification, and how confident it was, so admin reports can later show adoption and accuracy. Manually-corrected loans are not tagged as camera-identified - that's intentional, see §10 for the data model details.

Automatic WhatsApp reminders (Phase 6E)

Once a loan is in the system, the WhatsApp worker fires four reminders automatically. The cadence and recipient list is:

Offset	Severity	In-app	WhatsApp
T-2h (2 hours before due)	info	Site SK badge	Borrower
T-0 (due now)	warning	Site SK badge	Borrower
T+2h (2 hours overdue)	warning	Site SK badge	Borrower + every Site SK
T+24h (24 hours overdue)	critical	Site SK + Supervisor badges	Borrower + every Site SK + every Site Supervisor

The borrower's phone is taken first from the Employees master (for badge-scanned loans) and falls back to the phone number typed on the loan itself for older or manually-entered loans. The system remembers which reminders it has already sent for each loan, so restarting the reminder service mid-day never causes double-fires. Admins can mute any of the four events with the per-event WhatsApp toggles if a particular escalation gets too noisy.

4.5.1 Issue a Returnable Item - expander (expanded by default)

Field	Type	Purpose
Material / Tool Name	Text	What you're handing out
UOM	Text (e.g., Pcs, Set)	Unit
Quantity	Number (min 0.1)	How many
Borrower Name	Text	Person taking custody
Borrower WhatsApp Number	Text (optional, +966 ...)	For overdue alerts
Expected Return Date	Date (default: tomorrow)	When you expect it back
Expected Return Time	Selectbox: 04:15 PM, 06:15 PM, Custom Time...	Time-of-day expectation
Issue Item button	Validates name+borrower, writes to returnable_items	

4.5.2 Currently Borrowed Items section

Shows all borrowed (not yet returned) items at your site.

- Overdue red banner - when one or more items are past their expected return time
- Borrowed items table (styled HTML) with columns: ID, Material, UOM, Qty, Borrower, Phone, Given Time, Expected Return, Status (pill: green "On Loan" / red "[!] Overdue")
- Mark as Returned section:
 - Selectbox of borrowed items
 - [OK] Mark as Returned button - marks the loan returned, refreshes the figures, and confirms on screen

4.6 Entry Log -> Stock Count

For reconciling physical shelf count with system stock when they don't match (damage, expiry disposal,

miscount, found-extra).

4.6.1 1. Select Material to Count

Element	Purpose
Search by SAP Code or Description selectbox	Pick the material to reconcile
Help caption	Explains when to use this tab

4.6.2 2. Enter Count Details (only renders after picking a material)

Left column

Element	Purpose
System Qty card	Read-only dashboard showing current stock at your site (auto-fetched)
Counted Qty input	What you actually count on the shelf right now (defaults to system qty)
Variance preview banner	Color-coded: green " Found N extra" or red " Short by N" or grey "No variance"

Right column

Element	Purpose
Reason Code selectbox	9 options. Default depends on variance direction.
Notes textarea (300 char max)	Free explanation: "found behind shelf 3", "damaged in transit", etc.

Reason codes:

Code	Label
cycle_count	Cycle count correction
damaged	Damaged / unusable
expired_disposal	Expired - disposed
miscount_in	Miscount - found extra
miscount_out	Miscount - short
lost	Lost / unaccounted
theft	Suspected theft
return_to_supplier	<- Returned to supplier
other	Other (see notes)

4.6.3 Submit section

Element	Purpose
Amber warning banner	"Submitting this sends the count to your HOD for approval. No stock changes until they approve."
Submit Count for HOD Approval button	Disabled while the variance is 0. On success the count is filed as a stock adjustment awaiting HOD approval, a WhatsApp with the count details goes to the site HOD, and the submission is audit-logged.

4.6.4 Recent Adjustments at This Site (history)

Read-only table showing: ID, SAP Code, Material Name, Variance, Reason, Status, Submitted By, Submitted At.

4.7 Store Keeper - Use Cases

Use Case 1: Issue 10 Pipe Gaskets to PM Work

1. Log in -> Entry Log -> Consumption Log tab
2. Open Scan / Add New Item to Queue
3. Search "Pipe Gasket" -> pick from dropdown
4. Read the stock badge (e.g., " 156 Pcs"), confirm FEFO lot suggestion
5. Quantity: 10
6. Work_Type: PM Work
7. Issued_To: name of recipient
8. Tank_No / PR_Number: as applicable
9. Click Add to Grid
10. Repeat for any other items in this batch
11. Review the Staging Queue table -> edit if needed -> click Submit Grid to HOD

The row(s) are now in HOD's EOD queue. Stock won't decrease until HOD commits EOD.

Use Case 2: Receive a delivery of 200 units with expiry date

1. Entry Log -> Receipt Staging tab
2. Open Add Receipt to Queue
3. Pick material, fill Quantity (200), Supplier, Expiry_Date (e.g., 2027-06-11)
4. Lot_Number: leave blank (system will generate LOT-20260611-<SAP>) OR type the supplier's lot ID
5. PR_Number: link if applicable
6. Click Add to Receipt Queue
7. Click Submit to HOD for Approval when done staging

Use Case 3: Found 5 extra units of an item (physical count > system)

1. Entry Log -> Stock Count tab

2. Pick the item from the dropdown
3. Read the System Qty card (e.g., "95")
4. Counted Qty: 100
5. The variance banner turns green: " Found 5 extra"
6. Reason: Miscount - found extra
7. Notes: e.g., "Found in box behind shelf 3"
8. Click Submit Count for HOD Approval

HOD will approve in HOD Portal -> Adjustments. After approval, your live stock matches reality and a synthetic receipt row of +5 is posted to the ledger.

Use Case 4: Issue a tool to a borrower

1. Entry Log -> Returnable Items tab
2. Issue a Returnable Item
3. Material: Torque Wrench 1/2", UOM: Pcs, Qty: 1
4. Borrower Name, WhatsApp Number, Expected Return: today 06:15 PM
5. Click Issue Item

If the item is not returned by 06:15 PM, the borrower (and you) get a WhatsApp alert.

Use Case 5: FEFO override (the right bin is blocked)

1. Entry Log -> Consumption Log -> pick material with 2+ open lots
2. Scroll to FEFO panel - note the suggested lot (earliest expiry)
3. Open Pull from a different lot expander
4. Pick a different lot from the dropdown
5. Type a reason >= 5 chars (e.g., "FEFO bin blocked by pallet, clearing 1700")
6. Amber confirmation banner appears
7. Fill quantity + other fields
8. Click Add to Grid

Your HOD gets a real-time WhatsApp alert documenting the override.

4.8 Store Keeper - FAQ

Q: I can't see Live Dashboard, HOD Portal, or Reports. Did I lose access? A: No - Store Keepers only have access to Entry Log by design. If you need to view stock, ask your Supervisor or HOD.

Q: I submitted to HOD by mistake. Can I cancel? A: Not directly. Contact your HOD - they can reject the row in their EOD Commit review.

Q: The "Add to Grid" button is disabled or the form rejects my entry. A: Check three things: (a) is quantity > current stock? (b) is the expiry override needed but unchecked? (c) are all required * fields filled?

Q: I tried to issue 20 but the system says only 15 available. A: Believe the system. Either physically recount, or if there is genuinely 20 and the system disagrees, use the Stock Count tab to log a +5 adjustment.

Q: I don't have a barcode scanner. How do I quick-find items? A: Type a partial SAP code or description in the selectbox - it has type-ahead search. Or use the Recently-scanned pills if you've used the item recently.

Q: WhatsApp alerts aren't reaching the HOD when I submit. A: HOD's phone number must be set in their user profile. Tell the HOD to update it via Admin (or check with Admin directly).

Q: My borrowed item is overdue but I haven't received an alert. A: WhatsApp alerts are sent by a background service. Ask your Admin to confirm it is running - queued messages send as soon as it is back, so nothing is lost in the meantime.

Q: I uploaded an OCR image and it extracted wrong items. A: The OCR review grid lets you fix every row before staging. Pick the correct material from the suggested-matches dropdown, or type it manually.

Q: How do I see my recent activity? A: Submitted/approved items don't show on your screen. Ask HOD or Admin to filter the audit log by your username.

4.9 Printing a consumption form

Where: Execution Entries (/execution) -> Print a consumption form. Store Keeper, Supervisor and HOD.

The field fills these in by hand and photographs them; the app reads the photo. Everything about the form's design exists to make that reading reliable.

4.9.1 What is already printed, and what you write

The form prints every material for the system - name, component, SAP code and unit - so nobody ever writes a material name by hand. Reading handwritten names is the one thing the vision model is genuinely bad at, and a name that is already printed cannot be misread.

You write three things at the top - Date, Equipment / Tank No. and Area done (m²) - then, on every row, the quantity used and the lot / batch number for that material, and your name at the bottom.

[!] The lot is per row, not per form. One system draws several materials and each one arrives from a different batch, with its own certificate. Writing a single batch number at the top would be right about one material and wrong about the rest - and the certificate check at approval looks at the lot for each material, so a batch on the wrong line clears a check for material that was never used.

[!] The four small black squares in the corners are not decoration. They are what lets the app square up your photograph and show you the right row. Keep them in frame.

[!] The date box is blank on purpose. Forms are printed in batches and used today or tomorrow, so a pre-printed date would be wrong on half of them - and the date decides which day's progress your work lands on. The small date in the footer is when the blank was printed: it tells you if you are holding old paper.

[!] There are no spare rows. Only materials in the system's recipe can be recorded. Write 0 for anything you did not use; never add a material by hand, because there is no box for it and the app cannot match it.

4.9.2 [!] Rows that look almost identical

Some systems list the same product several times as separate components. LSC8 prints "Cumicrete PU MF 300 - 3mm" four times - as Comp-A, Comp-B, Comp-C and Comp-D, each with its own SAP code.

They are different materials with different quantities. Check the component letter, not just the product name.

4.9.3 The QR code

The square in the top-right holds the site, the system, the sub-activity, this sheet's own number, and which page of the form it is. It is read by a scanner, not by the AI, which is why none of those things can be got wrong.

[!] Photograph the whole page, including the QR. A photo that crops it out cannot be matched to anything, and you will be asked to retake it.

[!] A long form has more than one page, and each page has its own code. Photograph and upload every page. Upload them in any order - they all join the same entry. Until they are all in, the missing pages' materials read 0, and the app tells you which page it is still waiting for.

Do not submit an entry that is still waiting for a page. Those materials would be recorded as unused, and the figures would look entirely reasonable.

4.9.4 [!] Every download is a new sheet

Downloading the form twice gives you two different sheets with two different numbers - not two copies of one form.

That is deliberate. The app has to be able to tell "you printed a second sheet" from "you photographed the same sheet twice", and it does that by the number on the paper. If you print a spare, it is a genuinely separate form.

An HOD can see every form printed and not yet filed, and who printed it.

4.9.5 Printing several at once - and [!] never photocopying one

Set Forms to the number you need and press Download. You get one PDF containing that many forms, each with its own number and its own QR code, and each printed SHEET 3 OF 50 at the top so you can sort the pile and see if one is missing.

[!] Never photocopy a form. This is the most important sentence on this page. A photocopy repeats the QR code, and the QR code is how the app knows which sheet it is looking at. Two sheets carrying one code cannot be told apart: the second one you send in will either be refused as already filed, or worse, filed against the first one's tank. If you need fifty, print fifty here.

[!] The number is FORMS, not pieces of paper. A system with more than 18 materials prints as several pages - that is still one form, with one QR across its pages. The line under the box does the arithmetic for you and changes as you type:

> 50 forms x 2 pages = 100 A4 sheets

Check that line before you send it to a printer.

You can print up to 200 forms in one go. Above that, print in runs: a download that fails halfway leaves the app holding records for paper that never reached you, and those show as outstanding sheets until somebody clears them.

4.9.6 If the materials change after you print

If someone edits the system's recipe after your form was printed - adds a material, reorders them, changes a code - the app will refuse the photo and ask for a fresh form.

It is not being awkward. The app matches your handwriting to materials by row position, so a sheet printed against different rows would file your quantities against the wrong materials - and the numbers would look perfectly reasonable. Changing a material's rate is fine and does not invalidate printed paper.

4.9a Surface Shield material issued from the store - where its area is recorded

Phase 13.

The gap this closes. When a Surface Shield material is issued from the general Inventory, the ledger records that a drum left the shelf. It does not record what that drum covered - which system, which vessel, how many square metres. So the estimator could not see it and nobody could compare it against the recipe.

Every such issue now appears in a queue until somebody says what it was for.

4.9a.1 What the Store Keeper does

Nothing new, and one optional extra.

You already pick the Lining System Code before issuing a Surface Shields material - that has not changed. Beside it there is now an Equipment / tank box, filtered to the equipment that system actually applies to.

[!] Leave it blank if you do not know. You are handing over a drum; where it ends up is often decided in the plant. A guess here becomes a wrong attribution later, and the queue will simply ask. A blank is not a failure.

[!] You are never asked for the area. You cannot know it - the drum leaves before it is applied. That question belongs to the person who applied it.

4.9a.2 What the field does - the queue

Where: Execution Entries -> the Surface Shield consumption queue. Store Keeper, Supervisor and HOD.

The queue lists every Surface Shield consumption with no area recorded against it, oldest first. For each one you supply three things:

System Code	Pre-selected when the store keeper already chose one. Otherwise pick from the list - it offers only the systems whose recipe actually contains that material.
Equipment / tank	Filtered to the equipment carrying the system code you picked.
Area covered (m ²)	What that material actually covered.

[!] The system code is asked, never guessed. If the app cannot tell which system a draw was for, it offers you the candidates rather than choosing one. A guessed code compares the draw against the wrong benchmark - and a wrong benchmark looks finished, where a blank looks unfinished. Unfinished is safer.

[!] The queue reaches back. It is not an inbox that starts today: it holds every unattributed Surface Shield consumption, including issues from months ago and rows brought in by an Excel import. Expect it to be long the first time you open it, and work oldest first.

[!] Consumption filed from a printed form is NOT in this queue, and must not be. A photographed consumption form already recorded its system, its equipment and its area, and the area has already been credited. Adding it here would count the same drum against the same vessel twice.

4.9a.3 What happens to the numbers

Nothing is deducted. The stock left the shelf when it was issued; this records what it was for, not that it went.

Each submission is compared against the recipe: rate x area is what the system expected, and the difference is the variance. Every row then goes to the HOD - see §4.9a.4.

> [!] Your estimator figures do not move. Readiness, completion, achievable > area and the buy list are calculated from the Material Estimator's own > workbook figures and are untouched by any of this. What you gain is > visibility of what was actually drawn, beside the plan.

4.9a.4 [!] Every row goes to the HOD - the +/-10% band decides the order

Every Surface Shield consumption is reviewed, whatever the variance. There is no band inside which a row files itself.

What the +/-10% tolerance does is set priority. A draw more than 10% above or below the benchmark is flagged High Priority and sits at the top of the HOD's queue; the rest follow, oldest first.

[!] A variance that cannot be calculated is High Priority too. If the recipe has no line for that material in that system, there is no benchmark to compare against - and "we cannot work this out" is exactly the thing worth looking at. It is not treated as zero.

An admin can change the band. Doing so re-orders the queue and changes nothing about rows already filed - each one keeps the reading it was measured against at the time.

4.10 Filing a consumption form

Where: Execution Entries (/execution).

This replaced the old store-keeper-first flow on 27 August 2026. The record now starts with the paper you filled in.

You fill the form -> Store Keeper verifies -> HOD approves

4.10.1 Step 1 - photograph it

Upload the photo under Upload a filled form. JPG, PNG, HEIC or PDF.

[!] Photograph the whole page, including the QR code. The QR is what tells the app which form this is - a photo without it cannot be matched to anything and will be refused.

[!] Only forms printed from "Print a consumption form" belong here. A handwritten consumables sheet or a supplier's delivery note has no QR code and never will. Those are read in Entry -> OCR Import instead - and if you upload one here by mistake, the refusal will say so and link you to the right page.

Reading a page takes minutes, not seconds, and the card now tells you how many. You will see a live counter and the usual time for that kind of page: roughly 6 1/2 minutes for a printed consumption form, 3 1/2 for a 30-row handwritten sheet, 1 1/2 for a delivery note. A dense or badly-lit page takes longer and the card says so rather than pretending to be nearly finished.

You can leave the page; the read carries on and the result will be waiting.

> [!] If it says "This read was interrupted", it means the server process doing > the reading stopped - not that your page was bad. Waiting longer will not > help; press Read it again. This is the honest version of a problem that > used to look like the page simply never finishing: before 2026-09-02 the card > claimed "usually takes under a minute" and then showed a spinner with no > elapsed time, so a perfectly good six-minute read and a dead worker looked > identical, and people quite reasonably gave up on both.

4.10.2 Step 2 - check every figure

The app opens a draft with what it read. Beside each row is the crop of your photograph it read that number from, so you are checking against the paper and not against memory.

[!] Where the handwriting was not certain, the box is left EMPTY rather than guessed. Those rows are listed at the top and marked in gold with what was actually written. Type the real number in. A guessed figure would post straight to stock with nobody questioning it.

Then correct anything else, pick the equipment from the list if it was not read, and give the two reasons - material and manpower - which are required on every entry, even one with no variance at all.

4.10.3 Step 3 - the store keeper verifies

It goes to the store keeper, not to the HOD. They check your quantities against what actually left the shelf and may change them - every change costs a written reason, shows to the HOD in red, and you are told about it before the entry is approved rather than after.

Blasting and buffing skip this step: there is no material to verify.

4.10.4 Step 4 - the HOD approves

[!] Approval is what deducts the material and posts the area. Nothing before it moves a quantity, which is what makes correcting a figure safe at every earlier step.

The HOD sees the whole chain on each row:

Colour	What it is
grey	what the camera read
amber	what you filed
red	what the store keeper corrected
purple	what the HOD settled on

A row everybody agreed on shows one number and the word "agreed". The colours only appear where something actually changed - which is what makes a red one worth looking at.

[!] Rejection is final. A rejected entry cannot be revived; raise a new one from a fresh form.

4.10.5 [!] Store keepers: stop raising a separate issue

The execution entry is now the only way lining material leaves the ledger. Raising a material issue for the same drum as well would deduct it twice - and nothing would show it until somebody counted the shelf.

4.10.6 [!] When a certificate is missing

If a Surface Shield material has no test certificate or no quality clearance, the HOD's screen says so before they press approve, naming the lines.

The material has already been applied by then, so this is a paperwork gap rather than something anyone can prevent. The HOD may approve anyway - the button says "Approve WITHOUT clearance" and will not work until a reason is typed - and the Head of Qualities is notified every time.

Often the simpler fix is a corrected lot number: the check runs after the HOD's edits, so fixing the batch can clear it with no override at all.

4.10.7 Things the app will refuse, and why

The same sheet twice. Each printed form is filed once. Two people photographing one form, or one person retrying on a bad signal, produce different files of identical paper - the number on the sheet is the only thing that can tell.

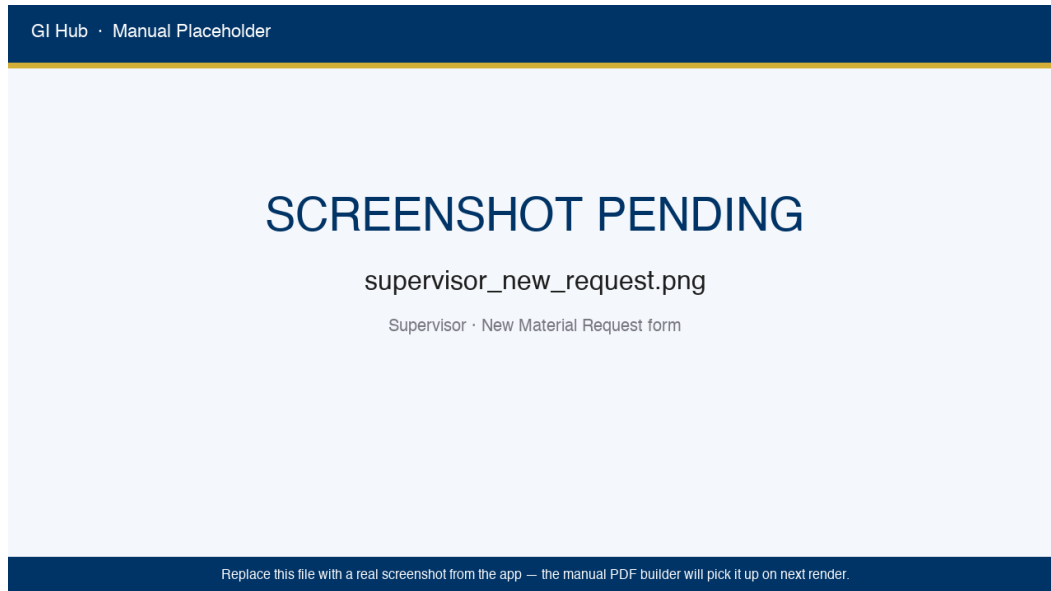
A form printed before the materials changed. If someone edits the system's recipe after your sheet was printed, the rows no longer line up. The app matches your handwriting to materials by row position, so your quantities would be filed against the wrong materials and would look entirely reasonable. Print a fresh form and copy the figures across.

A form printed for another site.

A photo it could not read. It says so rather than creating a blank entry - a blank entry that gets submitted is a consumption of zero, recorded silently.

5. Supervisor Manual

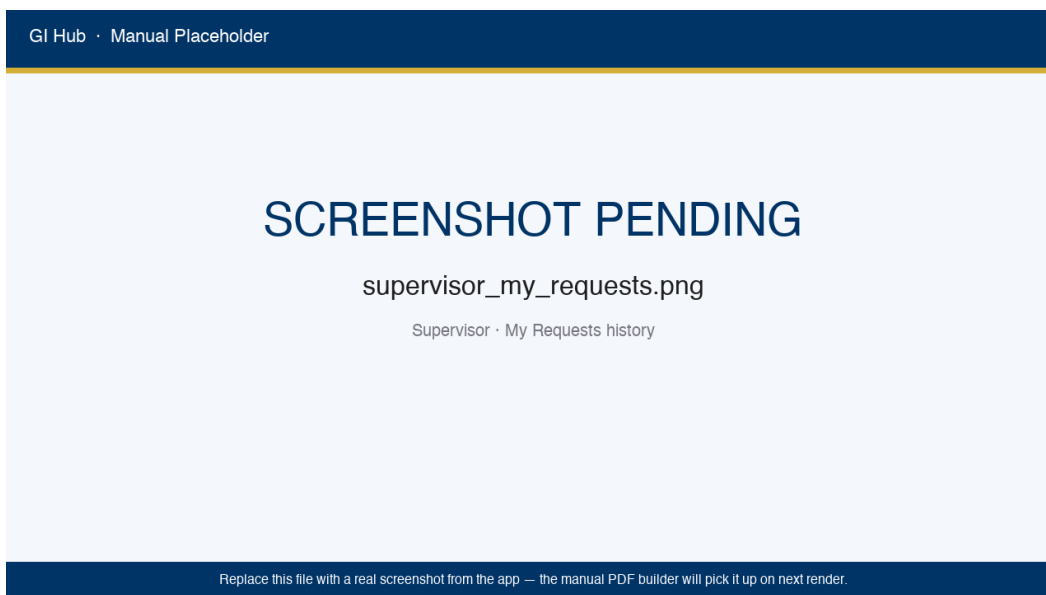
The Supervisor monitors a single site's stock, generates reports, and provides oversight. They cannot approve transactions (that's HOD).



Supervisor Portal - New Material Request form for workers

5.1 Pages visible

- Live Dashboard
- Supervisor Portal - Request Material for workers
- Entry Log (same interface as Store Keeper - see §4)



Supervisor Portal - My Requests history with cancel option



Supervisor Portal - Intent vs Actual variance report

5.2 Live Dashboard

The Live Dashboard is the at-a-glance view of every catalogue item with current stock, value, and status.

5.2.1 Brand header

Reads "Live Warehouse Stock Dashboard" - gold subtitle + today's date.

5.2.2 Hero metric strip (4 cards)

Card	Source	Tone logic
Catalogue items	Count of rows in inventory	Neutral

Card	Source	Tone logic
Total stock value	Sum of Current_Stock x Unit_Cost, all sites - formatted as SAR 1,234 / SAR 1.2M	Neutral. Delta: "standard cost - all sites" or "set Unit_Cost in Admin -> DB Editor" if 0
Below minimum	Count	Green=0, Amber<10, Red>=10. Delta: "all healthy" or "needs reorder"
Expiring / expired	Count	Green=0, Amber<10, Red>=10. Delta: "shelf-life clear" or "review HOD Portal"

5.2.3 Ask in plain English (AI search) - expander (when the local AI assistant is switched on)

Available when the local AI service is installed and running on the server:

Element	Purpose
Your question text input	Ask in ordinary words, e.g., "items below minimum at site B"
Search button	The local AI turns your question into a database lookup and runs it in read-only mode
Clear button	Resets the panel
Result table	Returned rows
Show the lookup the AI built	Reveals the read-only query the assistant produced, so you can check it before trusting the answer. It can only read - it is never allowed to change anything.

If the local AI service is not running, an amber card explains how an Admin starts it.

5.2.4 Burn alert banner

Appears at the top of the table area when there are items burning to zero within the configured window (default 7 days). Color-coded amber/red.

5.2.5 Main inventory grid

Column	Source	Notes
SAP_Code	inventory.SAP_Code	Primary key
Equipment_Description	inventory.Equipment_Description	Material name
UOM	inventory.UOM	Unit of measure
Total_Returned	Computed	Sum from returns table
Current_Stock	Computed identity math	Received - Consumed - Returned
Minimum_Qty	inventory.Minimum_Qty	Reorder threshold

Column	Source	Notes
Unit_Cost	inventory.Unit_Cost	Set via Admin -> DB Editor
Stock_Value	Computed	Current_Stock x Unit_Cost, rounded 2 dp
Status	Computed badge	OK / Low / Below Min / Empty, shown as a colour-coded pill

You can sort by any column - sorting by Stock_Value shows your biggest SAR exposure first.

5.2.6 Expanders below the grid

Expander	Content
Stock vs Minimum Threshold	Horizontal bar chart, sorted by criticality. Each bar = item; gold line markers = minimum threshold. Color: red=empty, amber=below min, orange=close, green=ok. Capped at 20 items.
Burn Rate Forecast (30-Day)	Plotly chart with daily-burn-rate bars + vertical line at the 30-day alert threshold. Colors: red <10 days, amber <30, green >30.
Top Consumed Items	Bar chart of top 10 items by 30-day consumption, blue->gold gradient.

5.3 Reports

Supervisor sees the full Reports module but with site scope locked to their site (cannot pick "All Sites"). See §8 for the full Reports reference.

5.4 Supervisor - Use Cases

Use Case 1: Daily morning check

1. Log in -> Live Dashboard
2. Read the 4 hero cards - note any below-minimum or expiring items
3. Sort grid by Stock_Value descending -> check no over-stocked items
4. Sort by Status -> spot Empty / Below Min items
5. Open Burn Rate Forecast expander -> check the next 30 days

Use Case 2: Generate end-of-month report for management

1. Reports -> Generate Report
2. Pick Monthly Summary
3. From date: 1st of month; To: today
4. Site: locked to your site

5. Format: PDF
6. Click > Generate Report
7. Review the preview (includes SAR-value columns: Issued_Value_SAR, Received_Value_SAR, Closing_Value_SAR)
8. Click v Download PDF

Use Case 3: Investigate why an item ran out

1. Live Dashboard -> search the item (or click its row)
2. Open Burn Rate expander - see the daily rate trend
3. Reports -> Burn Rate Analysis -> date range last 30 days -> generate
4. Filter the result table for the SAP code -> see daily breakdown
5. Reports -> Daily Consumption -> narrow down which Work_Type drove the spike

5.5 Supervisor - FAQ

Q: I can only see my own site. Why can't I see other sites' stock? A: By design - site isolation. Only Admin sees all sites. If you need to know another site's stock for a transfer, ask your HOD to file a Cross-Site Request.

Q: Total stock value shows SAR 0. What's wrong? A: Items have no Unit_Cost set. Ask Admin to enter costs in Admin -> Master DB Editor -> inventory table.

Q: AI search says "Ollama not running." A: Local AI is optional. An Admin has to install the local AI service and download the language model it uses. Without it, the system still works - you just type into the standard grid filters.

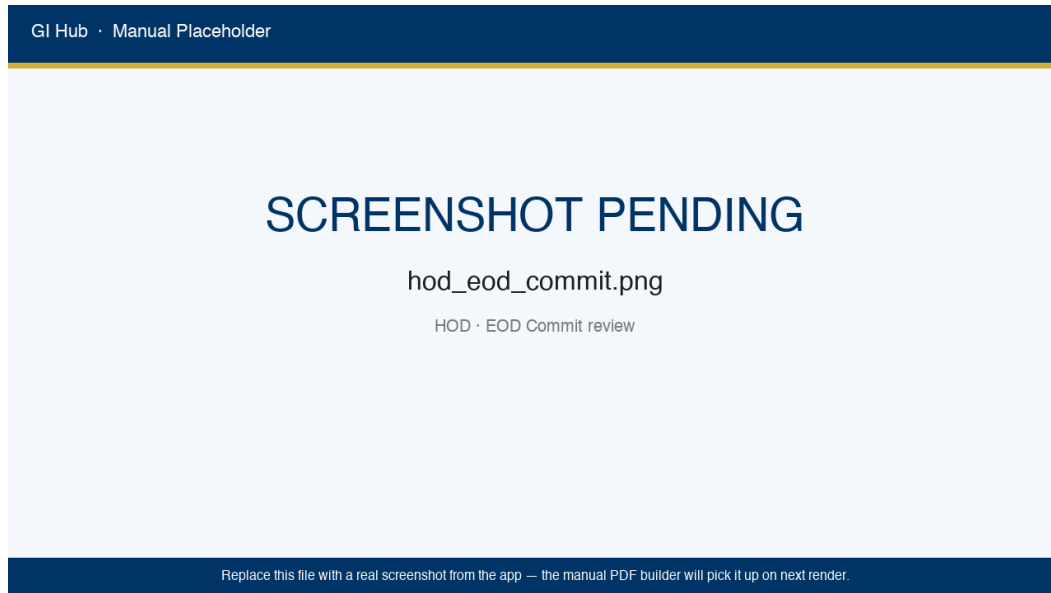
Q: The Stock_Value column is empty for some items. A: Those items have Unit_Cost = 0. The valuation is correct - they're tracked in qty but not in money.

Q: I can't approve any transactions. A: Correct. Supervisors monitor; HODs approve. Talk to your HOD.

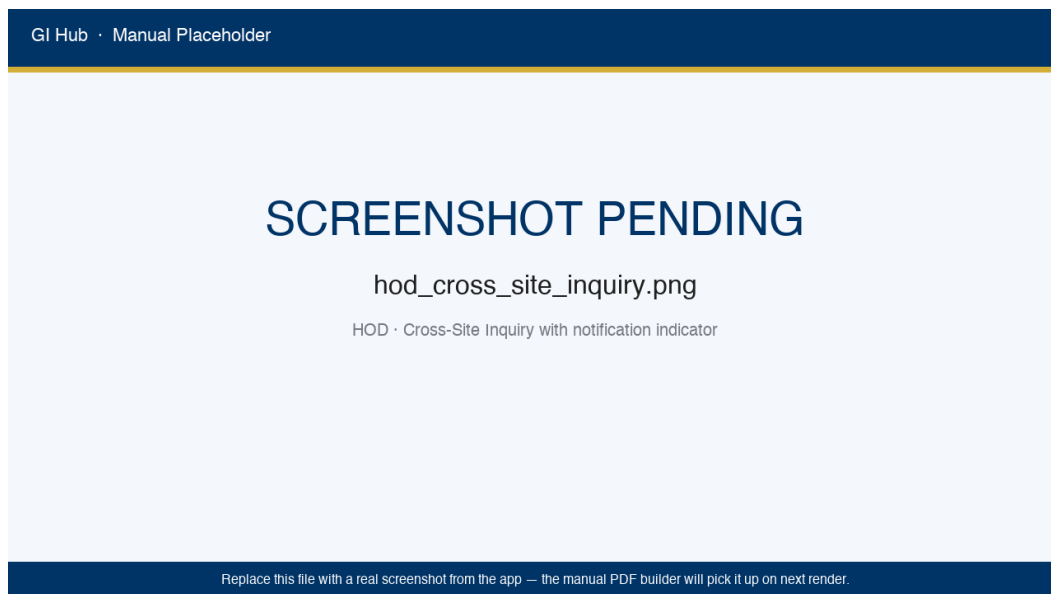
Q: Can I export the Live Dashboard grid? A: Yes - every table has a download button for CSV or Excel. For a formal, branded report use Reports and generate one.

6. HOD (Head of Department) Manual

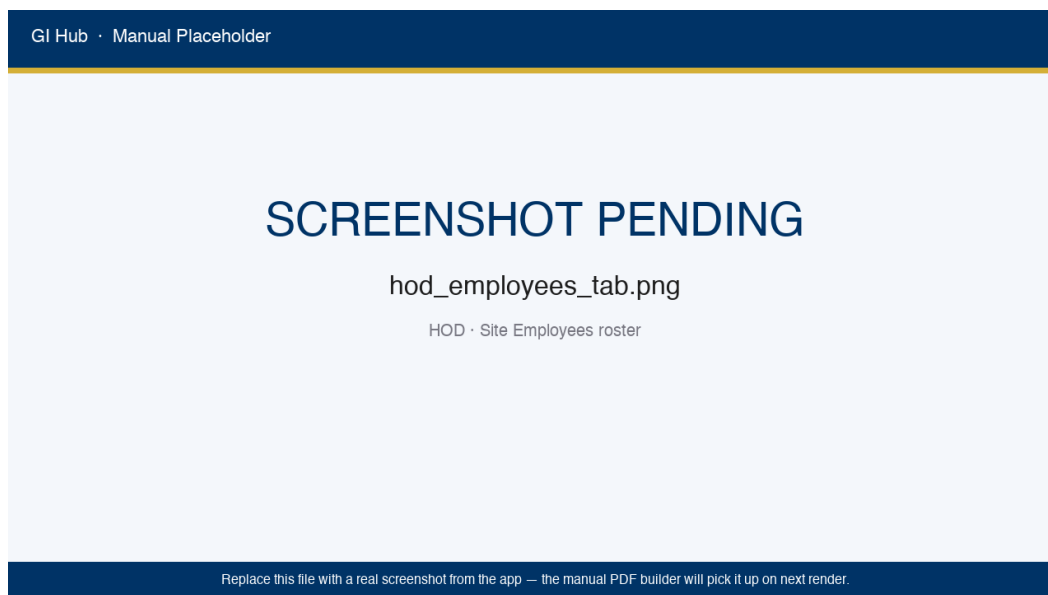
The HOD owns their site's inventory ledger. Every transaction flows through their approval. The HOD Portal has 16 tabs as of v3.0 - covering EOD, Cross-Site, Burn Rate, Receipts, Returns, Adjustments, PRs, Shelf-Life, Notifications, My Requests, Site Config, the new Employees tab (Phase 7A), DOC, QR Approval, DN Approvals, In-Transit.



HOD Portal - End-of-Day Commit review with negative-stock guard



HOD Portal - Cross-Site Inquiry with the Phase 7C "you are viewing" notification banner



HOD Portal - Site Employees roster with add/edit form (Phase 7A)

6.1 Pages visible

- Live Dashboard (see §5.2)
- Entry Log (see §4 - HODs can also stage entries)
- HOD Portal - detailed below
- Reports (see §8)

6.2 HOD Portal overview

6.2.1 Page header

Reads "HOD Management Portal" - purple subtitle + today's date.

- Page title: " HOD Portal" + "Managing Site: <SITE_ID>"

6.2.2 Hero metric strip (4 cards)

Card	Source	Notes
Site stock value	Total value of stock held at your site	Delta: SAR consumed in the last 30 days - site-scoped
Below minimum (site)	Count of items at your site below their reorder threshold	Green/amber/red
Expiring / expired	Count of short-dated and expired items at your site	Green/amber/red
Pending receipts to approve	Receipts at your site waiting on your approval	Neutral or amber

6.2.3 Tab strip (10 tabs)

1. EOD Commit
2. Cross-Site
3. Burn Rate
4. Pending Receipts
5. Adjustments
6. Purchase Requests
7. Receive Material
8. [!] Shelf-Life
9. Notifications
10. [OK] My Requests

6.3 HOD Portal -> EOD Commit

The single most consequential action in the system: committing the day's staged consumption to the permanent ledger.

6.3.1 Top section

Element	Purpose
Header	" End-of-Day Commit (<SITE>)" with intro caption

6.3.2 4-card stat strip

Card	Source
Total entries	Total pending rows at your site
Pending	Status = pending
[!] Flagged	Status = flagged
[OK] Approved	Status = approved

6.3.3 Filter pills

Pill	Action
All / Pending / Flagged / Approved / Rejected	Filters the table below to that status

6.3.4 Top action bar

Button	Action
[OK] Approve All Pending	Approves every pending row in one action. Disabled when there are none.
Commit EOD to Master	Opens the type-COMMIT modal. See §6.3.7

6.3.5 Main pending-issues table

Custom dark-themed HTML table showing every column from the staging row (joined with inventory for descriptions):

Column	Source
Date	pending_issues.Date
SAP	pending_issues.SAP_Code
Mat Code	inventory.Material_Code
Material	inventory.Equipment_Description
UOM	inventory.UOM
Qty	pending_issues.Quantity
Work Type	pending_issues.Work_Type
PR	pending_issues.PR_Number
Tank	pending_issues.Tank_No
Serial	pending_issues.Serial_No
Issued By	pending_issues.Issued_By
Issued To	pending_issues.Issued_To
Remarks	pending_issues.Remarks
Status	Colored pill (pending / flagged / approved / rejected / committed)

The table also reflects Lot_Number and FEFO_Override columns when present (visible by editing the row).

6.3.6 Row Actions panel (per-row Approve/Reject)

Below the table, top 20 actionable rows render as cards:

Element	Purpose
Card line 1	SAP - Material Code - Material name - Qty UOM
Card line 2	Date - Work Type - PR - Tank - Serial - By - To - Remarks (only fields with values)
[OK] button	Approve this single row
[X] button	Reject this single row

6.3.7 EOD Commit confirmation modal (type-COMMIT dialog)

When you click Commit EOD to Master:

Step 1 - pre-flight: negative-stock guard

Before committing, the system checks that nothing would push stock below zero. If anything would:

Element	Purpose
Red error banner	"Cannot commit - N item(s) would go negative"
Violation table	SAP - Material - Current Stock - Trying to Consume - Deficit
Hint banner	"Common fixes: receive inbound stock first, raise a stock adjustment, or reduce the consumption to fit available stock."
Close button	Cancels - fix the staging rows and try again

Step 2 - confirm (only if no violations)

Element	Purpose
Warning banner	"You are about to commit N pending row(s)..."
Type COMMIT text input	Must type exactly COMMIT to enable the button
Cancel button	Abort
Confirm Commit button (red)	Moves every approved row out of staging and into the permanent consumption ledger, refreshes the stock figures, and - if the commit leaves anything below its minimum - queues a post-EOD low-stock alert to the HOD.

A confirmation appears once the commit succeeds.

6.3.8 Flagged-items banner

If there are flagged rows: amber banner appears below the table reminding the HOD to verify with the store keeper before committing.

6.4 HOD Portal -> Cross-Site

For requesting material from another branch.

6.4.1 Top section

Element	Purpose
Intro caption	Explains the > 5-item escalation rule

6.4.2 Single-target inquiry flow (left/right columns)

Left column:

Element	Purpose
Select Target Branch selectbox	List of all sites except yours

Element	Purpose
Select Material selectbox	Type-ahead search across inventory
Quantity Needed number input	Min 1
Justification / Notes textarea	Why you need this transfer

Right column:

Element	Purpose
Live Stock at <target> header	Shows availability at the target site
Available Quantity metric	Live stock at target
Suggested Transfer Qty metric	Whichever is smaller - the quantity you asked for, or the quantity actually available at the target site
Add to List button	Adds to the cart

6.4.3 Your Request Cart section

Element	Purpose
Cart table (styled HTML)	Columns: Target Site - SAP - Description - Qty - Notes
Submit All Requests to Admin button	Creates a requests row per cart item, queues WhatsApp to all admins, AND (if >5 items) escalates to the target-site HOD via separate WhatsApp + audit log
Clear List button	Empties the cart

6.4.4 Incoming Cross-Site Requests panel (bottom)

Where a HOD sees requests addressed to their site from other HODs.

Element	Purpose
Intro caption	"Requests addressed to <site>"
Incoming table (styled HTML)	From - SAP - Material - Qty - By - Notes - When
Per-row Approve / Reject controls (top 10)	[OK] Approve and [X] Reject buttons
Approval action	Marks the request approved and writes an audit entry
Rejection action	Marks the request rejected, also audited

6.5 HOD Portal -> Burn Rate

Predictive analysis of which items will run out and when.

6.5.1 Compact horizontal-bar chart (top 10)

A sleek card with:

- Header row: "Monthly Consumption - Top 10" + legend chips (red=High / amber=Mid / green=Low)
- Per-row line: material name (truncated) - gradient bar (color reflects burn intensity proportional to max) - monthly figure with UOM - days-remaining badge (red if <=7d, amber if <=30d, green if >30d)

6.5.2 Burn alert banner (above the Plotly chart)

Auto-renders if items are projected to hit zero within the alert window.

6.5.3 Full Plotly burn rate chart

Daily-burn-rate bars with vertical line at 30-day alert threshold. Color-coded.

6.5.4 Detailed forecast table

Columns: SAP_Code - Equipment_Description - UOM - Current_Stock - Daily_Burn_Rate - Days_Remaining

Sortable + exportable.

6.6 HOD Portal -> Pending Receipts

Approve or reject receipts staged by Store Keepers.

6.6.1 Header + count banner

Amber banner: " N pending receipt(s) require your approval before stock levels update."

6.6.2 [OK] Approve All button

Approves every pending receipt at your site in one action. For each one, the system:

1. Works through every pending receipt in turn
2. Records the delivery (creating a lot record where an expiry date was entered)
3. Posts it to the permanent receipts ledger
4. Auto-closes the PR if it is now fully fulfilled
5. Removes the staged row
6. Refreshes the stock figures, writes an audit entry, and confirms with a success message

6.6.3 Receipt table

Columns: Date - SAP - Equipment_Description - UOM - Quantity - Supplier - PR_Number - Site_ID - Status pill

6.6.4 Per-row Reject controls (top 10)

Element	Purpose
Card line	SAP - Qty UOM - Supplier

Element	Purpose
[X] Reject button	Marks the row rejected rather than deleting it, so the history is preserved, and records your reason in the audit log

6.7 HOD Portal -> Adjustments

Approve physical-count reconciliations submitted by Store Keepers from the Entry Log -> Stock Count tab.

6.7.1 3-card stat strip

Card	Source
Pending	Count of pending adjustments at your site
Shortfalls	Variance < 0 count
Surpluses	Variance > 0 count

6.7.2 Pending adjustments table

Columns: # - SAP - Material - UOM - System - Counted - Variance (red for negative, green for positive) - Reason - By - When

6.7.3 Per-row Approve/Reject

Below the table:

Element	Purpose
Card line 1	SAP - Material - System X -> Counted Y - Variance
Card line 2	Reason label - submitted by - notes
[OK] button	Approves the adjustment and, in the same step, posts a matching correcting entry to the ledger - a shortfall is written as a consumption entry marked "stock adjustment"; a surplus is written as a receipt from supplier "stock adjustment". The adjustment keeps a reference to the ledger entry it created, so the two can always be tied together in an audit.
[X] button	Rejects the adjustment. The row is kept, marked rejected, so the audit trail stays complete.

6.7.4 Recent Adjustments History (last 30)

Columns: # - SAP - Material - Variance - Reason - Status pill - Submitted By - Approved By - Ledger Ref

6.8 HOD Portal -> Purchase Requests

Manage Purchase Requests (PRs) for your site.

6.8.1 Create New PR (manual entry) - expander

Field	Purpose
PR Number	Free text, required (e.g., 3001234567)
Material selectbox	Picks from inventory; auto-fills Material_Code, Material_Name, UOM via blue-tinted info card
Requested Qty	Min 0.01
Preferred Supplier	Optional
Estimated Cost (SAR)	Optional
UOM	Auto-filled, editable
Notes	Optional
Create PR Draft button	Creates the PR as an open request in draft state

6.8.2 Upload PR PDF (details read automatically from the file) - expander

Element	Purpose
File uploader (PDF only)	Upload Purchase Request PDF
Process Upload button	Reads the PR number and the material codes (anything in the GI-XXXXXXX format) out of the document, matches them against the material codes in the inventory master, and creates the PR lines
Result banner	Green success or amber warning (if some materials unmatched)

6.8.3 Current PRs table

Columns: PR No. - SAP - Material - UOM - Qty Req - Qty Pending - Supplier - Est. SAR - Workflow pill

Qty Pending = Requested Qty - everything already received against that PR number.

6.8.4a Submit PR(s) to Logistics Portal (NEW in v3.0)

A new expander sits between the PR list and the email/PDF block. It opens the procurement-chain path - handing off your PR to the in-app Logistics queue instead of (or in addition to) the legacy email path.

Element	Purpose
Multi-select	All open PRs at your site that haven't yet been submitted to Logistics
Submit Selected to Logistics button	The PR is marked as submitted to Logistics and appears

Element	Purpose
	in Logistics Portal -> Incoming PRs. A "PR submitted to Logistics" notification fires to the Logistics inbox, and to WhatsApp if that event is enabled.

The legacy email path (§6.8.4) still works - your team can use it for direct-to-vendor relationships that don't go through central Logistics. Both paths coexist. The email path is marked for future deprecation once procurement chain adoption reaches the agreed threshold.

6.8.4 Notify Logistics section

Shown whenever at least one PR is still open:

Element	Purpose
Select PR for Actions selectbox	Pick from open PRs
Draft Outlook Email button	Opens email client with HTML table of pending items. Mac: opens Mail.app via AppleScript with formatted HTML table. Windows: opens Outlook via COM with HTMLBody. Fallback: mailto: with monospace plain-text table.
Download PR PDF button	Generates a PDF record with columns: Material Code - Description - Req. Qty - Received - Pending - Status. Auto-saved metadata: Site, Generated By, Timestamp.

6.9 HOD Portal -> Receive Material

For HOD-direct receipts (when a HOD logs a delivery themselves, e.g., direct purchases or HQ deliveries).

6.9.1 PR linker

Element	Purpose
Link to Open PR selectbox	Filters the material list to items in that PR, or "None (Direct Purchase)" for free-form

6.9.2 Receipt form (st.form, clears on submit)

Field	Type	Purpose
Select Material	Selectbox	From inventory or filtered to PR items
Quantity Received	Number (min 0.1)	Units arrived
Delivery Date	Date (today)	When
Expiry Date	Date (optional)	Lot expiry trigger
Logistics extras	Text inputs	All non-system columns on receipts

Field	Type	Purpose
		(Vehicle_No, Driver_Name, DN_No, Supplier, Remarks, etc.)
Save Receipt button	Posts straight to the receipts ledger, and creates a lot record if an expiry date is set. Auto-closes the PR if it is now fulfilled, and refreshes the stock figures.	

6.9.3 Receipt History (last 50)

Below the form, a table showing recent receipts for this site: entry no. - Date - SAP - Material - Quantity - Supplier - PR_Number - Expiry_Date.

6.10 HOD Portal -> [!] Shelf-Life

Lots at risk of expiry.

6.10.1 3-card stat strip

Card	Source
Expired	Lots with days_left < 0
Critical (<=30d)	Days_left 0-30
Warning (<=90d)	Days_left 31-90

6.10.2 Action-required banner

When expired_n > 0: red banner instructing physical isolation + disposal.

6.10.3 Shelf-life table (top 30, sorted by days-left ascending)

Columns: SAP - Material - Lot - Qty - Expiry (color-coded) - Days Left - Status pill

6.10.4 Log Disposal button

Bulk-logs an audit event for the expired count. Note: this does NOT post a stock adjustment automatically - for proper inventory reduction, use the Adjustments flow with reason expired_disposal.

6.11 HOD Portal -> Notifications

Manual WhatsApp sends + alert threshold tuning.

6.11.1 Send Manual WhatsApp card

Field	Purpose
Recipient phone number	E.g., +966 5X XXX XXXX
Message textarea	Up to several hundred characters
Send WhatsApp button	Adds the message to the outbound queue; the background sending service picks it up from there. The send is audit-logged as a manual WhatsApp.

6.11.2 Alert Thresholds card

Three sliders. Each is remembered until it is changed again:

Slider	Range	Default	What it controls
Low stock alert (days of supply)	1-60	5	How few days of cover before an item counts as low
Burn-rate warning (days remaining)	1-60	7	How close to running out before the burn banner appears
Expiry warning (days before)	1-120	30	How far ahead of expiry an item starts being flagged

Save Thresholds button - saves the values and writes an audit entry recording the change.

> Note: The Notification Log table moved to Admin Portal -> WhatsApp Console for global visibility.

6.12 HOD Portal -> [OK] My Requests

Outbound cross-site requests YOU have made.

6.12.1 Outbound requests table (styled HTML)

Columns: # - To Site - SAP - Qty - Status pill - Created

6.12.2 Mark Incoming Transfers as Received

Shown whenever at least one of your requests has been approved:

Element	Purpose
Select Approved Request selectbox	List of approved request IDs
Confirm Delivery Received button	Marks the request fulfilled and refreshes the stock figures. This is where you confirm the transfer physically arrived.

6.15 HOD Portal -> DN Approvals (NEW in v3.0)

This is your approval queue for Delivery Notes inbound to your site. The Warehouse has prepared them, Logistics has approved the delivery date - now you confirm the content (what's actually arriving, in what qty).

6.15.1 Empty state

If nothing's pending: an empty-state card reads "No DN's awaiting your approval - They appear here after Logistics signs off the delivery date." No action needed.

6.15.2 Per-DN cards

Each pending DN renders as its own bordered container with two columns:

Left column:

- Header line: DN <number> - PO <number> - From Warehouse <id>
- Subline: DN Date - Vehicle No - Driver Name
- View lines expander: Read-only preview of every line item (Material_Code, Description, Qty, UOM, rl_bl_family, Lot_Number, Expiry_Date, Remarks)

Right column (Decide popover):

- Notes textbox (required if rejecting)
- [OK] Approve button (primary) - flips DN to pending_sk and mirrors lines into pending_receipts so SK sees them
- [X] Reject button - flips DN to rejected with your reason; Warehouse gets pinged to redo

6.15.3 When to approve

- DN qty matches what your site needs (matches the PR line you originally raised)
- RL/BL family is correct
- Lot + expiry are acceptable (not over-expiry)
- The originating PO is the right one (not a misroute)

6.15.4 When to reject

- Wrong qty (e.g. site asked for 100, DN says 200 - too much exposure)
- Wrong material (the Material_Code doesn't match the PR you raised)
- Lot has insufficient remaining shelf life
- Vehicle / driver details suggest a routing problem

Always include a clear rejection reason. The Warehouse user sees the reason verbatim in their bell inbox and on the bounced DN. Sloppy reasons cost a rebuild cycle.

6.16 HOD Portal -> In-Transit (NEW in v3.0)

A read-only window onto the procurement chain for your site. Use this tab when a user asks "when is X arriving?" or "why didn't Y arrive yesterday?". Three sub-tabs.

6.16.1 Sub-tab: Active in-transit

KPI strip at the top with counts per pipeline state:

- At Logistics - DN drafted by Warehouse, waiting on Logistics date approval
- Logistics approved - date confirmed, waiting on YOU
- Awaiting my approval - same as Logistics approved (highlighted gold)
- Pending SK receipt - you've approved, SK has it in their tab

Below the KPI strip, each in-transit DN renders as a card:

Element	Purpose
DN header	DN <number> (gold) + RL/BL chip (orange for RL, purple for BL)
Subline	PO <number> - Warehouse <id> - ETA <date> - <line count> line(s), <total qty> units
Status pill (right)	Colour-coded pipeline state
View lines expander	Read-only line preview
Request reschedule popover	Date picker (defaults to ETA + 3 days, min=today) + reason textarea + Submit button. Submits to Logistics.

The reschedule UI is deliberately frictionless:

- Date is pre-filled so a quick "+3 days" submission is one click
- min_value=today prevents accidentally picking yesterday
- Same-date submission warns instead of wasting a round-trip
- Caption " Goes to Logistics" makes the destination unambiguous

6.16.2 Sub-tab: My reschedule requests

KPI strip: Pending / Approved / Rejected counts.

Custom table showing your full reschedule history for THIS site:

Column	Source
PO No.	the linked PO
DN No.	the linked DN (if any)
From	current_date when you raised the request
Requested	the new date you asked for (gold, monospace, with arrow)
Reason	your justification (tooltip-truncated)
Status	pill - pending / approved / rejected
Decided by	Logistics user who handled it
Notes	decision notes from Logistics

6.16.3 Sub-tab: Force-closures affecting me

Read-only audit table showing every PR / PO / line that Logistics force-closed, scoped to your site. Where a closure record itself carries no site, the system traces it back through its parent PR or PO, so nothing goes missing from your list.

Column	Source
Type	"PR closed" / "PO closed" / "Line closed" badge
Target	The closed ref (PR number / PO number / line id)
PR No.	Linked PR (if applicable)
PO No.	Linked PO (if applicable)
Reason	Logistics' force-close reason
Closed by	Logistics user who closed it
When	Timestamp

50 most recent shown.

6.13 HOD - Use Cases

Use Case 1: Approve and commit the day's consumption

1. HOD Portal -> EOD Commit
2. Review the 4-card stat strip
3. Sort/filter using the filter pills
4. For each row: read the full detail card (line 1 = essentials, line 2 = all fields)
5. Click [OK] to approve or [X] to reject individual rows
6. Optionally click [OK] Approve All Pending for a clean batch
7. Click Commit EOD to Master
8. If pre-flight blocks: read the violation table, close, fix the offending rows (reduce qty, drop, or receive stock first), then re-open
9. If pre-flight passes: type COMMIT exactly, click Confirm Commit
10. Watch the balloons.

Use Case 2: Approve a Store Keeper's physical count adjustment

1. HOD Portal -> Adjustments
2. Read the pending table - variance and reason are color-coded
3. For each adjustment, decide: did the Store Keeper count accurately?
4. [OK] Approve -> system posts a synthetic ledger row (receipt or consumption), live stock updates immediately
5. [X] Reject -> audit-logged; Store Keeper sees a status update

Use Case 3: Request material from another site

1. HOD Portal -> Cross-Site
2. Pick target site -> pick material -> see live availability
3. Add to cart, repeat
4. If cart > 5 items: an automatic WhatsApp escalation to the target HOD will fire on submit. Document why in Justification.
5. Click Submit All Requests to Admin
6. Watch the [OK] My Requests tab - when status flips to "approved", you can mark it as fulfilled once received

Use Case 4: Approve an incoming bulk request from another HOD

1. HOD Portal -> Cross-Site -> scroll to Incoming Cross-Site Requests
2. Review the request table
3. [OK] Approve -> physical handoff begins
4. [X] Reject -> use when the request is unjustified or your stock is too tight

Use Case 5: Raise a Purchase Request

Option A: manual entry

1. HOD Portal -> Purchase Requests
2. Create New PR
3. Type PR Number, pick Material (auto-fills SAP, MatCode, UOM), Qty, Supplier, Est. Cost SAR, Notes
4. Click Create PR Draft

Option B: PDF upload

1. Upload PR PDF
2. Drop the PDF in
3. Click Process Upload
4. System auto-extracts PR number + items, creates pr_master rows

Use Case 6: Notify logistics about a pending PR

1. HOD Portal -> Purchase Requests
2. Select open PR in the email dropdown
3. Draft Outlook Email -> Mail/Outlook opens with HTML table pre-filled
4. Review/send
5. Also: Download PR PDF for your records

6.14 HOD - FAQ

Q: The EOD commit modal won't let me commit - keeps showing the violation table. A: Pre-flight is blocking because consuming the staged amount would create negative stock. Either: (a) reduce qty on the violating rows in the EOD table, (b) commit pending receipts first to top up stock, (c) raise a stock adjustment to correct any system-vs-physical discrepancy.

Q: A Store Keeper claims they submitted but I don't see anything. A: Check the filter pills (you may be filtering by Approved/Rejected only). Switch to "All" or "Pending".

Q: I approved an adjustment but the live stock didn't change. A: It should change within 30 seconds - the stock figures refresh on that cycle. Refresh the page. If it still doesn't, check the audit log for the adjustment-approved entry to confirm the correcting ledger entry was posted.

Q: I got a WhatsApp about a "bulk cross-site request" - what is it? A: Another HOD at a different site has requested more than 5 items from your site. Go to Cross-Site -> Incoming to review.

Q: My burn rate forecast looks wrong - items aren't appearing. A: Items with fewer than ~3 consumption events in the last 30 days have insufficient data and may be omitted. Increase the window via Settings if needed.

Q: I can't see a receipt I just approved. Did it work? A: Approving via "Approve All" commits all pending receipts at once. Check the Live Dashboard for updated stock. The receipt now lives in receipts table, not pending_receipts.

Q: The Outlook email button does nothing on my Mac. A: Make sure Mail.app or Outlook is installed and the system has permission to open it. The fallback is mailto: (plain-text monospace table) which works in any browser default mail handler.

Q: PR PDF upload didn't recognize some items. A: The extractor matches material codes in the format GI-XXXXXXX. Items not in your inventory's Material_Code field will be flagged in the warning. Add the missing items to inventory first.

Q: Cross-site request stuck in 'pending' - why isn't Admin approving? A: Admin sees all pending in Admin Portal -> Pending Requests. If urgent, queue a WhatsApp from Notifications. Or for >5 items, the target-site HOD can also approve directly.

Q: I want to see all FEFO overrides logged by my Store Keepers. A: Filter Admin -> Audit Logs by the FEFO override action. Every consumption entry that used an override also carries the override reason on the row itself.

7. Admin Manual

The Admin is the system owner. The Admin Portal has 11 tabs as of v3.0 - the original 10 (Overview, Pending Requests, Global Sites, Users, Master DB Editor, Audit Logs, WhatsApp Console, Settings, Access Control, Reports & Bugs) plus the new Logistics Oversight tab. Admins do NOT see HOD Portal (intentional - Admin uses Admin Portal for cross-site work). Admins CAN see the Logistics Portal and Warehouse Portal as shadow access - when entering the Warehouse Portal, a sidebar dropdown lets the admin pick which warehouse to view as.

7.1 Pages visible

- Live Dashboard (global - sees all sites combined)
- Entry Log (can stage entries themselves)
- Admin Portal - detailed below
- Reports (with "All Sites" filter unlocked)

7.2 Admin Portal overview

7.2.1 Page header

Gold subtitle + pulse status chip (green "All systems operational", or amber "Degraded - see Overview" when more than 20 items are low on stock across all sites).

7.2.2 Hero strip (3 cards)

Card	Source
Sites managed	Count of distinct sites
Pending cross-site requests	Transfer requests still waiting on a decision
Critical items (all sites)	Below-minimum count plus expiring/expired count, summed across every site

7.2.3 Tab strip (9 tabs)

1. Overview
2. Pending Requests
3. Global Sites
4. Users
5. Master DB Editor
6. Audit Logs
7. WhatsApp Console
8. Settings
9. Access Control

7.3 Admin Portal -> Overview

System health at a glance.

7.3.1 4-card KPI strip (technical)

Card	Source
DB size	How much space the database file takes up on disk (MB)
Users	Count of non-suspended users
Total transactions	Every consumption and receipt entry ever recorded
Audit events	All-time count of audit-log entries

7.3.2 4-card valuation strip (financial)

Card	Source
Total stock value	all sites
Biggest-value site	The single site holding the most stock value, and its share of the total
30-day consumption value	Value of everything consumed in the last 30 days
Pending receipts value	Placeholder - it reads the estimated cost carried on the Purchase Requests

7.3.3 Service Health card

Per-service row with a pulse dot + status + note:

Service	Up indicator	Source
Database	Reachable, with its current size	
WhatsApp Queue	The outbound message queue is reachable	Pending count or "queue clear"
Ollama / AI	The local AI service answers, when AI is switched on	"ready" / "not reachable"
Mail / SMTP	Informational	"Outlook + mailto fallback"

7.3.4 Database Stats card

8 row counts:

- Inventory items
- Consumption rows
- Receipt rows
- Pending issues
- Pending receipts

- Open PR lines
- WhatsApp queue size
- Audit events

7.3.5 Live Activity Feed (last 12)

Per-row card, drawn from the audit log:

- A red, amber or green severity icon, set automatically from the kind of action recorded
- Timestamp (gold monospace)
- Action name
- Target table pill
- "<user> - <details>" line

7.4 Admin Portal -> Pending Requests

Approve or reject cross-site material requests from HODs.

7.4.1 Pending requests data editor

Editable table with a Select checkbox column (every other column is read-only). It lists every cross-site request still waiting for a decision.

7.4.2 Admin Notes text input

Optional when approving; required when rejecting - the requester is shown the reason, so a rejection is never unexplained.

7.4.3 Action buttons

Button	Action
[OK] Approve Selected	For each selected row: looks up the material code and description, then marks the request approved. Each requester then gets one WhatsApp listing their items plus your notes, and each target site's HOD gets a "TRANSFER ORDER" WhatsApp with packing instructions.
[X] Reject Selected	Works through the selected rows the same way but marks them rejected. Notes are MANDATORY for rejection. Each rejected requester gets a WhatsApp with the reason.

7.5 Admin Portal -> Global Sites

Cross-site inventory viewer (read-only).

Element	Purpose
Site selectbox	"All Sites (Global)" or pick one
Inventory table	Live stock for the chosen site, or for every site at once

Useful before approving a cross-site request to confirm the target site really has the stock.

7.6 Admin Portal -> Users

Delegates entirely to the system which provides:

Section	Purpose
Pending registration requests	Approve / reject self-registered users
Active users table	All users with role, site, phone
Edit user form	Change role, site, phone, suspend/activate
Add user form	Create user directly (Admin shortcut)
Password reset action	Generates temporary password, audit-logged

7.7 Admin Portal -> Master DB Editor

This is the most powerful tab. Use with care. It lets an Admin view and edit the contents of any table, and change which columns that table has.

7.7.1 Table selector

Element	Purpose
Select Table dropdown	Lists every data table in the system, including lots, stock corrections, reported issues and settings.

7.7.2 Action radio

3 modes:

Mode A: View / Edit Data

Element	Purpose
Row count caption	"<N> rows in <table>"
Export as PDF button	Produces a PDF of the whole table
Data editor table	Edit any cell. Password hashes are masked as *****.
Print Label checkbox column	Only on inventory table - select items to print QR labels

Element	Purpose
Save Table Updates button	Clears the whole table and re-writes every row from what is on screen. The edit is audit-logged, and the stock and settings figures are refreshed.
Generate QR Labels for Selected button	Only on inventory - A print sheet for items where label checkbox is checked
Download QR Labels PDF button	Appears after generation

> [!] The Save action is destructive. It deletes all rows and re-inserts. If anything fails mid-write you may lose data. Industry best-practice is to make ledger tables immutable; this tab bypasses that - Admin discretion required.

Mode B: Add New Entry

Form generation depends on the table:

- For users: warns to use User Management tab instead (safer)
- For receipts: opens the full Logistics Receipt form (Site, Open PR linker, Material, Qty, Date, Expiry, Supplier, Remarks). It records the delivery the same way the Logistics portal does, and creates a lot record if an expiry date is entered.
- For any "transaction table" (has SAP_Code, isn't inventory): 2-section form - Section 1 picks material (shows MatCode + UOM info card), Section 2 dynamically generates inputs for every editable column.
- For inventory itself: dynamic 3-column form for all editable columns.

Mode C: Manage Columns

Element	Purpose
Add Column section	Type a column name and click Add - a new free-text column is appended to the table
Rename Column section	Pick a column, type the new name, click Rename
Drop Column section	Pick a column and delete it. This destroys the data in that column and cannot be undone - take a backup first, every time.

> The users table is protected - its columns cannot be changed here, because the sign-in system owns that table. Use the Users tab instead.

7.8 Admin Portal -> Audit Logs

Complete forensic record.

7.8.1 Filter row

Filter	Source
User selectbox	DISTINCT usernames + "All Users"
Action selectbox	DISTINCT action_types + "All Actions"
Target selectbox	DISTINCT target_tables + "All Targets"
Limit selectbox	50 / 100 / 500 / 1000

7.8.2 Search text input

Substring match across details + username + action_type.

7.8.3 Audit table (styled HTML)

Columns: // severity - Timestamp - User - Action (color-coded by severity) - Target pill - Detail (truncated, tooltiped)

Severity heuristic (from action_type):

- Critical (): any action containing FAIL, REJECT, DELETE, PURGE, EMERG, ROLLBACK, RESET, DESTRUCTIVE
- Warning (): SUSPEND, REVOKE, ROTATE, FLAG, DOWNGRADE, WARNING
- Info (): everything else

Expired-row red tint on critical events.

7.9 Admin Portal -> WhatsApp Console

Outbound message queue + manual sends + thresholds + event mapping.

7.9.1 Queue stats strip (4 cards)

Card	Source
[OK] Sent	Messages the queue has delivered
Pending	Waiting to be picked up
Processing	Being sent right now
[X] Failed	Could not be delivered

7.9.2 Send Manual WhatsApp card

Field	Purpose
Recipient phone	E.g., +966 5X XXX XXXX
Message textarea	The message to send
Send WhatsApp button	Adds the message to the outbound queue and audit-logs it as a manual WhatsApp

7.9.3 Alert Thresholds card (global)

The same three sliders as HOD Notifications - low-stock days, burn-rate warning days and expiry warning days. The Admin's values are the system-wide defaults.

7.9.4 Event -> Recipient (current wiring) - read-only summary

A reference table showing what auto-triggers exist in the codebase:

Event	Recipient	Role
Issue staging submitted	Site HOD	hod
Pending receipt submitted	Site HOD	hod
EOD committed	Site HOD	hod
Cross-site request created	All admins	admin
Cross-site bulk (>5 items)	Target site HOD	hod
Cross-site request approved	Requesting HOD	hod
Cross-site request rejected	Requesting HOD	hod
Returnable item overdue	Store Keeper	store_keeper
New access request	All admins	admin
Access request approved	Requesting user	store_keeper
Post-EOD low stock alert	Site HOD	hod

7.9.5 Outbound Queue Log (last 80)

Styled table of the 80 most recent outbound messages: Status pill - Recipient - Message (shortened, hover for the full text) - Queued at - Sent at

7.10 Admin Portal -> Settings

7.10.1 Dropdown Manager - Work Types (expander, expanded by default)

Element	Purpose
Current Work Types caption	Comma-separated list of the options currently configured
New Work Type Name text input	Add a new option
Add to Dropdown button	Saves the new Work Type and makes it available in the dropdown straight away

7.10.2 Maintenance Mode card

Element	Purpose
Enable maintenance mode toggle	Turns maintenance mode on and keeps it on until you turn

Element	Purpose
	it off
Status caption	"ACTIVE - Non-admin sessions will be told to come back later" or "Off"

The change is recorded in the audit history with who made it and when. Sign-in enforces it immediately - while maintenance is on, only Admin can sign in.

7.10.3 Database Backup card

Element	Purpose
Last manual backup caption	When the last manual backup was taken
Backup Now button	Copies the whole database into the backups folder, date- and time-stamped. Records the new backup time and writes an audit entry.

7.10.4 Site Management

Element	Purpose
Sites table (styled HTML)	Site Name - Code (first 4 chars upper) - Users count - Status (always "Active")
Add New Site expander	Type the site name and click the button - the site is registered and the action is audited

7.10.5 [!] Danger Zone (red bordered card)

Element	Purpose
Purge old draft issues card	"Delete every staged issue older than 30 days that is still sitting in draft" - approved and committed entries are never touched
Type PURGE to confirm	The button stays disabled until you type PURGE exactly
Run Purge button	Deletes those drafts and writes an audit entry

7.11 Admin Portal -> Access Control

7.11.1 Recent Sign-Ins (last 10)

Per-row card, drawn from the audit log and filtered to sign-in, failed sign-in and sign-out events:

- Pulse dot (green=success, red=failed)
- Username
- Action label

- Timestamp + details

7.11.2 Force Password Reset card

Field	Purpose
Target user selectbox	Pick any registered user
New password	Typed masked
Confirm	Must match
Amber warning	"User must log in again immediately."
Reset Password button	Checks the password is at least 8 characters and that both entries match, stores it as a one-way scramble, and writes an audit entry for the forced reset

7.11.3 Security Policy

Read-only 2-column grid showing:

- Passwords are stored as a one-way scramble, never as text
- Sessions are held in memory only and are never written to disk
- RBAC hierarchy (store_keeper < supervisor < hod < admin)
- WAL mode + busy_timeout=5000ms
- Password min length (8 characters)
- Audit retention (indefinite, manual purge only)

7.11a Admin Portal -> Logistics Oversight (NEW in v3.0)

Cross-site, read-only window onto the entire procurement chain. For actions, jump to the Logistics Portal (shadow access) - this tab is observation-only by design.

7.11a.1 KPI strip

Six cards at the top:

Card	Source
OPEN PRs	Count - awaiting PO issuance
OPEN POs	Count of Purchase Orders still open
ACTIVE DNS	Count of DNS in pipeline states (pending_logistics, logistics_approved, pending_hod, pending_sk)
VENDOR RETURNS	Count of vendor returns still open
RESCHEDULES	Pending reschedule decisions
FORCE-CLOSURES	Lifetime audit count

7.11a.2 Filters

Element	Purpose
Site dropdown	"All sites" or pick one - narrows every sub-tab
Warehouse dropdown	"All warehouses" or pick one - narrows DN view

7.11a.3 Six sub-tabs

Sub-tab	What's shown
PRs	Every active PR in the Logistics queue, filterable by site
POs	Every open PO with vendor, dates, total, status, source (manual/PDF)
DNs	Every active DN with warehouse, site, status, family
<- Vendor Returns	Open returns awaiting vendor acknowledgement
Force-Closures	100 most recent force-closure records with reason + closed-by
Reschedules	Pending reschedule decisions Logistics hasn't acted on

7.11a.4 What it's NOT

- This tab cannot create / approve / reject anything. All mutation happens in role-specific portals.
- Admins who need to ACT (e.g. approve a reschedule because Logistics is on leave) should switch to the Logistics Portal - admin has shadow access there.

7.12 Admin - Use Cases

Use Case 1: Set initial Unit_Costs for valuation reports

1. Admin Portal -> Master DB Editor
2. Select Table: inventory
3. Action: View / Edit Data
4. In the data editor, find Unit_Cost column (rightmost)
5. Type SAR cost for each item
6. Click Save Table Updates
7. Wait ~60 sec for caches to refresh, then check Live Dashboard hero card "Total stock value"

Use Case 2: Approve a self-registered new user

1. Admin Portal -> Users
2. Pending requests section -> review the request
3. Edit role/site if needed
4. Click [OK] Approve -> user can now log in. WhatsApp alert auto-fires to the requester.

Use Case 3: Backup the database before a risky operation

1. Admin Portal -> Settings
2. Backup Now -> confirms file path: backups/gi_database_YYYYMMDD_HHMMSS.db
3. Note the timestamp in "Last manual backup" field
4. Now perform the risky operation (bulk import, adding or dropping a column, etc.)
5. If something breaks: restore by copying the backup back over gi_database.db (with the app stopped)

Use Case 4: Investigate a suspicious series of FEFO overrides

1. Admin Portal -> Audit Logs
2. Action filter: FEFO_OVERRIDE
3. Search: e.g., a specific username
4. Review the detail column - see SAP, site, chosen lot vs FEFO-suggested, reason
5. If a pattern of unjustified overrides emerges, talk to the Store Keeper / HOD

Use Case 5: Configure WhatsApp alert thresholds

1. Admin Portal -> WhatsApp Console
2. Alert Thresholds card
3. Adjust sliders: low_stock_days, burn_alert_days, expiry_warn_days
4. Save Thresholds -> audit-logged. HOD threshold sliders read the same values.

Use Case 6: Approve a batch of cross-site transfers

1. Admin Portal -> Pending Requests
2. Review the editor table - sort by created_at, target_site, or item
3. Tick checkboxes for the rows you want to approve as a batch
4. Type Admin Notes (e.g., "Per Tuesday meeting, expedite to Site B")
5. [OK] Approve Selected -> triggers two WhatsApp dispatches per unique requester + target HOD
6. Watch the Notification log in WhatsApp Console to confirm sends

Use Case 7: Drop a deprecated column from a custom table

1. Admin Portal -> Master DB Editor
2. Select the table
3. Action: Manage Columns
4. Drop Column section -> pick the column -> Delete Column

Use Case 8: Generate QR labels for new inventory items

1. Master DB Editor -> inventory
2. View / Edit Data mode
3. In the leftmost "Print Label" column, tick items
4. Scroll to QR Code Label Generator section below
5. Click Generate QR Labels for Selected

6. Download QR Labels PDF
7. Print on label paper, stick to shelf bins -> store keepers can now scan them on mobile

7.13 Admin - FAQ

Q: How do I add a new site? A: Admin Portal -> Settings -> Site Management -> Add New Site. New sites are immediately available for user assignment.

Q: How do I rotate a user's password without their input? A: Admin Portal -> Access Control -> Force Password Reset. Pick user, type new password twice, click reset. They'll need to log in again with the new credentials.

Q: Audit log filter has too many users - how do I find a specific user's activity? A: Use the search text input at the bottom of the audit filter row - searches across user + action + details.

Q: WhatsApp queue is stuck - messages aren't being sent. A: The notification service runs separately from the application, so the application can be perfectly healthy while messages are not going out. Ask your Admin to check it. Messages queue up in the meantime and deliver once it is running again.

Q: My DB Editor save crashed mid-way and lost rows. A: Restore from the backups folder. The Save action clears the table and re-writes it, so it is not crash-safe. Always take a backup before bulk edits.

Q: Maintenance Mode is on but non-admins can still log in. A: The toggle is remembered, and sign-in respects it - during maintenance only Admin can get in. Confirm with your Admin if you are unsure whether it is currently on.

Q: Can I delete an audit log entry? A: No - they're meant to be permanent. If you genuinely need to free space, an Admin can purge old entries by date from the Master DB Editor - but that purge is itself audited.

Q: A cross-site request is stuck pending and the requester is asking why. A: Admin Portal -> Pending Requests. Confirm it's actually in the queue. If yes, approve or reject with notes. The HOD will get a WhatsApp.

Q: I want to set per-user 2FA. A: Currently unsupported in the security model (placeholder shown in Access Control). Future enhancement.

Q: How do I know if the local AI service is up? A: Admin Portal -> Overview -> Service Health card. If "Ollama / AI" pulses green, it's reachable.

Q: What's the difference between approving a Pending Receipt vs Pending Issue? A: Receipts ADD to stock (your team RECEIVED material). Issues SUBTRACT from stock (your team CONSUMED material). Both flow through approval. Issues commit via EOD (batch); Receipts commit individually via Pending Receipts tab.

Q: Stock_Value column shows nothing on the dashboard. A: Those inventory items have a unit cost of 0. Use the Master DB Editor to set their costs.

Q: A HOD set thresholds different from mine - whose wins? A: HOD and Admin both write to the same settings, so the most recent save wins. Agree a policy between you and stick to it.

Q: How big can the database grow before performance degrades? A: The current deployment runs comfortably at this site's user count, and the database it uses scales well beyond it. Growth in sites or users is a hosting decision rather than a rebuild.

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

The header reads "Reports & Analytics". Tabs:

1. Generate Report
2. Scheduled
3. AI Insights
4. Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	What it shows	Typical use
Daily Consumption	Every material issue in the window, grouped by work type	Reconciling a day's or week's usage against site activity
Monthly Summary	Per-item opening, issued, received and closing stock, with the SAR value of each	The month-end pack for finance
[!] Low Stock Alert	Materials below their minimum, with the shortfall	Building a reorder list
Burn Rate Analysis	30-day usage trend and days of supply remaining per item	Deciding what to order before it runs out
Inventory Valuation	Stock value per item, highest first,	Answering "what is on our shelves"

Report	What it shows	Typical use
	with the top-ten share	worth?"
Shelf-Life / Expiry	Lots grouped into expired, critical and warning	The weekly walk-round to pull short-dated stock
PR Status Report	Every purchase request with its stage and estimated cost	Chasing what has not arrived
[OK] FEFO Compliance	Whether issues took the oldest lot first, and where they did not	Proving shelf-life discipline to an auditor
Full Audit Report	The complete activity history for the date range	Investigations, and formal audit evidence

8.2.3 > Generate Report button

Builds the report for the type, date window and site you chose. The page shows a loading shimmer while it works.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	The report type and site, above the date range covered
v Download button	Saves the report in the format you picked - PDF, Excel or CSV
Summary cards	The headline figures for the report, filled in automatically
Bar chart	Shown for the Burn Rate and Daily Consumption reports
Preview table	The rows themselves, with status colours

8.2.5 A note about apostrophes in downloaded files

Sometimes a downloaded Excel or CSV file shows an apostrophe in front of a value you typed - for example '=A1 where you typed =A1. Nothing is wrong, and nothing has been lost.

Spreadsheet programs treat anything starting with an equals sign, a plus, a minus or an at-sign as a calculation to run, not as words to display. If that text came from something a person typed into the system, opening the file could make the spreadsheet run instructions nobody intended. The apostrophe tells the spreadsheet "this is text, show it exactly as it is".

- It only ever appears in front of text that begins with one of those symbols.
- Numbers are never affected. Quantities, prices and totals stay numbers, and every total in the file still adds up correctly.
- Ordinary words, notes and material names are untouched.
- If you need the plain text back, delete the leading apostrophe in the cell.

You will see this most often on notes and remarks fields, since those are the places people type freely.

8.2.5 Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st 06:00
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

- Report icon + label
- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- > Run Now button (manual trigger)
- Delete button

8.4 Reports -> AI Insights

Available when AI is switched on and the local AI service is running:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 standard checks and rewrites the AI commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	How sure the AI is of this finding
Right-side metric callout	Headline number with sub-label
Body paragraph	The AI's explanation, written from the figures the check returned
Recommendations list	Up to 3 numbered actions
Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 standard checks:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If the local AI service is not running, the tab explains what an Admin needs to start.

8.5 Reports -> Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
 - Re-email
 - Delete (audit-logged)
-

9. Automated Notifications - WhatsApp & Email & In-app Bell

The system automatically queues messages on key events. There are now THREE notification surfaces:

1. WhatsApp - messages queue and are sent by a background service, so a WhatsApp outage delays them but never loses them. There is a master on/off switch plus a per-event switch, so you can silence WhatsApp for one kind of event while leaving the in-app bell untouched for it.
2. In-app notifications bell (NEW in v3.0) - every event is also written to your personal inbox and surfaced by the sidebar bell described in §3.6. It ALWAYS fires, regardless of the WhatsApp toggle.
3. Email - the same Outlook / Mail app / mail-link and mail-server paths as before.

9.1 WhatsApp triggers (auto-fire)

Event	Recipient	Trigger location	Notes
Store Keeper submits issue batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits receipt batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits stock count	Site HOD	Entry Log Stock Count Submit	Includes variance + reason
Store Keeper performs FEFO override	Site HOD	Entry Log Add to Grid	Includes chosen vs FEFO-suggested lot + reason
HOD commits EOD	Site HOD (self)	post-commit	Only if low-stock items result
HOD cross-site request created	All Admins	Submit cart	Includes target site + count
HOD cross-site request, > 5 items	Target Site HOD	Submit cart	Bulk escalation
Admin approves cross-site requests (batch)	Requesting HOD	Approve Selected	Includes approved items + admin notes
Admin approves cross-site requests (batch)	Target Site HOD	Approve Selected	"TRANSFER ORDER - pack and ship"
Admin rejects cross-site requests (batch)	Requesting HOD	Reject Selected	Includes rejection reason
Returnable item overdue	Borrower + Store Keeper	Background scheduler	Time-driven, not in current scope
New self-registration	All Admins	Registration form	Includes username + role +

Event	Recipient	Trigger location	Notes
request			site
Self-registration approved	Requesting user	Admin User Mgmt	"ACCESS GRANTED - Welcome"
Manual send	Free	HOD Notifications / Admin WhatsApp Console	Recorded in the audit log as a manual WhatsApp
PR submitted to Logistics (v3.0)	Logistics role	HOD PR tab -> Submit PR(s) to Logistics	On by default
PO issued (v3.0)	Site HOD	Logistics -> Save PO	On by default
PO assigned to Warehouse (v3.0)	Warehouse users at that WH	Logistics -> Assign	On by default
Warehouse acknowledged (v3.0)	Logistics	Warehouse -> [OK] Acknowledge	Off by default - low value, kept quiet
Warehouse received goods (v3.0)	Logistics	Warehouse -> Record receipt	On by default
DN logistics approved (v3.0)	Site HOD	Logistics -> [OK] Approve DN	On by default
DN HOD approved -> SK (v3.0)	Site SK	HOD -> DN Approvals -> [OK] Approve	On by default; shares the DN-generated toggle
DN received at site (v3.0)	Logistics + Warehouse	SK -> [OK] Mark as Received	Off by default
Reschedule requested (v3.0)	Logistics	Warehouse / HOD -> Request reschedule	On by default
Reschedule decided (v3.0)	Requester	Logistics -> [OK] Approve / [X] Reject	On by default
Vendor return raised (v3.0)	Logistics	Any role -> <- Raise return	On by default
PR force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	On by default; critical severity
PO force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	On by default; critical severity
Delivery reminder T-2 / T-1 / T-0 (v3.0)	Logistics + HOD + Warehouse (per DN)	Daily sweep - see §9.3	Each of the three lead times has its own toggle

9.2 Email triggers

Event	Trigger location	Mechanism
Logistics PR follow-up	HOD PR tab -> Draft Outlook Email	Mac: opens the Mail app with a formatted table. Windows: opens

Event	Trigger location	Mechanism
		Outlook with a formatted table. If neither is available, it falls back to a plain-text mail link
EOD report email	Sent by the mail server	The mail server address and sign-in details are set up once by an Admin
Report delivery (scheduled)	Reports -> Scheduled	Uses the same mail server

9.3 In-app notifications bell (NEW in v3.0)

See §3.6 for the UI. Every notification is stored against the recipient and read back scoped to their role, their site and their warehouse - you only ever see your own. Each event carries a severity - info, success, warning or critical - and that is what colours the left border of the card.

9.4 Delivery reminder daily sweep (NEW in v3.0)

Once a day the system looks ahead at upcoming deliveries and sends reminders two days before, one day before, and on the day itself:

Watched	When	Severity
A Purchase Order's expected delivery date	T-2 / T-1 / T-0	warning / warning / critical
A Delivery Note's delivery date	T-2 / T-1 / T-0	warning / warning / critical

PO reminders ping: Logistics + originating Site HOD. DN reminders ping: Logistics + Site HOD + Warehouse user(s) at the receiving warehouse.

No duplicates - the sweep cannot fire twice for the same delivery on the same day. Two guards:

1. Every reminder sent is recorded against the document, its delivery date and how many days out it was, and the same combination can only ever be recorded once.
2. The system also stamps the date it last ran the sweep, so once it has run for the day the check is skipped entirely rather than repeated every minute.

Restarting the reminder service mid-day is safe. Re-running the sweep manually on the same day sends zero new notifications.

Customising the cadence - the two-day, one-day and same-day pattern is currently fixed. Making it configurable per site is on the backlog; ask if you need a different rhythm.

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (the unique identifier), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current Stock} = \text{Total Received} - \text{Total Consumed} - \text{Total Returned}$. It is never stored as a number anyone can type over; it is recalculated from the ledger every time it is shown.

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	Access tokens for the offline-capable mobile app
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered -> closed / force_closed / cancelled
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered / returned / closed / force_closed
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed / cancelled
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned / cancelled
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled

Table	What it stores	State machine
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	Unread -> read, stamped with the time it was read
delivery_reminders_sent	Record of which T-2 / T-1 / T-0 reminders have gone out	Write-once - the same reminder can never be recorded twice

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	The Material Code or Description contains "RL-", "RUBBER LINING" or "RUBBER-LINING"
BL	Brick Lining	It contains "BL-", "BRICK LINING", "BRICK-LINING" or "BRICK MATERIAL"
(none)	Neither family	The default when nothing above matches

The system works this out for you. An item is never tagged as both - if the wording matches both families, RL wins.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's

Term	Meaning
	transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

13. 2026-06 Feature Update - What Changed

This section documents the upgrades shipped in the 2026-06 release. Everything above remains accurate except where this section overrides it.

13.1 Field-level changes

Change	Where	Behaviour
All form fields mandatory	Entry Log (Consumption / Receipt Staging), HOD Receive Material, Admin Add Entry forms	Every text/number/select input is required. Validation lists missing fields on submit.
Expiry Date is optional	SK Receipt Staging, HOD Receive Material	Marked (Optional). Leave blank for non-perishable items.
Remarks, Tank No., Serial No., PR Number	Same forms	No longer optional.

13.2 Live Dashboard column order

The Live Dashboard table now renders in this order (when columns exist):

SAP_Code -> Material_Code -> Equipment_Description -> UOM -> Opening_Stock -> Receipt -> Consumption -> Return -> Closing_Stock -> Minimum_Qty -> Unit_Cost -> Stock_Value -> Category -> Status

- Opening_Stock is now a configurable column on inventory. Default 0; admin can edit in DB Editor.
- Identity formula updated to $\text{Closing_Stock} = \text{Opening_Stock} + \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned}$.
- Material_Code now appears after SAP_Code in every report (Daily Consumption, Daily Receipts, Monthly Summary, PR Status, etc.) and in the HOD Pending Receipts approval list.

13.3 Entry Log access - Store Keeper only

The Entry Log page is now visible only to the store_keeper role. HODs review submissions in HOD Portal; Admins use Admin Portal. The page is hidden in the sidebar for other roles.

13.4 EOD Commit - checkbox confirmation

The "Confirm EOD Commit" dialog no longer requires typing COMMIT. Tick the confirmation checkbox and click Confirm Commit. Cancel still drops all pending state.

13.5 Material Category

Every inventory item now carries a Category. Categories: Consumable, Equipments, Utilities, Maintenance, Others (default), Rubber materials, Tools, QC items.

- Admin Portal Add New Entry -> renders Category as a selectbox.
- Reports page -> Filter by Category dropdown alongside the SAR / cost-columns toggle. "All Categories" disables the filter.

13.6 Rubber MTC workflow

When a Store Keeper stages a receipt and the selected material's category is Rubber materials, the system shows:

- MTC Number text field (e.g. MTC-2026-001234)
- MTC Document file uploader (PDF / JPEG / JPG / XLSX)

Either field can be blank - the receipt still goes through. What changes is HOD-side visibility:

- HOD Portal -> Pending Receipts: a red banner lists rubber items received without an MTC.
- Click Draft Logistics Email to open a pre-filled email to the logistics team listing SAP, description, lot, qty.
- Sending (or clicking "Mark all as sent") flips the rubber rows to sent_to_logistics.

The logistics email recipient is configured once by an Admin, and falls back to logistics@generalindustries.net if nothing is set. On Windows it opens Outlook; on macOS it opens the Mail app; on Linux it opens whichever mail program is set as the default.

13.7 Document attachments - Entry Log + HOD DOC tab

Store Keepers can attach reference documents (PDF / JPEG / JPG / XLSX) on:

SK form	Doc number used	Notes
Consumption Log	Auto = DDMMYY of the date	Pick scope: "Whole entry (batch)" or "Specific date".
Receipt Staging	DN No. of the row (or manual override)	Falls back to DN-DDMMYY if no DN_No found.

Each file is stored inside the database itself - that is the authoritative copy, and it travels with a backup. A second copy is also written to an uploads folder on the server, arranged by site, document type and document number, purely so files can be browsed on that machine. Only the copy held in the database is portable.

HOD Portal -> DOC is a new tab with three sub-tabs: Consumption / Receipt / <- Return. Each shows period (From/To dates) and Doc Number text filters, with a per-file download button.

The <- Return sub-tab pulls from the new Return Items workflow (see §13.10), not from Returnable Items.

13.8 QR Label approval flow

The Admin DB Editor's QR generator (single-user, single-item) is unchanged. The new flow is two-step:

1. Store Keeper -> Entry Log -> QR Label Request (new tab)
 - Multi-select materials in one form.
 - Per-item label quantity.
 - Click Submit Batch for HOD Approval.
1. HOD Portal -> QR Approval (new tab)
 - Pending sub-tab: select rows via checkbox, then [OK] Approve Selected or [X] Reject Selected.
 - [OK] Approved sub-tab: Download QR Labels PDF for ALL approved generates one consolidated PDF.

13.9 Returnable Items - clarification

The Returnable Items tab is for temporary tool loans only (e.g. issuing a torque wrench to a worker who'll return it before EOD). It is not a way to return stock to the warehouse. No DN No., no document attachment.

For real returns (defective material going back to logistics), use the new Return Items tab - see §13.10.

13.10 Return Items workflow (NEW)

The new <- Return Items tab (between Receipt Staging and Returnable Items in the Entry Log) handles real returns to the warehouse / logistics.

Store Keeper flow

1. The material picker is restricted to materials received in the last 30 days at the user's site.
2. If multiple receipts exist for the same SAP code, the SK is asked which receipt is being returned (Date / DN No. / Received Qty).
3. The system shows a locked summary of the original receipt: Date, DN No., PR, Lot, Received Qty.
4. The SK enters:
 - Return Quantity (capped at the original Received Qty)
 - Reason (work-types dropdown)
 - Return DN No.
 - Attachment - mandatory: the Return DN + any photos
9. To return material older than 30 days, tick "Override 30-day window" - the picker widens to 12 months and an Override Justification field appears. This routes to HOD as an explicit override request.
10. Submit -> request lands in HOD Portal <- Returns tab. A WhatsApp ping goes to the site HOD if a phone number is on file.

HOD flow

1. HOD Portal -> <- Returns lists every pending return with a card per row.
2. Rows that required an override are highlighted in red and show the SK's justification.
3. [OK] Approve -> writes a row to the returns ledger (so Current_Stock reduces by the returned qty, the dashboard Return column ticks up, and the entry shows up in monthly / consumption reports). Then automatically opens the logistics email draft with item, qty, reason, and the original receipt's DN/PR/Lot context.
4. [X] Reject -> marks the request rejected. The SK sees this in their request history.

Returning material that Quality rejected (added August 2026)

When a Quality inspector rejects some or all of a batch, they no longer just tell you a number. The rejection is given a Return No that looks like QCR-20260813-41, and it is shown to the inspector, to you, and to your Head of Department.

1. Type or paste that Return No into the box at the top of Return Stock and press Fetch.
2. The form fills itself in: the material, the site, the lot, the quantity that was rejected, and the inspector's own reason.
3. Change anything you need to. You may return less than was rejected - some of it may already be out on the job, or still being discussed with the supplier. You cannot return more.
4. A Return DN number and a scanned document are always required for this kind of return, even at sites where supporting documents are otherwise optional. Rejected material does not leave the site without paperwork.
5. Submit as normal. Your Head of Department approves it, and the quantity then leaves stock.

Two things worth knowing:

- A Return No can only be used once. If you try it a second time you will be told which return already used it. This is deliberate - without it the same rejected quantity would be taken out of stock twice.
- You are not asked to pick a source receipt for these returns. The rejection already records where the material came from, and material inspected at a warehouse has no site receipt to point at.

If a recently received item is missing from the Source Receipt list

This was fixed in August 2026. The list used to be built from the delivery date written on the supplier's paperwork, so goods you received this morning against a document dated weeks earlier did not appear, while older items did. The list now also considers when the receipt was actually entered, and the most recently entered receipts appear at the top.

Dashboard / report impact

- The Return column on the Live Dashboard reflects approved returns (since returns is the source of truth).
- All existing reports (Daily Consumption, Daily Receipts, Monthly Summary, Audit) include returns via the same identity math.
- The HOD DOC tab <- Return sub-tab lets the HOD browse all attached return documents.

13.11 Per-site Work Type and Tank No.

HOD Portal -> Site Config lets the site HOD add or delete Work Types and Tank Numbers scoped to their site. Empty per-site lists fall back to the global defaults.

13.12 WhatsApp worker - startup fix

The notification service now starts instantly instead of taking tens of seconds, because it loads its delivery components only when the first message actually needs sending.

If startup appears to hang for more than a few seconds after an update, the update was probably applied without its accompanying dependency step. Ask your administrator to re-run the update procedure.

13.13 Category rename - Rubber Materials -> Surface Shields

The category that triggers the MTC workflow is now Surface Shields instead of "Rubber Materials". Everywhere the system used to look for Rubber materials, it now looks for Surface Shields. The behaviour is identical:

- SK selects a Surface Shields item on Receipt Staging -> MTC Number + MTC File uploader appear.
- Missing MTC = HOD sees the red banner on Pending Receipts with Draft Logistics Email.

[!] The certificate is never required to receive material, and never required to send it to site. The uploader appears at Receipt Staging because that is a convenient moment to file a document somebody is holding - not because the receipt is blocked without it. The one place a missing certificate stops work is issue to a worker. See §22.1 for the full rule and for who is expected to file the document.

Which category requires a material certificate is a single setting, so extending the requirement to another category is a one-line change for your administrator rather than a development task.

13.14 WBS Master + WBS-aware Entry Log + WBS Report

- HOD Portal -> Site Config -> WBS Numbers lets the HOD add, close, or re-open WBS numbers for their site. Each WBS carries an optional Description and an active / closed status.
- Entry Log -> Consumption Log and Receipt Staging show a WBS Number dropdown filtered to the SK's site. If the HOD hasn't added any WBS yet, the SK sees a warning and a free-text fallback so work isn't blocked.
- Reports -> WBS Report rolls everything up by WBS for a chosen date range, scoped to the user's site. Columns: WBS_Number, Consumption_Rows, Consumption_Qty, Consumption_Value_SAR, Receipt_Rows, Receipt_Qty, Receipt_Value_SAR. Sorted by consumption value descending.
- The WBS field is added automatically wherever it was missing on consumption, receipts and both staging queues. Entries made before the field existed show as "(no WBS)" in the report; new entries carry the WBS you picked all the way through from Store Keeper to HOD commit, untouched.

13.15 Site_ID badge in sidebar

Every signed-in user now sees their site code as a gold pill in the sidebar user card, next to the role label. This makes "wait, which site am I logged into?" impossible to get wrong, which matters when an

Admin shadows a site for support.

13.16 Live Dashboard - live-typing filter on key columns

The dashboard filter row no longer waits for Enter. Typing into any of the four searchable columns - SAP_Code, Material_Code, Equipment_Description, Category - narrows the table on every keystroke (~180 ms debounce). The numeric columns deliberately have no filter input: text filters on numbers don't help and they made the page laggy.

Header reads "Filters (live) - searchable on SAP / Mat Code / Description / Category".

Live typing needs an optional add-on to be installed on the server. Without it the dashboard falls back to ordinary search boxes - still fully working, they just need Enter.

13.17 Sidebar Hub Assistant - role-aware Q&A

A new sidebar expander Ask Hub Assistant lets any signed-in user ask plain-English questions about the system. The answer is written by the local AI service running on the server. The role filter is applied to the manual chapters BEFORE the question is answered, not merely to the instructions given to the AI - a Store Keeper's question is never shown the Admin chapter, so the assistant physically cannot answer about admin features even if asked.

Section visibility:

- Store Keeper: §1, §2, §3, §4, §11, §13
- Supervisor: SK list + §5, §8
- HOD: Supervisor list + §6, §9
- Admin: everything

The answer appears a few words at a time as it is written. Click Clear to reset.

13.18 GMT+3 (Asia/Riyadh) timestamps

Timestamps shown in the UI are now Riyadh local time. Background:

- New records are stamped in Riyadh time directly, because the server itself is set to that time zone.
- Times stored by the database itself are converted to local time when they are displayed.
- Affected views so far: Admin Pending Cross-Site Requests, Admin Audit Logs, Admin Live Activity Feed, Admin WhatsApp Console, HOD Cross-Site Incoming, HOD Pending Receipts. Other tabs still show UTC; tell the admin to file a ticket if any specific column matters and isn't converted yet.

A site in a different time zone is a single setting for your administrator - no code change is needed.

13.19 Self-hosting

Everything needed to run the system on your own hardware ships with it: the background services, the installer, the restart and uninstall procedures, the nightly backup, and a full setup playbook that takes

about 45 minutes end to end.

Section 17 covers what this means for you day to day. The procedures themselves are in the deployment runbook, for whoever administers the server.

13.20 WhatsApp notifications

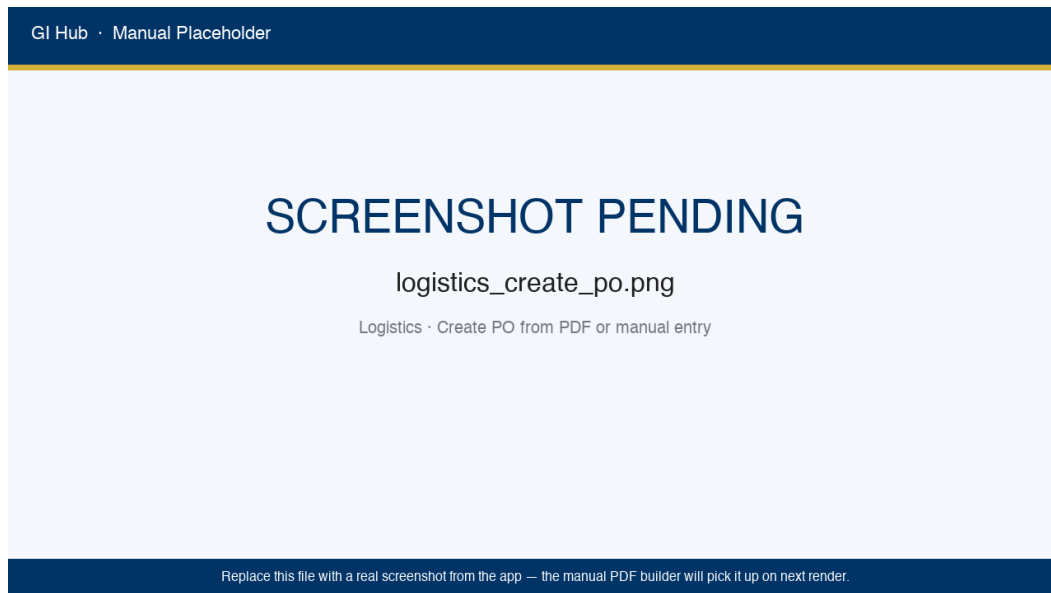
WhatsApp delivery is handled by a background service that runs independently of the application. Two things follow from that, and both are improvements:

- A WhatsApp outage never blocks work. Messages queue and send once the channel is back. Nothing is lost and nobody is held up.
- In-app notifications are unaffected. The bell in the header is filled by the application itself, so it keeps working even when WhatsApp does not.

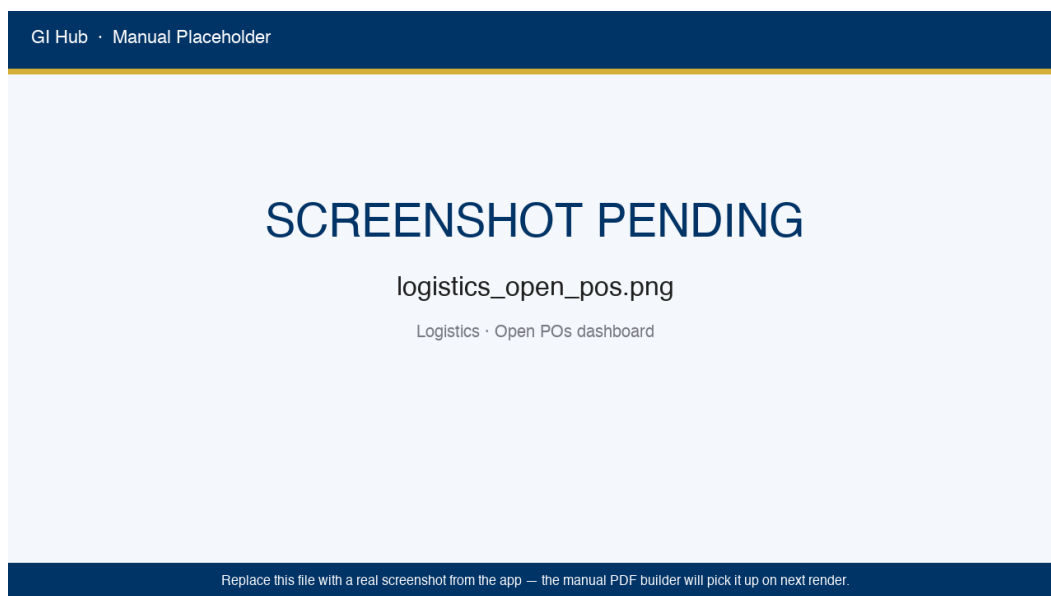
If messages stop arriving, see section 17.7.

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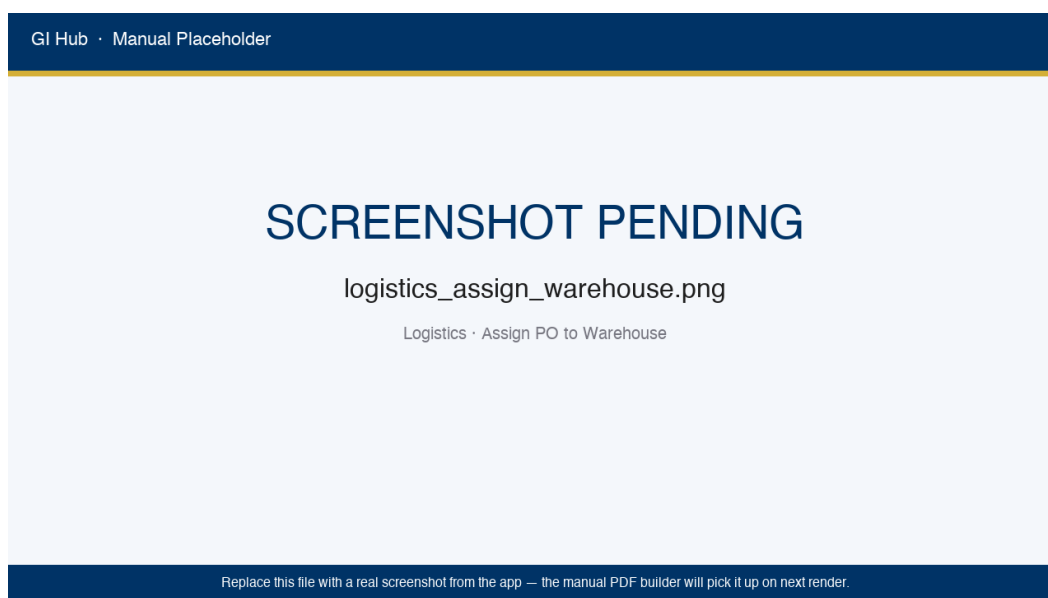
The Logistics Portal sits between Site HOD (who creates PRs) and Warehouse (which physically receives goods). Role-locked to {logistics, admin} - exact-role lock means no other role inherits access via the hierarchy. Eight tabs.



Logistics Portal - Create PO from PDF upload or manual entry



Logistics Portal - Open POs dashboard with status pills

*Logistics Portal - Assign PO to a Warehouse*

14.1 Pages visible to Logistics

- Live Dashboard (read-only - all sites)
- Logistics Portal (this page)
- Reports (incl. the 3 new procurement reports)

14.2 Tab 1: Incoming PRs

Site HODs submit PRs to Logistics from their HOD Portal -> PR tab. They land here.

14.2.1 Site filter

Dropdown of all sites + "All sites". Defaults to all.

14.2.2 Hero strip (3 cards)

Card	Source
OPEN PRs	Active queue count
TOTAL QTY	Sum across all open PR lines
SITES	Distinct sites currently submitting

14.2.3 Queue table

A table with columns: PR No. - Site - Lines - Total Qty - Submitted - Earliest Delivery - Status. Sortable, filterable, exportable.

14.2.4 Per-PR drilldown

Selectbox under the queue table. Pick a PR + Site combination and a section card appears:

Card row	Source
PR Number	the row
Site	the row
Lines	line count
Total Qty	summed across lines
Earliest Delivery	min of Delivery_Date across lines

Below: full line items grid with Material_Code, Material_Name, Requested_Qty, UOM, WBS_Number, Network, Plant, Delivery_Date, Supplier, Est_Cost_SAR.

14.2.5 Use this PR to create a PO button

Loads the PR into Tab 2 (Create PO). Selectbox state is preserved so you can switch tabs without losing the chosen PR.

14.3 Tab 2: Create PO

Two sub-tabs: Manual entry and PDF upload. Both create the Purchase Order the same way - the only difference is where the details come from.

14.3.1 Vendor picker

Selectbox of all active vendors `<code> - <name>`, plus a " Add new vendor" option that opens an inline expander with Vendor Code, Name, Address, Default Inco Terms, Default Payment Terms. Save -> re-renders the parent form with the new vendor pre-selected.

14.3.2 PO header form

Three-column layout:

Column	Fields
Left	PO Number * - PO Date - PO Type
Middle	PR Number - Quotation No. - Quotation Date
Right	Expected Delivery - Inco Terms - Payment Terms

Plus a second three-column row for Contact (vendor), Contact Email, Mobile, Our Reference, Your Reference, Our Email.

Vendor defaults (Inco / Payment Terms) auto-fill from the master row but remain editable.

14.3.3 Manual sub-tab: line items

If a PR was loaded via the Tab 1 shortcut, every PR line pre-fills with Include = True and zero Unit_Price + Total_Price (you set them). Otherwise an empty editable grid opens.

Editable columns: Include - Material_Code - Description - Qty - UOM - Unit_Price - Total_Price - WBS_Number - Network - Plant. Rows with Include = False are dropped on save.

14.3.4 PDF upload sub-tab

Element	Purpose
File uploader	Drop the PO PDF
Extract from PDF button	Reads the document and pulls out the header (PO Number, Vendor, Inco/Payment terms, Quotation refs, totals), the line items (Sr. No, Material_Code, Description, Qty, UOM, Unit_Price, Total_Price) and the PO Annexure delivery schedule (Shipment N / Material Group / Date)
Review extracted PO	Editable preview of every extracted field. The header form pre-fills. The line items grid pre-fills. Edit anything before saving.
Delivery schedule	If Annexure parsed, a table shows shipment_no - material_group - target_date
Save PO (from PDF)	Saves the PO, its line items and its shipment schedule. The original PDF is kept attached to the PO itself, so the source document is always available for audit.

On PO numbers with X-masking: the sample PDF has the last 4 digits masked as XXXX for security. The extractor preserves whatever is on the page verbatim. In production, vendors send full 10-digit numbers and those pass through unchanged.

14.3.5 Side-effects on save

- A new Purchase Order is created, open
- One line is created per item, each auto-tagged with its RL/BL family
- The matching PR lines - same PR number, same site - are marked as now covered by a PO, so they leave your Incoming PRs queue
- The site HOD gets a "PO issued" notification in-app, and on WhatsApp if that event is enabled

14.4 Tab 3: Open POs

Browse + drill into every open PO.

14.4.1 Filters

Site dropdown - Vendor dropdown - PR Number exact-match textbox.

14.4.2 KPI strip

OPEN POs count - Total value SAR - PENDING (not yet delivered) count.

14.4.3 Per-PO drilldown

Select a PO -> section card with PR Number, Vendor, Inco/Payment Terms, PO Date, Expected, Source (manual/pdf_upload), Status, Total. Below: items grid with an RL/BL family chip per row, the parsed

delivery schedule (if any), and a list of warehouse assignments.

14.5 Tab 4: Assign to Warehouse

Route a PO (or a subset of items) to a warehouse for receiving.

14.5.1 PO + Warehouse pickers

Two-column row: PO selectbox - Warehouse selectbox (shows <id> - <name>).

14.5.2 Items selector

Editable grid showing every PO line with an Include checkbox (default True). Disable lines you don't want to route - they stay with this PO for a future assignment.

14.5.3 Header

Expected Delivery date + Notes (visible to Warehouse).

14.5.4 Assign to Warehouse button

- If every line is included, the assignment is recorded as covering the whole PO - which reads more cleanly in an audit than listing each line
- If only some lines are included, the assignment records exactly which PO lines were routed
- Warehouse users at the chosen warehouse get a "PO assigned to warehouse" notification in-app, and on WhatsApp if that event is enabled
- PO header's Expected_Delivery auto-fills with this date if not set
- Audit log entry: ASSIGN_PO_TO_WAREHOUSE

14.6 Tab 5: Reschedules

Incoming reschedule requests from Warehouse / Site HOD.

14.6.1 Empty state

If no pending requests: empty-state card.

14.6.2 Per-request card

Element	Purpose
Header line	PO <number> - DN <number> - From <current_date> -> <requested_date>
Subline	requested_by + role + reason
Decide popover	Decision notes textbox + [OK] Approve / [X] Reject buttons (reject requires reason)

Approval automatically pushes the new date everywhere it matters: the PO's expected delivery date, the

warehouse assignment's expected delivery date, and the Delivery Note's date if one is linked.

14.7 Tab 6: Force-Close

Use sparingly. Force-closing notifies Admin + originating Site HOD immediately with the reason.

14.7.1 Target radio

PR (entire) - PO (entire) - PO line (single item)

14.7.2 Reason textbox

Mandatory, minimum 3 characters. Stored word-for-word on the force-closure record and in the audit log.

14.7.3 Target picker (changes per radio)

- PR: dropdown of open PR numbers in Logistics queue
- PO: dropdown of open POs
- Line: PO dropdown -> then line dropdown showing only open / partially_delivered lines

14.7.4 Force-close button

Single confirm-and-execute. Behind it:

- PR target: the PR is closed and marked force-closed in the Logistics queue
- PO target: the PO and all of its lines are marked force-closed
- Line target: only that one PO line is marked force-closed
- A force-closure record is written, carrying your reason
- Notifications go out to every Admin and to the HOD of the site the document came from, at critical severity

14.7.5 Recent force-closures (audit)

A table of the last 50 closures. Read-only.

14.8 Tab 7: <- Vendor Returns

Raise a return to the vendor against a PO. Returning a line REOPENS the PO so it shows in your active queue again.

14.8.1 PO picker + scope

PO selectbox (all POs incl. closed). Then radio: Whole PO or Single line. If single line, a line picker appears.

14.8.2 Return details form

- Return quantity (number, > 0)

- Reason (textarea, mandatory)
- Expected resupply (date picker, defaults to today + 14)
- Notes (optional)

14.8.3 <- Raise vendor return button

- A return record is created, showing it was raised by Logistics
- The PO line's returned quantity goes up, and the line reopens as either partly delivered or open
- If the PO had already been closed, it reopens as partly delivered and its closure details are cleared
- A "vendor return raised" notification goes to the Logistics inbox, and to WhatsApp if that event is enabled

14.8.4 Open returns table

A table of all open returns. Read-only.

14.9 Tab 8: History

Read-only archive. Two sub-tabs: Closed POs and Closed PRs.

- Closed POs: status in closed / force_closed / cancelled
- Closed PRs: any PR that is itself closed, or that Logistics has closed, force-closed, or already turned into a PO

For force-closures with the reason history, use Tab 6 -> Recent force-closures table.

14.10 Logistics - Use Cases

Use Case 1: Issue PO from a fresh PR

1. Incoming PRs -> pick PR -> Use this PR to create a PO
2. Create PO -> Manual entry -> pick / add vendor, fill header, edit Unit Price per line
3. Save PO -> balloon animation -> notification fires to Site HOD

Use Case 2: Issue PO from a vendor's emailed PDF

1. Receive PO PDF from vendor (sample format: Create PO -> PDF upload)
2. Drop the PDF -> Extract from PDF
3. Review extracted header + line items + delivery schedule
4. Correct anything the parser missed
5. Save PO (from PDF) - original PDF is archived

Use Case 3: Route a PO to a warehouse

1. Assign to Warehouse -> pick PO + warehouse
2. Disable lines you want to route separately later

3. Pick Expected Delivery, add a routing note
4. Assign - warehouse user is pinged

Use Case 4: Handle a Warehouse reschedule ask

1. Reschedules -> review the request (reason text + current vs requested date)
2. Decide -> notes + [OK] Approve
3. The PO's expected delivery date updates automatically; Warehouse and the requester get the decision notification

Use Case 5: Force-close a stale PR

1. Force-Close -> radio = PR (entire)
2. Write the reason (e.g. "Project cancelled by site management 2026-06-15")
3. Pick PR -> Force-close PR
4. Admin + Site HOD see a critical notification in their bell

14.11 Logistics - FAQ

Q: A PR appears in Incoming PRs but I can't issue a PO against it - Material_Code is wrong. A: PR lines come from the Site HOD's inventory catalogue. If the Material_Code is wrong, the SK created it wrong upstream. Don't fix in your PO - bounce the PR back via WhatsApp / chat, ask the site to re-submit. There's no in-app "reject PR" button (yet) because PR rejection should be a conversation, not a click.

Q: PDF extraction picked up wrong qty. A: The Review preview is editable. Always check Qty / Unit_Price / Total_Price before saving. The extractor is calibrated against the General Industries sample layout; other vendor templates may need template additions.

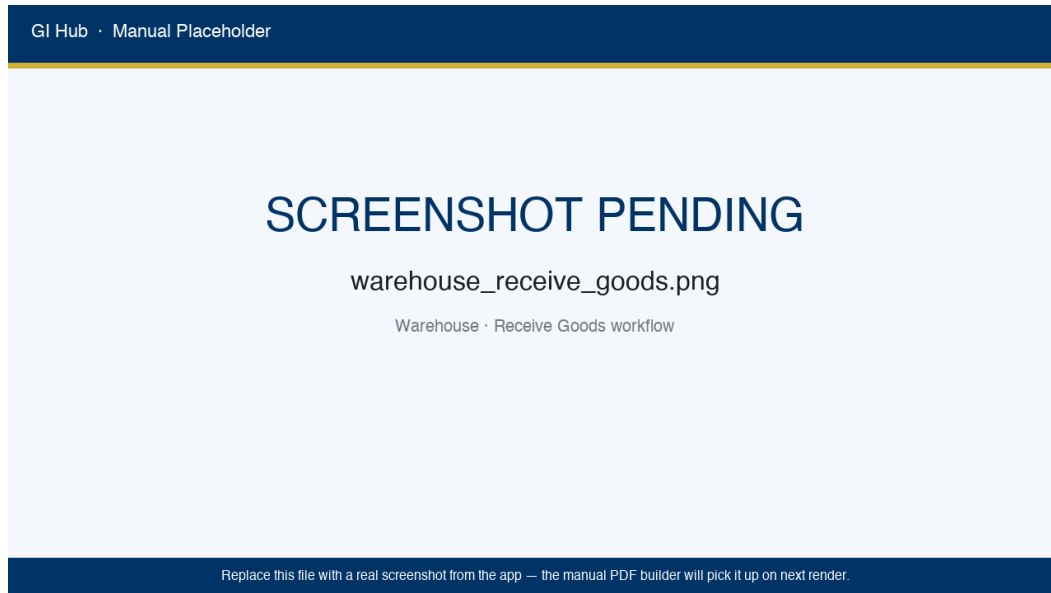
Q: I want to assign a PO to multiple warehouses. A: Two separate Assign actions - one per warehouse. Each assignment records its own set of PO lines, so the two never overlap.

Q: Why can't I see prices in Tab 4 / on assignment cards? A: You CAN see prices in Tab 3 (Open POs drilldown). Tab 4 (Assign to Warehouse) shares the read with Warehouse users, so prices are hidden there for consistency.

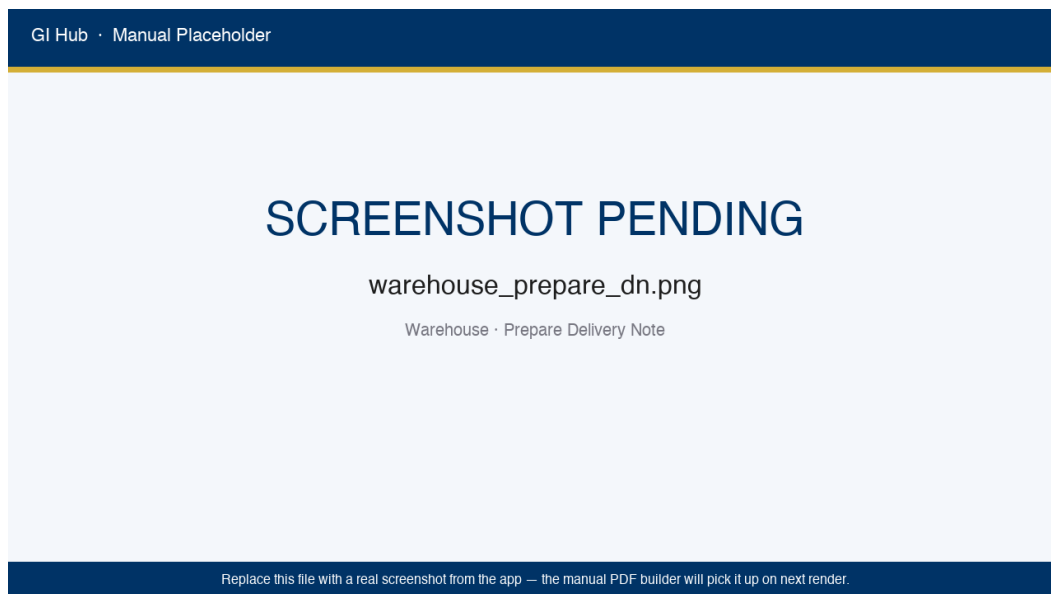
Q: Force-closure undo? A: Not yet. Once force-closed, you'd have to either raise a Vendor Return (reopens the line) or admin-edit via Master DB Editor. A 24-hour undo window is on the v3.0 backlog.

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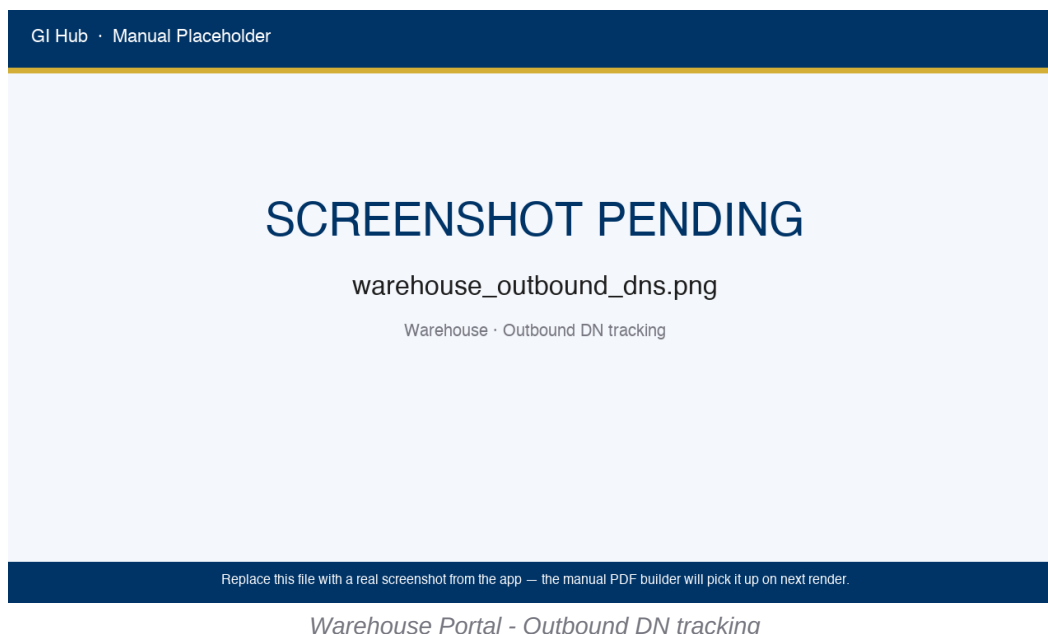
The Warehouse Portal is the physical-receiving and DN-preparation side. Role-locked to {warehouse_user, admin}. Six tabs. Prices are completely hidden in every view - three independent enforcement layers guarantee Unit_Price, Total_Price, Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are never visible to a warehouse user.



Warehouse Portal - Receive Goods workflow



Warehouse Portal - Prepare Delivery Note with RL/BL strict separation



15.1 Pages visible to Warehouse User

- Live Dashboard
- Warehouse Portal (this page)
- Reports
- Locator - which rack a material is on, and the reverse: scan a rack and see everything that should be on it, which turns a stock count into a checklist (see §21.8)
- Assets - serialised tools and equipment, each with its own record, status and location history, including a map link where a position was captured (see §21.7)

15.2 Sidebar warehouse resolution

- A Warehouse User is tied to a single warehouse on their own user record. The portal works out which one from their profile.
- An Admin shadowing the portal gets a " Shadow warehouse" picker in the sidebar listing every active warehouse.
- If neither applies, a red error card appears: " Your account is not bound to a warehouse. Ask Admin to set your Warehouse ID in Admin Portal -> Users."

15.3 Page title

" Warehouse <ID>" in gold, with a muted "(prices hidden - Logistics-only)" caption next to it as a permanent visual reminder.

15.4 Tab 1: Incoming Assignments

POs Logistics has routed to this warehouse.

15.4.1 Hero strip

Card	Source
AWAITING ACK	Assignments routed to you that you have not acknowledged yet
ACKED	Acknowledged - you've seen them, waiting on the goods
IN/RECEIVED	Partly or fully received

15.4.2 Assignments grid

A table with columns: Assign # - PO No. - PR No. - Vendor - Dest Site - Expected - Assigned - Acked - Status. No price columns.

15.4.3 Acknowledge action

A picker listing every assignment still awaiting acknowledgement. Choose one, click [OK] Acknowledge, and it moves to acknowledged and pings Logistics with a "warehouse acknowledged" notification (this one is off by default in the WhatsApp toggles to keep the noise down).

15.5 Tab 2: Receive Goods

Record qty actually received at this warehouse against an acknowledged assignment.

15.5.1 Assignment picker

A picker listing every assignment you have acknowledged but not yet fully received.

15.5.2 PO snapshot card (no prices)

KV row	Value
PO Number	the PO
Vendor	<code> - <name>
Inco / Payment	the terms
Expected	the date

Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are stripped from the header dict before render. Never visible.

15.5.3 Receive grid

Editable grid showing every line on the assignment:

Column	Source
id	po_items.id (disabled - read-only)
Material_Code	line code

Column	Source
Description	line description
UOM	unit
rl_bl_family	family chip (RL / BL / blank)
Qty	ordered (read-only)
Delivered_Qty	already received cumulative (read-only)
Open_Qty	computed Qty - Delivered + Returned (read-only)
Receive Now	EDITABLE - type the qty you physically received this event

15.5.4 Record receipt button

- Validates: at least one row has a Receive Now quantity above zero
- Over-deliver guard: what has been delivered, less anything returned, plus what you are recording now, may NOT exceed the quantity ordered. If it does, the whole batch is rejected with a friendly message naming the line.
- On success: each line's delivered quantity goes up, and the line is marked delivered once the net delivered amount covers the ordered quantity, or partly delivered otherwise
- The assignment moves to received once every line on the parent PO is delivered, and to partly received otherwise
- The PO itself is updated to match
- A "warehouse received goods" notification goes to Logistics with the line count

15.6 Tab 3: Prepare DN

Build a Delivery Note for a site. RL/BL strict separation is enforced here - a DN cannot span both families. If you try, the action is rejected with: "Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."

15.6.1 Source pickers

- PO No. picker - assignments you have acknowledged, partly received or fully received
- Destination Site selectbox

15.6.2 Items grid

Editable grid with read-only inventory columns (Material_Code, Description, UOM, rl_bl_family, Qty, Delivered_Qty, Returned_Qty) and editable shipping columns:

Column	Purpose
Ship Qty	What you're sending on this DN (must be > 0 to include)
Lot_Number	If you tracked the lot at receive time
Expiry_Date	Lot expiry

Column	Purpose
Remarks	Free text

15.6.3 DN header

Three-column row: DN Date - Vehicle No - Driver Name - Driver Phone - Prepared By (auto-filled with your username) - Remarks.

15.6.4 Save DN draft button

- Validates: at least one row with Ship Qty > 0
- Over-ship guard: for each line, what you may ship is what was delivered to you, less anything already returned, less anything already committed on other live Delivery Notes. Ship quantity cannot exceed that.
- RL/BL strict-separation check: if the items span both families, REJECTED
- On success: a new Delivery Note is created as a draft, with one line per item shipped. The DN header records the RL/BL family where the items carry one.
- Toast: DN <number>

15.6.5 DN numbering convention

DN-<WAREHOUSE_ID>-<YYYYMMDD>-<seq> - seq resets per (warehouse, day). Example: DN-WH-A-20260616-003.

15.7 Tab 4: Outbound DNs

Track every DN this warehouse has prepared.

15.7.1 Status filter

Multi-select of every DN status. Defaults to the active ones - draft, awaiting Logistics, awaiting HOD and awaiting Store Keeper.

15.7.2 DN grid

A table of every matching DN with full state metadata.

15.7.3 Submit-to-Logistics action

Whenever the filter shows a draft, a "Submit a draft to Logistics" section appears with a picker of your drafts and a Submit to Logistics button. Submitting moves the DN into the Logistics approval queue and sends them the DN-approval notification.

15.7.4 Per-DN drilldown

Select a DN -> section card showing PO, Site, Status, RL/BL family, Vehicle/Driver, and the full signature trail (Logistics decided by/decision, HOD decided by, SK received by). Items grid below.

15.7.5 Request reschedule expander

If the date you targeted isn't going to work, raise a reschedule from here. Same flow as the HOD's In-Transit tab - you cannot pick a date in the past, and a reason is mandatory.

15.8 Tab 5: <- Returns from Site

Raise a return when a site flags defective material from a DN this warehouse delivered.

15.8.1 DN picker

Selectbox of DNs in received / pending_sk / hod_approved states.

15.8.2 Items grid

Editable grid with a Return Qty column. Filling > 0 on a line includes it in the return.

15.8.3 Reason textarea

Mandatory.

15.8.4 <- Raise return to vendor button

- One return is raised per affected line
- For each line: a return record is created showing it was raised by the Warehouse, the DN line is flagged as returned, the parent PO line's returned quantity goes up, and that line reopens as either partly delivered or open
- If the PO had already been closed, it reopens
- Notification fires to Logistics

15.9 Tab 6: History

Read-only. Two sub-tabs: Completed DNs (status in received / rejected / cancelled) and Closed assignments.

15.10 Warehouse - Use Cases

Use Case 1: Receive a vendor delivery

1. Incoming Assignments -> [OK] Acknowledge the assignment when you see it
2. Goods physically arrive
3. Receive Goods -> pick the assignment -> type the actual received qty per line -> Record receipt

Use Case 2: Ship to a site (single family)

1. Prepare DN -> pick PO + destination site

2. Type Ship Qty per line - keep all RL or all BL
3. Fill DN header (Vehicle, Driver, Date)
4. Save DN draft -> status = draft
5. Outbound DNs -> select the draft -> Submit to Logistics

Use Case 3: Ship RL + BL to the same site on the same day

You need TWO DNs:

1. Prepare DN with only RL lines -> save + submit
2. Prepare second DN with only BL lines -> save + submit

The system rejects any attempt to combine them on a single DN.

Use Case 4: A site rejects part of a DN - raise to vendor

1. <- Returns from Site -> pick the DN
2. Type Return Qty on the offending line
3. Reason: "Site reported defective surface coating"
4. <- Raise return to vendor - PO + po_item reopen; Logistics sees the return in their tab

Use Case 5: Request a reschedule

1. Outbound DNs -> drill into the affected DN
2. Request reschedule for this DN -> pick new date + reason -> Submit

15.11 Warehouse - FAQ

Q: Why don't I see prices anywhere? A: Warehouse role is intentionally blind to commercial data. Three independent enforcement layers strip every monetary field. If you genuinely need a price (e.g. to file a damage claim), ask Logistics or Admin.

Q: My over-ship guard rejected a DN. I just received everything from the vendor. A: Probably another DN is already drafted (or submitted) shipping a slice of that line. Check Outbound DNs filter for live DNs against the same PO. If one was abandoned, ask Logistics to reject/cancel it so the qty frees back up.

Q: RL and BL are going to the same site on the same truck. Can I combine? A: No. Two DNs. The strict separation is by design (different testing standards, different storage requirements). The site receives them as two separate DNs in their Receipt Staging queue.

Q: Where did the "Receive Goods" assignment go? It was here this morning. A: Once every line on the parent PO is fully delivered, the assignment moves from partial to received and stops appearing in Tab 2 (which filters to acknowledged + partial). Check the assignment grid in Tab 1.

Q: Can I edit a DN after submitting to Logistics? A: Not directly. Ask Logistics to reject it (Outbound DNs shows the rejection if it happens) - then it returns to draft for you to edit.

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number 4003951, Network 4003951-PROJ-A, Plant GI-PS01, Delivery_Date Day 3.
Day 1 - 8:15 AM	Site HOD	Opens the Submit PR(s) to Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	A "PR submitted to Logistics" notification lands in the Logistics inbox with a red badge, plus a WhatsApp ping if that event is enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL. Clicks Use this PR to create a PO.
Day 1 - 9:30 AM	Logistics	Switches to Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Number 4720033030, Unit_Price per line, Total_Price computed. Save PO.
Day 1 - 9:30 AM	(auto)	The PR lines are marked as covered by a PO and disappear from the Logistics queue. The site HOD gets a "PO issued" notification with the

When	Who	What
		vendor and PO number.
Day 1 - 10:00 AM	Logistics	Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.
Day 1 - 10:00 AM	(auto)	A "PO assigned to warehouse" notification lands in the WH-A users' inbox, plus a WhatsApp ping if that event is enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK] Acknowledge assignment #N. Goes about their day.
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL line. Record receipt. Both lines are marked delivered.
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/BL strict separation). Saves DN-WH-A-20260617-001 for RL only.
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line -> DN-WH-A-20260617-002.
Day 2 - 5:40 PM	Warehouse user	Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	Both DNs move to awaiting-HOD-approval. The site HOD gets the "DN approved by Logistics" notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via DN Approvals tab. Approves both.

When	Who	What
Day 2 - 6:00 PM	(auto)	Both DNs move to awaiting-Store-Keeper, and a matching draft receipt appears in the site's Receipt Staging for each. The site SK gets the incoming-DN notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new Incoming Delivery Notes from Warehouse expander is open at the top, showing both DNs. Inspects both, confirms physical match, clicks [OK] Mark as Received on each.
Day 3 - 8:45 AM	(auto)	Two receipts are posted to the ledger, each carrying its DN number, source warehouse and originating PO. Both DNs are marked received, the staged drafts are cleared, and the Live Dashboard shows the new stock immediately.

16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

- DN-002 is marked rejected, and the reason is stored with it
- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed

- Site HOD's In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> Outbound DNs -> drill into DN-001 -> Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

16.4 What stops a duplicate PR or a double order

Added 2026-08 (Phase 8). Three rules run underneath the procurement chain. None of them changes how the screens are used; they change what happens when two people, or one impatient person, do the same thing twice.

16.4.1 A PR number is reserved, not guessed

Every PR number is claimed in a registry before the requisition's lines are written, so two HODs pressing Create PR in the same second get different numbers. If a number is somehow taken, the system takes the next one rather than failing. Renaming a PR moves its reservation with it and is refused if the new name is already in use - by a registered PR or by one that was imported without ever being registered.

16.4.2 Each step happens once, in order

A requisition moves site_draft -> submitted -> in_po, and each move is checked against the state the row is actually in - not the state the screen was showing when it loaded. Consequences you will see:

- Submitting twice does nothing the second time. It used to accept lines that were already submitted, rewrite their timestamp and fire a second notification at Logistics.
- A PO cannot be raised twice over the same lines. The second attempt is refused with a message naming the lines, rather than silently succeeding.
- A PR may still carry several POs. Partial fulfilment splits one requisition across vendors or deliveries, so the lock is per line, not per PR.

16.4.3 A retry is not a second order

Submit PR, Create PO, Assign PO and Rename PR each carry a key minted when the form opens. Press the button twice, lose the connection and retry, or double-click - the system recognises the repeat and replays the first answer instead of doing the work again. Sending the same key with different details is refused, because that is a client bug rather than a retry.

[!] The buttons disappear rather than grey out. Once a PR is submitted, the Submit button is gone - not disabled. A greyed-out button invites a second click and a support call about why it "isn't working"; an

absent one says the step is done.

17. Operations & Hosting - the after-launch chapter

This chapter is for whoever is responsible for keeping the system running once it is live. Everything before it describes how to use the application; this chapter describes how it is hosted, what protects it, and what to do when something goes wrong.

It is written for a manager, not an engineer. Nothing here asks you to type a command. Where a task genuinely needs technical work, this chapter says what the task is, who performs it and how long it takes. The step-by-step procedures live in the technical runbooks listed in section 17.8, kept separately for whoever administers the server.

16.5 WBS numbers and work types (HOD)

Where: HOD -> WBS & Work Types (/hod/wbs). HOD and Admin only.

Every issue and receipt can carry a WBS number - the cost centre the material is charged to. Until 2026-08-25 the column was blank on every entry in the system, and the reason was not a missing feature: the table, the rule and the endpoints all existed and there was no screen to reach them. This page is that screen.

16.5.1 Nothing is enforced until you turn it on

Both rules are conditional. A site with no WBS numbers is never asked for one; a site with no work types keeps a free-text Work Type box. Adding the first row turns the rule on for that site, and the banner at the top of the page always says which state you are in.

That is deliberate: switching the requirement on is an act you take, recorded in the audit log, not something a software release does to your site overnight.

16.5.2 The two tabs

WBS Numbers - the cost centres themselves. Add them, and close ones that are finished. Closing a number stops it being offered; it does not touch entries already posted against it.

Work Types - the canonical list your Issue form offers, and the WBS each one charges to. A work type with no WBS is perfectly legal - it simply will not fill one in automatically.

Import from history reads the work types your site has actually used and offers them one at a time, with how often each was used. Nothing is pre-loaded on purpose: see the next section for why.

16.5.3 [!] One work type, one spelling

The system stores work types by a normalised spelling - lower case, trimmed, with repeated spaces collapsed. Civil and civil are the same work type and the second one is refused, naming the first.

This matters because the live ledger already contains both spellings of four work types (civil/Civil,

coating/Coating, In yard/In Yard, others/Others). Left as separate entries they would take different WBS numbers, and the report would split in two with nothing on the page to explain why.

What it does not do is merge things that merely look similar. Arrangement and Site Arrangement stay two work types, as do Blasting and Sweep blast. Whether those are really the same work is a judgement about the work - it is yours to make here, not a rule the software should apply behind your back.

Whatever casing you choose is what gets stored on every entry from then on.

16.5.4 Where the WBS on an entry comes from

In order, first match wins:

1. the number picked on the form - a human chose it, and it is never overridden. The map is a default, not a correction.
2. the work-type map for that site.
3. the equipment master's WBS (sme_equipment.WBS_No), where the SME workbook carries one.
4. nothing - and that stays legal.

The audit line for the entry records which of these supplied the number, because the first question about a wrong cost centre is always "who chose this?".

16.5.5 [!] Two names that are not work types

SUPERVISOR_REQUEST and STOCK_ADJUSTMENT appear in the Work Type column but are markers the app writes itself, not descriptions of work anybody did. They cannot be added to the list, cannot be given a WBS, and are never refused by the strict dropdown - stock adjustments keep working exactly as before.

16.5.6 Reporting

Consumption, Daily Consumption and the dedicated WBS Report all print the number; rows without one read (no WBS) rather than showing an empty cell.

[!] WBS is applied forward only. Setting up a map today does not restamp entries posted last month. Historical corrections are an Excel sync, not a side effect of a settings change.

17.1 How the system is put together

Four layers sit between a user and your data, and each has a single job.

Layer	What it is	What it does for you
1. The user's device	A phone, tablet or laptop - browser or installed app	Where work gets entered. No company data is stored on the device
2. The security edge	A commercial protection service sitting in front of the system	Checks the person's company email address before they reach the login page, absorbs attacks, and renews the security certificate automatically

Layer	What it is	What it does for you
3. The application server	The server that runs GI Hub	Runs the application, holds the database, sends the notifications
4. Nightly backup	An encrypted copy held off the server	A second copy of everything, taken every night, kept for 14 days

Two properties of this arrangement are worth understanding, because they are what most of the security answers in section 17.5 rest on:

- The connection runs outward, not inward. The server opens a connection to the security edge; the edge never opens one to the server. There is no open door for anyone to find - someone scanning the internet for our system sees nothing at all.
- Your data stays on your own server. The security edge forwards encrypted traffic it cannot read. The backup copy is encrypted too. No outside party holds a readable copy of your stock figures, prices or documents.

17.2 Bringing a new installation live

Standing up a fresh installation is a one-off technical job for whoever administers the server. It is listed here so you know what it involves, what to ask for, and what must be true before the address is shared with staff.

#	Step	Who	Roughly
1	Install the application and its services on the server	Administrator	30 min
2	Connect the server to the security edge and point the company web address at it	Administrator	15 min
3	Set the email allow-list so only company addresses can reach the login page	Administrator	10 min
4	Connect the WhatsApp notification channel	Administrator	10 min
5	Complete the hardening checklist below	Administrator	15 min
6	Take a first backup and confirm it landed	Administrator	5 min
7	Change every default password in the application	You	10 min

Before the address is shared with anyone

These five are not optional. Each one closes a specific, realistic way of losing control of the system.

1. Disk encryption is switched on on the server. A stolen or salvaged disk is then unreadable rather than a complete copy of your business.
2. Every default password has been changed. The accounts the system creates on first start are publicly documented in this manual - see section 2.4.
3. The email allow-list is in place. Until it is, the login page is visible to the whole internet.
4. The server does not sleep or lock. A sleeping server is an outage.
5. A first backup exists and has been checked. An untested backup is not a backup.

17.3 Day-to-day running

In normal operation there is nothing to do. The system runs as a set of background services that restart themselves if they stop, and the backup runs overnight without being asked.

Four routine needs come up, and all four are one request to your administrator:

What you need	What to ask for	Impact on users
Confirm everything is healthy	A service status check	None
Investigate a reported problem	The recent activity logs	None
An immediate extra backup	A one-off backup, before risky work	None
Apply a new version	An application restart after the update	About 3 seconds

Applying an update

Updates are additive and reversible, which is why the downtime is measured in seconds rather than minutes:

1. The administrator applies the new version.
2. The application restarts. Anyone mid-form keeps their work - they may see a brief reconnect.
3. Any new database structure the version needs is applied automatically before the first person can sign in. Nobody runs a database update by hand.

Rolling back. If a release misbehaves, the administrator can return to the previous version in about a minute. Your data is unaffected: the system only ever adds to its database structure and never removes anything, so an older version reads newer data perfectly well. You do not lose the work entered between the upgrade and the rollback.

Take a backup first when

- A major version upgrade is being applied.
- Bulk edits are about to be made through the Admin Portal's data editor.
- The system is moving to different hardware.

17.4 Security and data safety - what to tell management

These are the questions that come up in a management or client review, with answers you can give directly.

Where does our data physically live?

- On our own server. The database and every uploaded attachment - delivery notes, material certificates, photographs - sit on storage we control. No third-party service holds a copy of your inventory.
- The disk is encrypted. A stolen or salvaged disk is unreadable without the key, rather than a complete copy of the business.
- The nightly backup is encrypted too, and is held somewhere the server itself cannot reach - so a problem on the server cannot corrupt the backup.

Who can reach the application?

There are two locks on the door, not one:

1. The security edge checks the email address first. Anyone not on the allow-list never reaches our server, never sees the login page, and leaves no trace on our system at all.
2. The application then requires a username and password. Passwords are stored using a one-way scramble, so even someone holding the database file cannot read them back.

Sensitive accounts can add a second factor, and a session that sits idle signs itself out rather than waiting to be found.

How is the connection secured?

- Every byte travels encrypted, using the same standard as online banking.
- The security edge forwards encrypted traffic and re-encrypts it on the way to our server. The contents of your records are not readable by it.
- The provider holds the recognised independent certifications for this class of service - SOC 2 Type II, ISO 27001 and PCI DSS.

What is the attack surface?

- No open ports. Our server never accepts an incoming connection from the internet; it makes an outgoing one instead. Someone scanning for our system finds nothing to attempt.
- Every action is permission-checked on the server, not just hidden in the interface. A user cannot reach another role's function by guessing an address, and a read-only account is refused every change centrally rather than screen by screen.
- Every consequential action is recorded with the user's name, the time and what changed. Admin can review or export that history at any point.

What happens if something fails?

Failure	Impact	Recovery
Power loss at the server	System offline while it lasts	A battery unit covers short cuts; otherwise it restarts by itself when power returns
Internet outage	Remote users cannot reach it; staff	None needed - it resumes on its own

Failure	Impact	Recovery
	on the local network still can	
Security-edge provider incident	Remote access blocked	An alternative route can be switched on in about 5 minutes
Server hardware failure	Offline until replaced	Restore the last backup onto replacement hardware: about 30 minutes, losing at most one day
Disk corruption	Database unusable	Restore from any of the last 14 nightly copies: about 5 minutes
A record deleted by mistake	One record lost	Restore a copy alongside the live system and re-enter it
A member of staff misuses their access	Limited to what their role can reach	The audit trail names the user, the action and the exact time

The two rows worth acting on before you need them: a battery unit for the server removes the most common cause of an outage, and a second backup destination (section 17.5) removes the only single point of failure left in the recovery plan.

What does it cost to run?

Running costs are dominated by the server itself; everything around it is either free at this scale or a few dollars a year.

Item	Cost
Web address	About \$8 a year
Secure connection to the server	Free, unlimited traffic
Email allow-list, up to 50 users	Free
Backup storage at this data volume	Free
WhatsApp and email notifications	Included
AI assistant	Free - it runs on our own server, so no per-question charge and no data leaves the building
Server and electricity	The main line item, and it depends on the hosting choice

Non-negotiables

Skipping any one of these defeats the security model rather than weakening it.

1. Disk encryption stays on.
2. The server stays in a physically secured location.
3. Default passwords are changed on day one - they are published in this manual, in section 2.4.
4. The email allow-list is in place before the address is shared.
5. The server is not used for anything else. Keep it single-purpose.
6. Operating-system security updates are applied within a week of release.

17.5 Backup and restore

What runs automatically

Every night at 02:00 the system takes a complete copy of itself, without going offline and without anyone starting it:

1. A consistent snapshot of the database - taken safely even while people are using the system, so there is no maintenance window.
2. A copy of every uploaded attachment: delivery notes, material certificates, photographs.
3. Snapshots older than 14 days are removed, so the backup area does not grow without limit.
4. Optionally a second copy to another destination, if one has been configured.

Restoring

Restoring is a technical procedure your administrator performs. What matters to you is the shape of it:

Question	Answer
How far back can we go?	Any night in the last 14 days
How long does a restore take?	About 5 minutes for the database; a little longer if attachments are also being restored
Is the system usable during a restore?	No - it is stopped for those few minutes, because the database must not be open while it is replaced
How much work would we lose?	Everything entered since the snapshot being restored. Restoring last night's copy loses today
Can a single deleted record be restored on its own?	Not directly. The practical route is to restore a copy alongside the live system, read the record out of it, and re-enter it

Where the copies live

One nightly copy goes to encrypted cloud storage. That is one copy in one company's hands, which is better than none but is not two independent failure domains.

If the data matters enough to survive a bad week - and inventory, procurement and valuation records normally do - ask your administrator to add a second destination in a different place. Any of these work, and all of them are cheap:

Option	Cost	Set-up
A second cloud storage provider	Free at this data volume	About 15 minutes, one-off
An external drive kept on site	Cost of the drive	Minutes, one-off
A drive held off site, rotated	Cost of two drives	Minutes, plus the discipline to rotate them

The recommended arrangement is the nightly cloud copy plus one other provider: three independent things would then have to fail on the same night before data is genuinely gone.

17.6 Controlling who can reach the system

Access is controlled in two independent places, and it is worth being clear about which does what:

Control	What it decides	Who changes it
The email allow-list, at the security edge	Whether a person reaches the login page at all	Administrator, in the security dashboard
The user account, inside the application	What that person can see and do once signed in	You, in Admin Portal -> Users

Both must be right. An allow-listed email with no user account cannot sign in; a user account whose email is not allow-listed never sees the login page.

Adding or removing someone

When a new person joins, two things must happen: their company email is added to the allow-list, and you create their account with the correct role and site. Neither works without the other.

When someone leaves, do both, in this order:

1. Deactivate their account in Admin Portal -> Users. This takes effect immediately and is the one that matters - it ends any session they have open.
2. Ask for their email to be removed from the allow-list. This takes effect globally within about a minute.

Changing the allow-list needs no restart and interrupts nobody.

A note on rate limiting

Repeated failed sign-in attempts are already slowed down by the application itself. Ask your administrator to add a limit at the security edge as well - it stops that traffic before it ever reaches your server, which is both faster and cheaper than handling it inside the application.

17.7 When something is not working

Find the symptom, try the first column, and escalate if it persists. None of these need you to type anything.

What you are seeing	Most likely cause	What to do
The web address shows an error page instead of the login screen	The application or its connection to the security edge has stopped	Ask your administrator for a service status check. This is almost always a restart
A valid company email is refused at the login page	That address is not on the allow-list	Ask for it to be added - takes about a minute to take effect
Someone signs in but sees "your account has no site"	The account was created without a site or warehouse assigned	Admin Portal -> Users -> set their site (or warehouse, for warehouse staff)
The application works but WhatsApp	The notification channel has become	Ask your administrator to re-connect

What you are seeing	Most likely cause	What to do
messages are not arriving	disconnected	it. Nothing is lost - messages queue and send once it is back
Notifications arrive late in a batch rather than immediately	The recipient has the evening-summary preference switched on	That is intentional. Change it in their profile if they want them immediately
The AI assistant says it is offline	The assistant service is not running	Ask your administrator to start it. Every other part of the system is unaffected
Times on records look wrong by a fixed number of hours	The server's time zone is not set correctly	Ask your administrator to correct it and restart
Backups have stopped appearing	The backup destination is full or disconnected	Ask for the backup log to be checked. Treat this as urgent - you are running without a safety net

Escalating to the developer

Three things make a report actionable, and without them the first reply will only ask for them:

1. What you expected to happen, and what happened instead.
2. The exact time, and which site and user account it happened on.
3. A screenshot, including any error text in full.

Your administrator can attach the relevant service logs. Please do not delete or re-enter the affected record before it has been looked at - the audit trail is usually what identifies the cause.

17.8 Technical runbooks

The procedures behind this chapter are documented for whoever administers the server. They are deliberately kept out of this manual, which is written for people who use the system rather than run it.

Runbook	Covers
Deployment guide	Server setup, the security edge, going live
Debugging guide	Diagnosing faults, reading the logs
Native apps guide	Building and distributing the Windows, macOS and Android apps
Architecture reference	How the system is built, and why
Standard operating procedure	Day-to-day operating rules for every role

18. Material Estimator (SME) Manual

> Access: Material Estimator page - exact-locked to HOD + Admin. HOD is scoped to their own site; Admin gets a sidebar site picker. SK / Supervisor / Logistics / Warehouse never see this page.

18.1 What it is

The Smart Material Estimator (SME) is a planning portal for Rubber-Lining / Brick-Lining work. It answers: "For the equipment and lining systems at this site, how much material do we need, what's already available, and what must we still procure?" It is read-only - planning here never writes a consumption, a receipt or a stock figure.

> The estimator is a separate pool from the warehouse, deliberately. > Every quantity the estimator uses comes from its own material baseline and from nowhere else. A warehouse issue does not reduce what the estimator says is available, and it is not supposed to: the plan answers "what does this project need and what has been bought for it", while the warehouse answers "what is on the shelf today". Netting one off the other would silently move every readiness figure on every tab. Where physical draw matters, it is shown beside the plan as an Actual Physical Balance - adjacent, never merged. See section 21.

It works from three sets of master data, all maintained on the Master Data tab:

Master data	Scope	What it holds
Equipment	Per site	Each tag, its location, its surface area and which lining system applies
Recipe	Shared	How much of each material one square metre of each lining system needs
Materials	Shared	The estimator's own quantity baseline - what has arrived and what is still on order

Locations and equipment types come from the system settings, so the drop-downs stay consistent across every tab.

18.2 Tabs (8)

Tab	Purpose
Dashboard	KPI strip + per-location material balance + a stock-only view of SME-tracked items + a procurement view (what to buy, per code/location).
Priority	Drag-to-rank equipment so the allocation engine fills the highest-priority tags first when stock is short.

Tab	Purpose
Session / Selective Order	Pick specific equipment, enter the SQM to execute, and get the exact material list + a combined procurement total with a GRAND TOTAL row.
Location Report	Material requirement broken down by location, with per-location colour schemes; plus an "All Equipment" view.
Equipment Report	Per-equipment three-section breakdown (requirement / available / shortfall), exportable as a multi-sheet workbook.
Execution Plan	Critical-items card + procurement-priority + production-detail blocks for executing a work package.
Total Overview	Whole-site rollup: 6-card KPI strip + master table + per-system-code drill-downs.
Master Data	Add, edit and remove Equipment, Recipes, the SME Materials baseline, and Locations/Types. Changes are written only to the estimator's own records - never to the warehouse inventory ledger.

18.3 How SME consumption reaches the ledger

The SME does not issue stock itself. Day-to-day consumption is entered on the Store Keeper -> SK Consumption page, in the SME Multi-Material Entry grid: pick an equipment tag -> system codes -> SQM per system -> auto-computed materials -> submit a batch. That flows through the normal end-of-day commit route - HOD approval, then the EOD commit - and the SME's completed-SQM figure advances at that moment. So the estimator's "available" figures stay in step with real ledger movements.

18.4 Key rules (for support)

- SME inventory is isolated from ERP inventory. The SME baseline lives in its own seed table; quantities shown are derived, not stored. Do not expect SME edits to change the Live Dashboard stock and vice-versa.
- Master-data grids order by an explicit key (not row order); if a grid says "No records found" but data exists, it's a known historical bug class - refresh / report it.
- The initial SME baseline is loaded per site by an Admin as a one-off setup step.

19. Man-Hours & Manpower Tracking Manual

> Access: Man-Hours page - exact-locked to HOD + Admin. HOD is scoped to their own site; Admin gets a sidebar site picker. (NEW - 2026-06.)

19.1 What it is

Man-Hours tracks labor the way the Material Estimator tracks material: log who worked, where, and for how long; record SQM produced; define the required man-hours for a scope; then compare estimated vs actual to surface over-runs. It is fully site-scoped and isolated - it reads the Material Estimator's Equipment / System-Code / Location lists for its dropdowns without ever changing them, and writes only to its own man-hours records. Nothing here touches inventory, the end-of-day path, or the SME data.

19.2 Tabs - all thirteen

The page has thirteen tabs. Only about six fit across a laptop screen; the rest are reachable through the "... " button at the end of the tab bar, or by adding ?tab= to the address (/manhours?tab=planner, ?tab=session).

#	Tab	What it is for	Added
1	Employees	the per-site labour roster	original
2	Daily Timesheet	who worked, where, for how long	original
3	Estimator	the required man-hours for a scope	original
4	Estimate vs Actual	the variance dashboard	original
5	Scorecard	material and labour side by side, per equipment	2026-07
6	Employee-wise	one person's timeline, date by date	original
7	[OK] HOD Approval Queue	execution entries waiting on the HOD	Phase 6
8	Actual vs Benchmark	reported work against the workbook norms	Phase 6
9	Reason Audit Log	why an over-consumption happened, and who said so	Phase 6
10	Surface Prep Progress	blasting coverage per	Phase 6

#	Tab	What it is for	Added
		equipment	
11	Manpower Planner	how many people to finish by a deadline	Phase 7
12	SME Session	the SME session costed in labour (\$19.5)	Phase 8

19.2.1 Employees

Per-site labour roster. Add or update a worker with Employee Code, Name, Designation, Worker Type, Shift and Company.

Worker Type is GI or NON_GI - your own staff against supplied (subcontractor) labour, e.g. DMC. Supplied workers live here even though they are not ERP app-users. *(The workbook's own words are "OWN" and "Supply"; the import maps both spellings onto GI / NON_GI.)*

Shift is Day or Night. It is what the Manpower Planner reads to decide whether a two-shift plan is a fact or a proposal.

The tab also carries the overtime thresholds, which the HOD owns: overtime begins after 8 net worked hours for a GI worker and 10 for a non-GI worker. They are settings, not constants - changing one applies to new timesheets only and never re-splits hours already posted.

[!] Designation is what maps a worker to a role. A blank or unrecognised Designation means the worker is reported as unmatched and is not counted as available in the planner. That is deliberate: "nobody wrote down that they are masons" and "there are no masons" call for completely different actions.

19.2.2 Daily Timesheet

Two ways to enter actuals, plus team-SQM:

- Excel upload - upload an attendance workbook in the standard site attendance format (the one with an "ADD EMPLOYEE" sheet and a "SAR" sheet). The system shows you what it read before saving anything, then on import you choose:
- Replace rows for these dates (recommended) - deletes this site's existing timesheets for the dates in the file, then inserts (safe to re-upload a corrected file - no duplicates), or
- Append - adds the rows as-is. Unknown workers are auto-created into the roster. Location / Equipment / System import blank (the source file doesn't carry them) and are assigned afterward here.
- Manual per-day batch grid - pick a Work Date, Equipment Tag, System Code (drop-downs from the Material Estimator's equipment), and a break (default 60 min). The grid lists active workers; tick Worked and set In/Out (default 07:30-16:30). On save, one timesheet row is written per worker.
- Team SQM - record the SQM the crew completed on that date/tag/system and choose even (equal per worker) or by-hours (pro-rata on hours) distribution; each worker's share is recorded as their allocated SQM for the day.

Hours math: Total = time out minus time in, less the break. Normal hours are the Total capped at the worker's own overtime threshold - 8 for GI, 10 for non-GI, both set on the Employees tab - and overtime

is the remainder, never less than zero. A full 12-hour shift is 11 worked hours plus an hour for lunch, so it splits 8 + 3 for a GI worker and 10 + 1 for a non-GI one. The source file's own hour columns are ignored - hours are always recomputed.

19.2.3 Estimator

Define the required man-hours for a scope: pick Equipment Tag + System Code, enter Estimated man-hours (and optionally Estimated SQM -> yields a man-hours-per-SQM norm) + a basis note. One estimate per Tag/System (re-saving updates it). Auto-draft proposes estimates for every scope that has none, using a norm you supply; you edit the drafts before saving.

19.2.4 Estimate vs Actual

The variance dashboard. KPI strip (scopes tracked / over-consuming / total actual man-hours) + a table where over-runs >10 % are highlighted red and on-or-under-budget rows green. Variance is Actual minus Estimated, and Variance % is that variance divided by the estimate. A "Where the most man-hours went" table ranks the top consumers. An expander lets you record a reason for an over-consumption (saved per Tag/System and shown back in the table).

19.2.5 Scorecard

Material and labour for the same equipment, side by side. For each Tag/System: SQM done, material consumed against the SME expectation, man-hours spent against the estimate, and a reconciliation note when the two disagree about how much work has been completed. It is the tab to open when the Estimator says one thing and the timesheets say another.

19.2.6 Employee-wise

The "where did each person work, date by date" view. Pick an employee (or All) and a date range; get a clean, date-ordered list of every tag/system they worked with hours and allocated SQM - the un-cluttered per-person timeline.

19.2.7 [OK] HOD Approval Queue

Execution entries posted by Store Keepers and Supervisors, waiting on the HOD. Each row shows the area claimed, the material it implies and the man-hours booked against it. Approve, or reject with a reason - the reason is compulsory and lands in the Reason Audit Log (19.2.9).

19.2.8 Actual vs Benchmark

What was actually reported against what the workbook's norms say it should have taken - material and man-hours together, per entry and in total. Totals sum the absolute figures and derive one percentage from the sums; they are not the average of the per-entry percentages, which would weight a 2 m² entry the same as a 2,000 m² one.

19.2.9 Reason Audit Log

Every over-consumption reason and every rejection reason, with who wrote it and when. Append-only: a reason is a record of what somebody believed at the time, so it is never edited in place.

19.2.10 Surface Prep Progress

Blasting coverage per equipment. Coverage is prep area divided by the equipment's own area, and it can exceed 100 % - a surface can be re-blasted, and clamping the figure would hide rework instead of showing it.

19.2.11 Manpower Planner

Answers one question: to finish this equipment by the deadline, how many of each role do I need, how many do I have, and what should I hire? Pick equipment (multi-select), optionally system codes, optionally include surface prep, then give a deadline as Target days or as hours per person - they are the same quantity, hours = days x 11, and giving both is refused.

Three sections: workload (area remaining -> man-hours), gap (per role: need, have, assign - collapsible, with the jobs that asked for each role), and strategy (where overtime comes from and how to remove it).

[!] Nights buy time, not a smaller payroll. Nobody works both a day and a night shift, so a two-shift plan needs the same total headcount - the natural reading, "two shifts, so half the people", under-hires by half. What running nights buys is calendar time: the site delivers (day crew + night crew) x 11 man-hours per day instead of day crew x 11, so the page shows days with the day crew alone, days with both, and the saving between them. Shifts default to 2 when anyone in a required role is on nights; the HOD can force two anyway.

[!] The two shifts are not the same size, and the split is read from your roster. This site runs a day shift of about 20 against a night shift of about 80 - different equipment, different tasks. The plan divides each role in the proportion your roster actually runs, never by dividing by the number of shifts: on those numbers an even split understates the night crew fourfold and overstates the day crew by the same. Where the roster cannot say, the page says so rather than blending a guess into a number:

Basis shown	What it means
(nothing)	the split came from this role's own day/night counts
site ratio	this role has nobody rostered, so the site's proportion was used
assumed	two shifts were forced with no night crew anywhere - this is a guess

A role with day workers and no night workers is not a valid basis: it means there is no night crew yet, and reading it as a proportion would put 100% of a forced two-shift plan on days and make the option do nothing at all.

[!] Idle roles do not add capacity. Hiring fifty blasters does not change a masonry plan. Normal capacity, the overtime and the hire-to-clear advice all count only the roles the job actually needs - the roster panel still shows the whole payroll beside them, and the gap between "we have 150 people" and "capacity is 100" is the point.

[!] It changes nothing. The planner is advice. It never assigns anybody, never edits the roster, and never writes to a timesheet.

[!] A stacked surface is blasted once. Two lining systems on the same physical location share one

surface, and prep is charged for it once - the plan shows the deduplicated area beside the gross so a figure printed before this rule can still be reconciled.

19.2.12 SME Session

The SME planning session costed in labour - see §19.5.

19.2.13 Efficiency by Day

How much manpower a job actually took, day by day - see §19.6. This is where you compare one piece of equipment against another.

19.3 Loading an attendance file in bulk

Attendance is normally uploaded through the Man-Hours page itself, which is the recommended route: it shows you what it parsed before anything is saved.

For a very large historical backfill, your administrator can load the same file directly on the server. It uses the identical parsing and import logic as the page, so the result is the same either way, and it offers two options worth knowing about:

Option	What it does	When to ask for it
Preview only	Parses the file and reports what it found, saving nothing	Always, the first time a new file format appears
Replace by date	Existing lines for the dates in the file are replaced	Re-loading a corrected file for a period
Add without replacing	Lines are added alongside what is already there	Loading a period that was never loaded before

Replace by date is the default, because loading a corrected file twice is a far more common mistake than deliberately wanting two copies of the same day.

19.4 Key rules (for support)

- Its own data, kept apart. Man-Hours never changes inventory, SME data, or the end-of-day ledger.
- Site-scoped. HOD sees only their site; Admin switches sites with the sidebar picker. Every timesheet and estimate is stamped with the site it belongs to.
- One timesheet line per (employee, date, equipment-tag, system-code). A worker split across two tags in a day has two rows.
- Equipment-Tag / System-Code / Location drop-downs come from the Material Estimator's equipment master - keep that seeded for a site or the lists will be empty.

19.5 SME Session Plan - labour for what the material allows

Added 2026-08-24 (Phase 8).

The Material Estimator knows what the stock on site can build. This page knows what a piece of work costs in labour. This tab joins them, and answers the question a morning meeting actually asks:

Column	Means	Hire against it?
We can do now	the part the material physically on site supports	Yes - this is the actionable number
Overall total	the whole remaining job, materials no object	for context
Blocked by material	the difference: the size of the delay	No - see below

How to get here

Material Estimator -> Session Builder -> add equipment and set the priority order -> Session Report For MP&H. The button is disabled while the session is empty, because costing nothing is not a report.

The session travels in the address bar, so the resulting link can be pasted to somebody else and shows them the same report. Nothing you do here changes the planning session you left open in the Estimator.

[!] The Blocked column deliberately shows no headcount

You cannot deploy people against material that has not arrived. A headcount printed beside blocked work is a headcount somebody hires against, and those people are idle when the delivery lands. The column shows the man-hours and crew-shifts - how big the delay is - and the materials responsible. The per-role To assign figure is measured the same way: against what can start, never against the overall.

Reading it

- Priority order matters. The cascade spends stock top-down, so re-ordering the session changes which jobs can start. The overall total never moves - it is the same work either way.
- "What is in the way" lists every short material, how much survives the open purchase orders, how many jobs it holds up, and how many of those it is the bottleneck for. Buy the bottlenecks first.
- No material is ever labelled with a number of man-hours. Several materials can be short on one job while only the scarcest limits it, so such a figure would add up to more than the delay.
- A system marked "no recipe - unmodelled" is not blocked by stock; nobody has written down what it consumes. Chasing procurement for it will not help.
- Surface prep is not in this report. Blasting uses no recipe line, so the material model has no opinion on it - use the Manpower Planner.

The material picture is up to a minute old

Working out what the stock allows is the heaviest calculation in the system, and none of it changes when you change the target days - so it is reused for about a minute and the page tells you (cached 12s ago). Changing Target days is therefore instant. Press the ? button beside Cost it to re-read stock after a receipt has been posted. The roster is never reused: hire a night worker and the very next answer plans two shifts.

Exports

Excel gives four sheets - Summary, Per job, Per role, Blocking materials (plus "Read this first" when there are warnings). CSV carries the Summary only. PDF prints all of it.

20. Auditor (View-Only) Manual

Added 2026-08-03.

19.6 Efficiency by Day

Where: Man-Hours -> Efficiency by Day.

Answers the question the rest of the portal cannot: for the same lining system, how much manpower went into Equipment A against Equipment B?

19.6.1 Why man-hours per m², and not hours

A 400 m² tank will always show more hours than a 40 m² vessel. That says nothing about how well either was done. Man-hours per square metre is the only figure on this page that survives a change of size, which is what makes two jobs comparable at all.

The cards at the top give you that number per job, best first. The chart is there to show you how it got there.

19.6.2 [!] The line is the RUNNING figure, not the day's

The bars are the hours booked each day. The line is cumulative - all the hours so far divided by all the area so far.

That matters, and it is the easiest thing on the page to misread. A day-by-day figure swings far too hard to follow, and on a day with no area it does not exist at all. The running figure settles as a job goes on, and it is the number you would quote in a meeting.

It also means setup is included. A tank that reads 6.6 MH/m² after a fortnight of scaffolding is telling you what the job really cost. Its good days might read 2.2 on their own - that is the same tank, with the setup left out.

19.6.3 [!] Days with hours and no area

Mobilisation, scaffolding, curing and inspection all book hours against zero square metres. On those days:

- the line breaks. There is no area to divide by, so there is no figure. A zero would read as "this crew achieved nothing per metre", which is a claim about them rather than about the day.
- the hours still count towards the running figure. They are part of what the job cost.
- the day is listed underneath, with whatever the timekeeper wrote in the Remarks box on the timesheet.

[!] Where nothing was written, the table says "no reason recorded". The app does not guess at "scaffolding" - only the person who was there can say, and a guess in that column would become the record.

19.6.4 Comparing across lining systems

If your selection covers more than one lining system, the page warns you. A tile lining and a coat are different work with different benchmarks, so their man-hours per m² are not comparable - filter to one system to compare equipment against equipment.

It still draws the chart. You may have asked for exactly that view, and refusing to show it would teach nothing.

19.6.5 If the chart is empty

It fills in from two places: the Daily Timesheet (hours) and the team SQM recorded against a day's work. A site that has entered neither will see a sentence saying so, not a broken chart.

20.1 What the Auditor role is for

The Auditor is for anyone who must see the numbers without being able to move them - an internal auditor, a client-side QA reviewer, a finance reviewer, a new manager still learning the system, or a consultant given temporary access.

It is a real role like any other, with one difference that is absolute:

> An Auditor can read everything its level reaches and can change nothing, > anywhere, ever.

That is not a UI convention. It is enforced on the server for every request.

20.2 What an Auditor can see

Area	What it shows
Dashboard	Every KPI, chart and category breakdown, across all sites
Stock	Current stock by site, minimum levels, expiry
Records	Inventory, Receipts, Consumption, Returns, Lots, Purchase Orders, Purchase Requests
Reports	Every report, downloadable as Excel, PDF or CSV
Lining Coverage	The read-only coverage analysis
Documents	The SOP and User Manual PDFs, QR/label sheets
Account	Own security settings (password-manager 2FA, phone number)

An Auditor is not site-locked. Every other role below Logistics sees only its own site; an auditor sees all of them, because an auditor confined to one site cannot audit the business.

20.3 What an Auditor cannot do

Everything that changes data. The Data Entry forms, Master Data editors, Bulk Excel Import, HOD

Approvals, the Logistics and Warehouse portals, the Supervisor portal and the Admin console do not appear in the sidebar at all - they are not hidden buttons on a visible page, the pages are simply not reachable.

On pages an Auditor can open, actions that would change something are disabled rather than hidden, with a tooltip explaining why:

> Your account is view-only (Auditor) - this action changes data.

On the Reports page, for example, Excel / PDF / CSV downloads work normally (they only read), while Archive, WhatsApp, New schedule, Run now and Delete are greyed out.

A "View only" tag sits permanently in the header beside the user name, so there is never any doubt about which kind of account is signed in.

20.4 What an Auditor CAN still do

Because they are read-only, not crippled:

- Download every report in all three formats.
- Use the CmdK / Ctrl-K command palette to jump to any page or material.
- Ask the Hub Assistant questions (it answers from the Auditor's own manual chapters - see §20.6).
- Manage their own account: enable or disable two-factor authentication and change their own phone number. These touch only their own user record.

20.5 The security model (for Admins)

Worth understanding before you hand the role out.

The restriction is enforced once, centrally, rather than screen by screen. Any request that would change data from a view-only account is refused before it reaches the feature at all, and the user is told plainly that their account is view-only and the action is not permitted.

Why that design matters to you: a permission check added screen by screen is only as reliable as the person who remembers to add it, and the one that gets forgotten fails open - the change succeeds and nothing looks wrong. Checking centrally inverts that, so a feature added next year is closed to Auditors from the moment it is written. 126 of the system's 143 data-changing actions are blocked; the 17 that are not are sign-in, sign-out, staying signed in, the user's own two-factor and phone-change steps, and a handful of read-only calculations that are technically submissions only because the figures are too large to fit in a web address.

Two consequences to be aware of:

- An Auditor probing an address that does not exist is told "not permitted" rather than "not found". The check runs first, so it cannot be used to map the system.
- Blocking is by role, not by page. Even if an Auditor somehow reached a write page, every button on it would still fail server-side.

20.6 Creating an Auditor account

Admin Portal -> Users -> Add user, and pick Auditor (view-only) in the Role dropdown. Leave the Site blank - Auditors are global by design, and the form does not ask for one.

Users can also request the role themselves on the registration screen; it still needs Admin approval like any other request.

20.7 The Hub Assistant for Auditors

The assistant answers from chapters 1, 2, 3, 8, 9, 10, 11, 12, 16 and 20 - orientation, the reports module, notifications, the data model, the glossary, the FAQ and the cross-role procurement walk-through. That is deliberately the set matching the pages an Auditor can open.

Ask it about a report, a status code or what a field means and it will answer. Ask it how to post a consumption entry and it will tell you that is outside the read-only Auditor view - because the operational chapters for roles that can post entries are never loaded into an Auditor's context in the first place.

21. 2026-08 Feature Update - What Changed

Everything in this chapter is live. Grouped by what you would notice.

21.1 Reports and exports look the same everywhere now

Every Excel export - not just the Material Estimator's - now carries the branded layout: the GI logo, a "Report Generated / Generated By" band, a title bar and a framed, filterable table with the header frozen. Previously only three SME workbooks looked like that and everything else came out as a bare grid.

> If you have automation reading these files: the header row is now row 6 > and data begins on row 7 (rows 1-4 are the logo and meta band, row 5 the > title bar). Anything that assumed a header on row 1 needs adjusting.

PDF reports no longer overlap or cut off text. Columns are now measured against their actual content and share the page fairly, and long values wrap onto more lines instead of being truncated. Previously every column got an equal slice of the page regardless of need, so a long material description was drawn on top of the next column while Date and UOM wasted most of theirs - and anything past 24 characters was silently deleted.

QR badge and bin labels were fixed the same way: a long employee name used to run past the edge of its sticker onto the one beside it.

21.2 The Material Estimator understands your order data correctly

Ordered_Qty is the TOTAL quantity procured for the project, and Available_Qty is the part of it that has already arrived - available is a subset of ordered, not a separate pile beside it.

The estimator used to add the two together, which double-counted every unit already on the shelf. On real data that understated the buy list by 22,951 units across 22 of 30 materials, and on GI-8005763 - where all 143,000 ordered units had arrived - it read 286,000 and reported nothing to buy against a demand of 152,685, hiding a 9,685-unit shortage completely.

What you will see in the UI:

- Second-tier quantities now read "Pending Delivery" - the part of the order still on its way - never just "Ordered".
- Second-tier coverage reads "When delivered".
- Reports carry a Total Procured column beside them.

Feasibility is unchanged and still judges physical stock only: a tank cannot be built with a purchase order.

21.3 A view-only Auditor role

See §20.

21.4 You are signed out after 30 minutes of inactivity

Leave the app untouched for 30 minutes and it signs you out and returns you to the login screen. A warning appears two minutes before with a button to stay signed in, and any real activity - a click, a keystroke, a scroll - resets the clock silently.

This matters most on shared site terminals, where an unattended session was previously good until the browser was closed.

21.5 Search anything with CmdK

Press CmdK (Mac) or Ctrl-K (Windows) anywhere in the app.

The palette already jumped to pages; it now also searches your inventory live. Type a SAP code, part of a material name or a material code and the matching items appear under a Materials heading - pick one and you land on its Material Intelligence card. Results are scoped to what your role may see.

21.6 The Hub Assistant is faster and more accurate

The assistant used to be handed its entire allowed portion of this manual on every single question. For an Admin that was the whole ~180 KB document, which is slow to process; for everyone else it was the first 800 characters of each chapter, which is the wrong 800 whenever the answer is further down.

It now finds the passages that actually match your question and reads only those - a 97.7 % smaller prompt for Admins, and noticeably better answers for everyone, because the relevant paragraph is now present in full instead of being cut off.

A parsing bug was fixed at the same time: numbered comment lines inside the worked examples in the Operations chapter were being mistaken for chapter headings, which silently replaced chapters 1-4 - Introduction, Roles & Permissions, Login, and the Store Keeper Manual - with fragments of a restore procedure. Ask "how do I log in?" today and you get §3.1, the login screen.

The role filter is unchanged and still applied before retrieval: a Store Keeper's context physically cannot contain an Admin chapter.

21.7 Serialised tool and equipment tracking

Two hammers share one item code, so scanning either label used to resolve to the same inventory line - the system could not tell you which one was in your hand or where the other had gone. Assets solves that.

Each physical item gets its own record, identified by its serial number. A scan now resolves three ways, and the order is deliberate:

What the label carries	What you get
A serial number or asset tag	That exact item, immediately - there is nothing to choose

What the label carries	What you get
Only the item code, and several units exist	A "which one is in your hand?" list to pick from
Only the item code, and no units are registered	The ordinary material view - the system does not invent an asset

Recording where something is. Open the item, tap Move, and either pick a rack or describe the place in words - not everything lives on a shelf, and "loaded on truck 4771" is a real answer. If the device can supply coordinates, they are captured alongside the update, never in front of it: a declined permission or a yard with no signal still records the move, with the coordinates simply absent. A recorded position becomes a map link on the item.

Status covers both what condition a tool is in - working, not in use, under repair - and where it sits in the stores workflow. The full history of every move is kept and is never overwritten, so "where has this been" is a question with an answer.

> Coordinate capture needs a secure connection. Over plain HTTP the browser > refuses to supply a position - the move still saves, without coordinates. If > your team is on the installed app or the company web address, this is already > the case.

> Your spreadsheet seeds this; the app owns it. A location typed in the > Consumption Log creates the asset on the next sync. From then on, what somebody > recorded in the app wins: a re-sync will not move a tool, revert its status, or > erase a captured position.

21.8 The warehouse rack locator

"Which rack is this in?" now has an answer. Search by material name, by code, or by scanning a label, and the system tells you the shelf to walk to - ordered so that the first result is where to go first, even when a material is stocked in three places.

- A material you stock but nobody has located comes back marked not located, rather than being left out. Silence would read as "we do not stock it", which sends somebody out to buy what is already on a shelf.
- Scanning the rack itself answers the reverse question - everything that is meant to be on that shelf. That turns a stock count from a hunt into a checklist.
- The locator is built into the CmdK search, so it is two keystrokes from anywhere in the application.

21.9 The Material Estimator handles messy tank references

The Consumption Log records a tank reference typed by hand, and the same vessel appears written four different ways. The system now recognises those as one vessel automatically - that alone accounted for 58 of 103 rows.

Where a reference is genuinely ambiguous, it refuses to guess. TNK-091 exists on both TRAIN J and TRAIN K; either answer would look completely plausible in every report afterwards, so those rows wait in a queue for someone who knows. Resolve the reference once on the Actual Consumption tab and every row waiting on it is tagged.

21.10 For administrators - local tooling

Two maintenance scripts ship with the system, documented in the administrator runbook: one puts the database and the public connection to sleep and wakes them again for a laptop that is not always on, and one takes and restores database snapshots, with an option to run nightly at 02:00.

> The previous nightly backup had been failing silently for 25 consecutive > nights - it pointed at a script that a repository reorganisation had moved. > Anyone relying on it had no local backups at all. The replacement is verified > by actually restoring into a scratch database on every change, which is the > only check that means anything.

21.11 Downloaded files now protect you from your own spreadsheet

If you type something that begins with an equals sign, a plus, a minus or an at-sign into a text field - a remark, a note, a reason - you may notice a small apostrophe in front of it when you open the downloaded Excel or CSV file.

That is deliberate, and nothing has been lost.

Spreadsheet programs treat a value beginning with one of those symbols as a calculation to run, not as words to show. That is fine when you write the formula yourself. It is not fine when the text arrived from somewhere else: a note typed at one site, exported by a manager and opened on their computer, would be run by their spreadsheet without anyone choosing to. The apostrophe tells the spreadsheet to show the text exactly as it was written instead.

What this means in practice:

- Numbers are never touched. Quantities, prices, areas and every total in every report stay real numbers, and every total still adds up.
- A plain number that happens to be negative, such as -5, is left exactly as it is - it is a number, not a calculation.
- Ordinary text - material names, supplier names, ordinary notes - is untouched.
- If you want the plain text back in your own copy, delete the apostrophe in that cell.

You are most likely to see this on remarks and reason fields, because those are the places people type freely. It applies to every download the system produces: reports, master-data exports and the Material Estimator workbooks alike.

21.12 Phase 9 - what changed for you

Five things changed at once, and one of them changes how you work every day.

21.12.1 [!] Consumption is filed from paper now

This replaces the store-keeper-first flow entirely. Full detail in §4.10.

You fill a printed form -> Store Keeper verifies -> HOD approves

- Print a form per lining system (§4.9). It lists every material for that system, so nobody writes a material name by hand - that is the one thing the reader is genuinely bad at.
- Fill in the date, equipment and area at the top, and a quantity and a lot number on every row.
- Photograph the whole page including the QR code and upload it.
- Check every figure. Where the handwriting was not certain the box is left empty rather than guessed, and those rows are marked.
- The store keeper verifies your quantities next, not the HOD. Their corrections show to the HOD in red, and you are told before it is approved.

[!] Approval now deducts the material as well as posting the area. Store keepers: stop raising a separate material issue for lining work - the execution entry is the only writer now, and doing both deducts the same drum twice.

[!] Rejection is final. Raise a new entry from a fresh form.

[!] Each printed sheet is used once, and a sheet printed before its system's materials changed will be refused. Both rules exist because your handwriting is matched to materials by row position.

21.12.2 WBS numbers, at last

The WBS # column was blank on every entry in the system - not because the feature was missing, but because there was no screen to reach it. There is one now: HOD -> WBS & Work Types (§16.5).

- Add your WBS numbers, then map each work type to one.
- An issue that leaves the WBS box blank gets the number its work type maps to.
- Work types are now a fixed list you curate, which is what stops civil and Civil becoming two cost centres.
- Nothing is enforced until you add the first row. Turning it on is your act.
- WBS is applied forward only; historical entries are not restamped.

21.12.3 A new chart: Efficiency by Day

Man-Hours -> Efficiency by Day (§19.6). For one lining system, how much manpower went into Equipment A against Equipment B - normalised on man-hours per m², which is the only figure that survives a change of size.

[!] The line is the running figure, not the day's, and it breaks on a day with hours and no area. Both are deliberate; §19.6 explains why.

21.12.4 The Manpower Planner tells you what nights actually buy

Three changes to the arithmetic:

- Nights buy calendar time, not a smaller payroll. The total headcount is unchanged - nobody works both shifts - but the plan now shows how many days the night crew saves.
- The day/night split follows your roster, not a division by two. This site runs about 20 on days against 80 on nights; halving the requirement understated the night crew fourfold.
- Hiring idle trades no longer flatters the plan: capacity counts only the roles the job needs.

21.12.5 "Labor" is now "Manpower"

The page, the menu entry and every heading now say Manpower - the word everyone here actually uses.

> If you have automation reading the API: the field names did not > change. Done_SQM_Labor and Labor_Variance_Pct are still spelled exactly > as they were. Only what you see on screen moved, so nothing you have built > against those keys needs touching.

21.13 Photo reading - the two failures that are fixed

If you tried to read a photographed Daily Consumption log or the new printed consumption form and it did not work, this is what was happening and what has changed. Nothing about how you use it is different - you still photograph the page and wait.

21.13.1 "The log came back empty, or said it could not be read"

The reader was being cut off part-way through a long sheet. It read your page correctly, got as far as about row 13 or 14 of a 30-row log, and ran out of room - and because the answer was incomplete, the system threw away all of it, including the thirteen rows it had read perfectly.

This is also why Delivery Notes always worked: a note is four or five items and always finished comfortably. It was never that the reader could not handle your handwriting or the free-form table.

Now: it is given enough room for a full 30-row sheet, and if a page is ever still too long, the rows it did read are kept and only the unfinished one is dropped. On the operator's own test photo this went from nothing at all to all 30 rows.

21.13.2 "The new PDF just hangs, then fails"

A full page of handwriting genuinely takes the reader several minutes. The system was giving up after four - often within seconds of it finishing - and then telling you "the vision model is not reachable", which sent people off to check a service that was running perfectly.

Now: it waits long enough for a real page, and if something genuinely is wrong it says which of the two it is: still working, or actually unreachable.

21.13.3 Two smaller things you may notice

- A quantity the reader could not make out now arrives BLANK, not as 0. It used to show a confident zero, which reads like a number somebody wrote and nobody re-checks. A blank box is a question, and the row will not submit until you answer it.
- A photo that produced nothing now says so. Previously a failed read could finish looking merely "empty", which is indistinguishable from a photo of a blank form. If the page could not be read, you are told to retake it.

21.13.4 If a photo still will not read

Nothing has changed about the advice, but it is worth repeating: get the whole page in frame, keep the lighting even, and hold the camera square to the paper. The Paste tab works with the AI switched off

entirely and is always available. Typing the entry in by hand is never wrong.

21.13.5 [OK] Ditto marks now come through (2026-09-02)

Where: Entry -> OCR Import (/entry/ocr), the Consumption log lane.

If you have been retyping the Name, Tank No. and Product Name columns after every photo read, that was a real fault and it is fixed.

Most people write a ditto mark - " or ? - instead of repeating themselves down a column, and on a typical sheet that is most of the page. The reader was seeing those marks perfectly well and quietly returning nothing for them, because to a reader "same as above" is not a word to transcribe. The step that was supposed to copy the value down was looking for the mark, found an empty cell instead, and left it empty. On the sheet this was diagnosed against, that was 19 of 30 tank numbers, 14 of 30 names and 8 of 30 product names.

Now: a dittoed cell is filled in from the row above, whether the reader returns the mark or returns nothing.

Two things worth knowing about how it behaves:

- A cell that was filled in for you is marked [?]. If the reader gave us nothing and we worked the value out from the row above, the row carries an information marker saying so. A cell where you actually wrote a ditto mark is not marked, because that one is what the paper says rather than what we concluded. It never blocks anything - it just lets you see the difference.
- Rows you never wrote on stay empty. The S.No. column is printed on all thirty rows whether or not anybody used them, so if you filled in ten rows, rows 11 to 30 come back blank. They are not filled in with the last person's name.

Nothing about how you use the page changed: photograph it, press Validate against the spec, review, stage.

22. Quality, Safety, Employees & Procurement (QSEP)

Everything in this chapter went live in August 2026. It adds four things that used to live on paper or in somebody's memory: who checked the material, who is wearing the safety gear, where a person works, and what the scanned purchase paperwork actually said.

Read section 22.1 first even if only one of the four concerns you - the QC approval rule changes what happens when a Store Keeper tries to issue certain materials, and that affects the whole site.

22.1 Quality Control - the new role and the issue block

What a QC does

A Quality Control inspector checks material that carries risk if it is wrong. Today that means the Surface Shields category - 36 materials out of the 466 in the master list. Nothing else in the catalogue is affected by any rule in this section.

> ### The rule, in one line > > Material without an MTC CAN be sent and dispatched to site. It CANNOT be > issued or consumed at the site. > > Receiving it is fine. Putting it on a Delivery Note and driving it to site is > fine. Handing it to a worker is not. If you have been told otherwise, that > came from a document written before 2026-08-12 - the rule below is what the > system actually does.

There are two separate gates. Both apply at the same moment - the instant before material goes to a worker - but they are satisfied by different people, which is what makes them worth telling apart:

Gate	When it applies	What it demands	Who is stopped	Who fixes it
Material Test Certificate	When a Store Keeper issues material to the field	A certificate on file covering that material at that site	The Store Keeper at the issue form	Logistics (or the warehouse, or the SK)
QC approval	The same moment	A QC must have inspected and approved that quantity at that site	The Store Keeper at the issue form	The QC

Material may be received, and may travel to site, with neither. That is deliberate and was decided by the operator. Holding a truck at the gate for paperwork that is sitting in somebody's inbox costs more than it saves - and worse, refusing to record a delivery does not stop it arriving; it only stops the system knowing about it, so real material sits in a yard invisible to the stock report and to everyone planning around it. What material may not do is reach a worker's hands unchecked. So both gates bind at issue.

When a controlled item is received without a certificate, Logistics is notified automatically to chase it from the supplier. Nobody has to remember.

You should almost never upload the same certificate twice

The certificate follows the material down the chain. Whoever has it files it once:

- Logistics attaches it to the purchase order line - before the goods have even shipped, and without knowing which site will end up with them;
- the warehouse attaches it to the Delivery Note as the material leaves;
- the Store Keeper uploads it at the site, from the Receive form or from the banner on the Issue form.

Any one of those clears the material for issue at the destination site, and the issue form tells you which document cleared it and where it came from. A certificate filed for one site does not clear another - it attests to one batch from one production run.

If you find yourself uploading a second copy of a document the company already has, stop and report it. That is a fault, not a workflow.

Raising an inspection

An inspection is opened automatically when controlled material is received. The QC does not create it by hand. It appears in Quality -> Inspections with a status of pending.

Deciding an inspection

Open the inspection and enter the quantity you approve. The status follows from the number, not from a separate choice:

You approve	Status becomes
Nothing (zero)	Rejected
Less than the submitted quantity	Partially approved
The whole submitted quantity	Approved

Rejected material stays in stock. It is not automatically sent back to the vendor and it does not disappear. It sits where it is, unusable, and is blocked from issue until somebody decides what happens to it. That was an explicit instruction: an automatic vendor return removes the evidence before anyone has looked at it.

What the inspector sees, and the Return No (added August 2026)

The inspection screen now shows the material's name, with the SAP and material codes underneath it. Before this it showed only the codes, which meant judging a material by a number rather than by what it is.

If a Material Test Certificate is on file, the certificate number appears with an Open certificate link that opens the actual document - in the queue and again in the inspection dialog. Previously it only said that a certificate existed, with no way to read it, which meant approvals were being made against documents nobody had opened.

When an inspector rejects any quantity, the system issues a Return No such as QCR-20260813-41. It is announced to the inspector on the spot, listed against the inspection, and sent to both the site's Store Keeper and its Head of Department. The Store Keeper types it into Return Stock and the return form fills itself in - see §13.10.

> This does not change the rule above. The Return No is an invitation for a > person to raise a return; nothing moves, and nothing is sent back to a vendor, > until the Store Keeper posts it and the HOD approves it.

The daily missing-certificate alert (added August 2026)

Every morning the system checks for Surface Shield material that is in stock with no Material Test Certificate on file, and tells the people who can do something about it:

Where the material is	Who is told
In a warehouse	Logistics, the Warehouse User, and the warehouse's QC
At a site	That site's Store Keeper, HOD and QC, and Logistics

Logistics appears on both lists because they are the only people who can obtain the document from the supplier. One message per location lists everything outstanding there, rather than one message per material.

The alert repeats every morning until the certificate is uploaded, and then stops on its own. That repetition is intended: this is a standing condition, not a one-off event, and the material cannot be issued to anybody while it lasts.

What the Store Keeper sees when the block fires

The issue form refuses with a message that names the actual numbers, so nobody has to guess which of three situations they are in:

- No inspection at all - "no quality inspection exists for it at this site. It cannot be issued to the field until a QC has checked the material and its certificate."
- Inspected but nothing approved yet - names how many inspections are still pending.
- Partly approved and used up - "QC has approved 40 and 40 is already issued or staged, leaving 0 - not enough for 10. Ask the QC to inspect the remaining stock."

The clearance is pooled per material per site, not per individual pallet. What is counted against it is everything already issued or staged awaiting HOD approval, from the date of the first inspection onward - issues that predate quality control entirely are not counted, or the site would be frozen forever by its own history.

> This does not change the FEFO or over-issue behaviour. Those remain > warn-and-record, as they always have. The quality block is about whether the > material is fit to use, which is a different question from whether the shelf > count agrees.

Creating a QC account

Quality -> QC Accounts, available to an HOD, a Warehouse User or Logistics. You create a QC inside your own scope: an HOD creates one for their site, a Warehouse User for their warehouse. Set a site or a warehouse, never both.

Moving a QC to another site is a request, not an action - the HOD raises it and an Admin decides. A QC whose remit could be changed by the person whose work they inspect is not an independent check.

22.2 PPE - issued through the ordinary form

There is no separate "Issue PPE" page, and this is deliberate. Safety boots leave the store the same way a drum of resin does: through Entry Log -> Issue. The form recognises that the item is PPE and asks for the extra details on the spot.

Select a PPE item and three fields appear:

Field	When it is required	Why
Employee ID number	Always	PPE is tracked against a person, not a site
Safety Approval document	Unless a usable-time rule says otherwise	Somebody signed for this issue
Reason for early replacement	Only if the worker's current item has not expired yet	A boot replaced after three weeks is either a defect or a leak

The stock movement and the PPE record are written together. There is no second ledger and no separate PPE stock - the quantity leaves through the normal issue path, so stock levels, burn rate, reports and every other number stay correct without knowing anything about PPE.

The messages you will see

- "...is PPE - name the employee receiving it (their ID number), so it can be tracked against them."
- "worker 30816 is not in the employee master" - the person must exist on the roster first.
- "worker 30816 is at site RIYADH, not CNCEC - transfer them first if they have moved" - see 22.3.
- "Ahmed already holds this item, issued 2026-07-02 and good until 2026-10-02 - give a reason for replacing it early."

Usable time

Safety & People -> PPE Usable Time (Store Keeper or HOD) is where you say how long an item lasts: 90 days for gloves, 365 for a helmet, and so on. A rule can be global or set for one site, and the site's rule wins where both exist.

> [!] Until a rule exists for an item, two things follow. The issue will > always demand a Safety Approval document, and the item gets no expiry date > - which means it will never appear in the forecast below. If the PPE Forecast > looks empty, this is almost always why.

An expiry is a suggested replacement date, not a restriction. Nothing stops a worker using gear past it, no alert is sent to anybody's phone, and the worker is not blocked from anything.

The 15-day forecast

Safety & People -> PPE Forecast lists PPE expiring in the next fifteen days and tells you what to order:

> suggested = expiring - what is on the shelf - what is already on order

If the answer is zero because thirty units are already on an open purchase order, that is the correct answer and not a broken screen. The list shows the names of the people whose gear is expiring,

because a column of quantities cannot be sanity-checked by a human and a column of names can. A 90-day issue rate is shown beside the suggestion as a cross-check, never folded into it.

22.3 Employees - one person, one number

The employee ID number is the person. It is unique across the whole company, not per site, and everything else hangs off it.

Safety & People -> Employees shows the roster. An HOD can transfer someone to another site, and the transfer takes effect immediately - no approval step, because a person who has physically moved has already moved.

PPE history follows the person, not the site. Transfer a worker from one site to another and the boots they are wearing are still on their record. This falls out of keying on the ID number rather than being a feature bolted on top, which is why it cannot quietly stop working.

Admins get a timeline for any employee: every site they have worked at, with dates, and everything they currently hold. Use it when somebody asks "who had this, and when".

Employees -> Data Quality lists records that cannot be used and says why - a missing ID number, a duplicate, an inactive worker still holding gear. Check it before a PPE rollout rather than discovering the gaps one refused issue at a time.

22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork

Delivery Notes draft themselves

When a warehouse receives goods against a purchase order, the system now prepares the Delivery Notes the sites are waiting for, grouped so that R/L and B/L material never share a note.

They are created as drafts. A person still adds the vehicle and driver and submits them, because the system cannot know that a truck exists. A notification tells the warehouse the drafts are waiting.

If a draft cannot be created - an over-shipment, a missing certificate - the goods receipt still stands. The stock genuinely arrived; losing that record because a convenience failed would be far worse. The reason is written to the audit log and the clerk cuts the note by hand.

Auto-drafting can be switched off in Settings.

Assigning a purchase order to a warehouse (changed August 2026)

A purchase order goes to one warehouse. The Purchase Orders list now shows a Warehouse column naming where each order was sent, and once an order has been assigned the Assign button is replaced by an assigned tag.

Previously the button stayed on every row for ever, and the list gave no indication that an order had already been routed. Pressing it a second time created a second assignment and told a second warehouse to expect the same goods, with nothing anywhere reporting a problem.

- Assigning the same warehouse again - a double-click, or an old tab - is accepted and changes nothing. No duplicate, no second notification.
- Assigning a different warehouse is refused, and the message names the warehouse that already holds the order. Re-routing an order another warehouse is already expecting is a decision for a person to make, not something the system will do quietly.

Shipping a delivery note (changed August 2026)

Pressing Ship now asks for two things before the delivery leaves:

- The delivery note number printed on the physical document - the carrier's or the supplier's number. This is not the DN number the system generated; the point is to connect the two, so that a delivery can be reconciled later without a telephone call.
- A scan or a photograph of the signed note. The camera button opens the phone camera directly, which is the practical option at a loading bay.

Both are required. If either is missing you are told which one, separately - one is typing and the other is scanning, and being sent to redo the half you had already done is its own small waste.

A Material Test Certificate is still optional here and works exactly as before. A certificate covers the material and is inherited from the purchase order or the delivery note down to the receiving site, so nobody uploads it twice. A delivery note covers this shipment and is inherited by nothing. They are different documents answering different questions.

Once shipped, the document number and a download link appear beside the delivery for everyone who touches it - the warehouse, Logistics, the HOD, Quality and the receiving Store Keeper. The Store Keeper is the one holding the paper copy while the truck waits, so this is where it matters most.

> Deliveries shipped before this change show "not shipped yet" in that > column. There is no document for them because none was ever captured, and > inventing a number would be worse than saying so.

Marking a reschedule urgent

A delivery reschedule can be marked urgent. An urgent one goes out immediately instead of waiting for the 4 p.m. evening digest. Use it sparingly - a channel where everything is urgent is a channel nobody reads.

Reading a scanned PR or PO

Upload the supplier's paperwork and the system reads the lines off it, so nobody retypes forty rows.

The file is stored before it is read, and this matters more than it sounds: the document that defeats the reader is precisely the one somebody will need to look at later. A failed read still leaves you the document.

There are two ways a document gets read, and the system chooses by whether the file actually contains text - not by whether it is a PDF:

The file	What happens
A PDF exported from a system	Text is read directly. Fast, exact.

The file	What happens
A PDF that is a photograph or a scan of a signed page	Read by the vision model. Takes longer; you poll for the result.

A printed-signed-and-scanned purchase order contains no text at all, only pictures of text. Before this change such a file was reported as read successfully with zero lines found - which is worse than an error, because nothing on screen distinguished "this order has no items" from "I could not read a word of it".

Once read, confirming the document links it to the purchase request or order you created, so the original scan is one click from the record. Offering a purchase order scan to a purchase request is refused by name - the two are not interchangeable.

> A quantity the reader could not make out is left blank, not zero. A zero on > a purchase order looks like an answer and would be ordered as one.

22.5 Assets - one hammer, one row, anywhere in the company

A serialised asset is now unique by its part number and serial number across the entire company, not per site. The same hammer cannot be registered twice by two sites who both think they have it.

Moving an asset between sites is a request approved by the HOD of the site it is leaving - the site giving something up is the one that needs to agree. On approval the asset's site changes and its old rack assignment is cleared, since a shelf in one yard means nothing in another. The move is recorded as a movement, so "where has this been" still answers correctly for the leg between the two sites.

Trying to re-register a serial that exists elsewhere is refused with a message that says where it actually is and what to do about it.

22.6 Passwords

The minimum password length is 8 characters, and a password must now also contain an uppercase letter, a number and a special character. This replaced a flat 12-character minimum with no composition requirement.

The rule is applied identically everywhere a password is set - admin creation, password reset and self-registration. Self-registration previously enforced a much weaker rule than the admin screens did.

The Request Access page now offers Quality Control in its role list.

23. Quality Oversight (Head of Qualities) Manual

> Access: Quality -> Quality Oversight - exact-locked to Head of > Qualities (qc_hod) + Admin. The role reads across every site; it has > no site of its own. (NEW - 2026-08.)

23.1 What the role is for

A Head of Qualities watches Surface Shield material across the whole company: what has arrived without a certificate, where it is being used, and what is sitting still long enough to expire. It is an oversight role, not an operating one - it reads, and it sends messages.

What it can do

- See every Surface Shield purchase order and whether each line has an MTC.
- See the certificate register across all sites and warehouses.
- See which sites are consuming which controlled material, and how recently.
- See lots that have not moved, or that are close to expiry.
- Escalate - ask a site QC, a warehouse or Logistics to obtain a certificate, inspect something, or take stock that would otherwise expire.
- Set the stagnation and expiry thresholds.

What it cannot do, and this is the point

- It cannot approve or reject an inspection - that is the site or warehouse QC inspector.
- It cannot receive, issue, adjust or transfer stock.
- It cannot approve a delivery note, raise a purchase request or create a purchase order.
- It cannot open the HOD Portal, the Estimator, Man-Hours, the Logistics Portal or the Admin Portal.

> Why it is not simply "a senior QC". Cross-site reach and the power to > decide are different things. This role sees everything and decides nothing - > which is what lets it be handed out to somebody who needs the whole picture > without also handing them the ability to release material.

23.2 Everything here is Surface Shield only

Every figure on every tab is filtered to the controlled category - Surface Shields unless an admin has changed mtc_required_category. The filter is applied in the database, on every query, not by the page.

That is deliberate: the account reads across every site, so without the filter it would also be a company-wide window onto PPE, tools, consumables and the prices on every purchase order. The category is the boundary of the role.

23.3 The seven tabs

23.3.1 Overview

The KPI strip: uncertified materials, how many sites and warehouses are affected, stagnant lots, lots expiring soon, lots already expired, and open escalations. If anything is uncertified you get a banner naming the places and a Chase a certificate button.

23.3.2 Surface Shield POs

Every Surface Shield line on every purchase order, with MTC: on file / missing on each row. A missing one carries a Chase button that opens the escalation form pre-filled with the PO, the material and the site.

23.3.3 MTC Register

Every certificate on file for controlled material - number, material, lot, where it is, quantity, who uploaded it and when.

23.3.4 Where It Is Used

Per site and material: how much has been consumed, over how many draws, and the date of the last one. A site that drew 2,000 kg last March and nothing since is a different conversation from one drawing steadily, so the last-used date is a column rather than a filter.

23.3.5 Stagnation & Expiry

Three sub-tabs - Expired, Expiring, Stagnant.

> Two different problems wear the same number. A lot received and never > touched and a lot used until March and then abandoned have the same idle > days and completely different causes. The Idle column says which: the second > is marked (never used).

Could move to lists sites already drawing that material, most recent first, excluding the one holding it. It is a contact list, not a transfer - moving stock is somebody else's authority.

23.3.6 Escalations

The comms log. Every escalation you have raised, who it went to, what it was about, and whether it is still open.

> It is a log, not a notification. "Send the site QC a reminder" is easy; > what that cannot answer is how long has this been chased, and by whom. > Uncertified material is a standing condition, so the second and third chase > are the ones that matter - and they only exist if the first was written down.

Closing one requires a note. The note is the record of what actually fixed it, and re-closing an already-closed escalation is refused rather than overwriting it.

23.3.7 Settings

Stagnant-after days (default 90) and expiry-warning days (default 60). These are your policy, not a system constant - changing them must not need a release.

23.4 Escalating: you must name one place

An escalation targets exactly one site or one warehouse. Neither, or both, is refused.

> A message aimed at everywhere is one nobody owns. An escalation that no > particular person is responsible for reading is worse than none, because it > looks like the problem was raised.

Choose who should act - site/warehouse QC, the warehouse, or Logistics - and what you are asking for: a certificate, an inspection, or redistribution. The recipient gets it in their notification bell and on WhatsApp, and it is written to the log at the same moment.

23.5 The daily alert

Once a day the system sweeps for controlled material on hand with no certificate. Sites and warehouses get their own alert about their own material; the Head of Qualities gets one aggregated message covering every location.

> Why a separate message rather than a copy of theirs. A notification is > visible when its site matches yours, and a Head of Qualities has no site - so > the per-site alerts are, correctly, invisible to them. One message naming six > locations is also the right shape for the job: six messages saying one thing > is how somebody responsible for all six learns to ignore them.

23.6 Creating the account

Admin Portal -> Users -> Add user -> role Head of Qualities. The account takes no site - it is cross-site by definition, and binding it to one would contradict the reason it exists. It cannot be self-registered.

Document end

This manual covers every page, tab, button, table and field in the General Industries Hub: the site workspaces (sections 4 to 7), the Reports module (section 8), the Logistics and Warehouse portals and the procurement chain (sections 14 to 16), hosting and operations (section 17), the Material Estimator (section 18), Man-Hours and Manpower Tracking (section 19), the Auditor role (section 20), the asset-tracking and warehouse-locator features added in 2026-08 (section 21), and Quality, Safety, Employees and Procurement (section 22).

For day-to-day operating procedure across all roles, see the Standard Operating Procedure document, which is shorter and organised by task rather than by screen.

Printed and PDF copies

Every role has its own booklet containing only the chapters that apply to them - a Store Keeper's booklet does not include the Logistics portal, and is a fraction of the length. Admin Portal -> Settings offers each of them as a download, and the full manual alongside.

If you are handing the system over to a new team, print the role booklets rather than this complete

document. People read twenty relevant pages and skip two hundred.

24. Phase 10 - Security, Training and the Board Brief

Four things changed. One of them will stop you signing in the way you always have, so read 24.1 even if you read nothing else.

24.1 Two-factor authentication is now required for some roles

If your role is Admin, Logistics, HOD, Head of Qualities or Auditor, your account now needs an authenticator app as well as a password.

Nothing changes for Store Keepers, Supervisors, Warehouse Operators or Quality Control inspectors. They may still turn 2FA on voluntarily from Security, and it is a good idea, but nobody will make them.

24.1.1 The 14-day grace period

You are not locked out on the day this ships. For fourteen days after the change goes live, a mandated account signs in exactly as before and sees a banner naming the date after which it will not. On and after that date, a mandated account with no authenticator cannot get a session at all - it is sent to the setup screen instead.

Your administrator sets both the list of roles and the date, so if the deadline is unworkable for your site, ask them rather than working around it.

24.1.2 Setting it up - three minutes

You need a phone with an authenticator app: Google Authenticator, Microsoft Authenticator and 1Password all work.

1. Go to Security (it is in the Account group at the bottom of the sidebar) and choose Set up 2FA.
2. Re-enter your password. This is deliberate: it means somebody who has borrowed your open laptop cannot attach their phone to your account.
3. Scan the square code with the app. If your phone cannot scan it, copy the text code underneath instead.
4. Type the six-digit number the app shows and confirm.

From then on, signing in asks for your password and then that six-digit number. The number changes every thirty seconds; typing the previous one is fine for a few seconds after it rolls over.

24.1.3 If you are past the deadline and not set up

You will not see the normal screens. You will see the setup panel instead, and you can complete it right there - password, scan, six digits - and then sign in again. You are not locked out; you are held at the door until the second lock is fitted.

24.1.4 If you lose your phone

Ask an Admin to reset your 2FA. There are no printed backup codes on purpose: a backup code is a second password that lives on a piece of paper, and this company has an administrator who can be reached. The reset is recorded in the audit log with who did it and when.

24.1.5 Five wrong codes

Five wrong codes in fifteen minutes and the account pauses for a few minutes. It throttles, it never locks - nobody has to be called, and the pause clears on its own. Getting it right immediately ends the pause.

24.2 Training - the new tab, and what it does to the OCR upload

Training now sits in the Account group beside Security. Everybody has it; what it contains depends on your role.

24.2.1 Watching a tutorial

Open Training. Each module shows a title, whether it is required for your role, and - once your administrator has published the videos - a player.

Pick your language. Tutorials are recorded in English, Tamil, Tanglish (Tamil spoken in Latin script) and Arabic. Choose whichever you follow most easily; the system records which one you watched, and no language counts for less than another.

Watch it. When you have seen at least 90%, the "I have watched and understood this" button becomes available. Pressing it records your name, the date and the version you watched.

> You cannot acknowledge a video you have not watched, and the button stays > disabled until you have. A register full of tick-boxes pressed on arrival is > worse than an empty one, because somebody would eventually produce it as > proof of training that never happened.

If a module says "Not published yet", the videos have not been uploaded. There is nothing to watch and nothing to acknowledge; that is not your problem to solve.

24.2.2 The soft gate on the consumption-form upload

The first time you press Photograph or upload on the Execution page without having completed the OCR tutorial, a panel appears offering two choices.

Choice	What happens
Watch now	Takes you to Training. Nothing is uploaded.
Watch later & continue	Your upload proceeds immediately. The deferral is recorded.

> [!] Nothing is ever blocked. You are standing in a plant holding a filled > form; a video must not stop you filing it. "Watch later & continue" carries > straight on with the upload you asked for - the click is not wasted and you > do not have to press it twice. > > What "Watch later" is not is invisible. It is counted, and your HOD sees the > count next to your name. The point is that somebody can have a conversation > with you, not that the system can refuse you.

Printing a blank form is never gated at all - the tutorial is about filling one in, so it has no business standing between you and a sheet of paper.

24.2.3 "Watch it" - the assistant sends you to the right second

When you ask the Hub Assistant a question and one of the published tutorials shows that exact step, a small Watch it button appears under the answer. It carries the tutorial's name and a timestamp.

Pressing it opens Training with that video already wound to the second where the step is on screen. You do not have to scrub through ninety seconds looking for it.

What you see when you arrive: the tutorial you were sent to, on its own, at the top of the page, with a note saying which second it starts at. Your other modules are folded away behind "Show my other training" - the one you asked about is not buried in a list.

It starts playing with the sound off. Turn the sound on with the player's own controls. Browsers refuse to auto-start a video with sound, and refuse it silently, so a page that tried would simply sit there looking broken.

If the tutorial has not been published on your system, the page says so in those terms - that the step exists and the recording is not on this server - rather than the general "not published yet". The assistant promised you a video; the page owes you an explanation of where it went.

If the link names a tutorial your role does not cover, the page says that too, at the top, and shows your own modules underneath.

Three things worth knowing:

- You still get the written answer. The button is never instead of it. If you are standing at a counter with somebody waiting, read the answer; the video is there for later.
- You are only ever offered a tutorial your role may watch. A Store Keeper is not shown a Head of Department's walk-through, however the question is phrased - the same rule that decides which chapters of this manual the assistant may answer from decides which videos it may point at.
- Sometimes there is no button, and that is deliberate. If nothing matches your question closely enough, nothing is offered. A link to the nearest video is worse than no link, because you would spend ninety seconds discovering it was the wrong one.

If the videos have not been published on your system yet, no buttons appear anywhere and the assistant behaves exactly as it always has.

24.2.4 For HODs - the compliance tab

An HOD or Admin opening Training sees a second tab, Team compliance.

It lists everybody whose role requires the module, not merely the people who have opened it - so somebody who has never touched it appears as "Not started" rather than being absent from the list. That absence is the thing worth seeing.

Each person shows Not started / Started / Acknowledged, and how many times they have deferred.

24.2.5 When a tutorial is re-recorded

If the workflow changes and your administrator publishes a new version, everybody must watch and acknowledge it again. Your old acknowledgement is not deleted - "watched version 1 on that date" stays true and stays in the record - it simply no longer counts for the new version.

24.3 Site-Wide Valuation and 30-Day Burn

Who: HOD (own site), Auditor and Admin (all sites).

A one-page, board-ready PDF answering two questions: what the material on the shelves is worth, and how fast it is being consumed. Available as a download alongside the Executive Summary.

It shows: stock value, the 30-day consumption value and its per-day rate, months of cover at that rate, a breakdown per site, and the highest-value lines consumed in the window.

> [!] This is not the Burn Rate Forecast. The HOD Portal has carried a > Burn Rate Forecast chart for a long time and it answers a different question: > how many days of stock is left before we run out, in units, per material. > This report answers what is the stock worth and what is it costing us, in > SAR, across the site. If you want to know when to reorder, use the Burn Rate > Forecast. If you are writing a board paper, use this.

24.3.1 [!] "Not Valued (N items)" - read this before quoting the number

A material with no unit cost on the inventory master is NOT counted as worth zero. It is counted separately and named.

This matters more than it sounds. If un-costed drums were multiplied by a cost of zero and added in, a site holding four hundred of them would report exactly the same value as a site holding none - a number that is arithmetically correct and completely false, and one a board would act on.

So the report shows:

- the value of the priced lines,
- "Not Valued (N items)" beside it, with how many units that covers,
- the coverage percentage - how much of the list carries a price at all,
- and a footnote saying the total is a floor, not a total.

If the coverage figure is low, the valuation is not wrong - it is incomplete, and it says so. The fix is unit costs on the inventory master, which is Admin work, not a change to the report.

24.3.2 Why the estimator's numbers sit in their own table

The Material Estimator's figures appear at the bottom under "stated separately", and they are never added to the value above them.

They measure different things. The tables above are the live ledger of what is physically on the shelf today; the estimator seed is a frozen snapshot of what the estimate said the project would need. Adding them together would count the same material twice and produce a bigger number that means nothing.

24.3.3 "Months of cover" and when it is blank

Months of cover is the stock value divided by the daily burn. When nothing was consumed in the window it shows n/a, not zero and not infinity - a site that has consumed nothing has no meaningful runway, and printing one would be inventing a figure out of a division by zero.

The per-day rate divides by the full 30 days, not by the days on which something was actually consumed. A site that worked eight days out of thirty would otherwise look four times busier than it is.

24.4 Day and Night shift on execution entries

When you open an execution entry there is now an optional Shift choice - Day or Night.

It is optional, and leaving it blank does not block anything. Setting it does one useful thing: at 07:00 each morning the system checks which materials are staged for today's day shift and have no Material Test Certificate, and warns Logistics, the store keeper, the HOD, the QC and the Head of Qualities before the crew reaches the counter.

Entries filed before this existed have no shift recorded and are simply not covered by that check. Nothing guesses which shift they were.

> Why not work it out from the clock? Because an entry is filed when > somebody reaches a desk, not when the work happened. A night crew filing at > 06:40 would be counted as day shift on the strength of a timestamp nobody > looked at.

Since 2026-09-02, the shift can also come off the paper. Crews already write it beside the date - 25/08/26 (Night) - and the system now reads it, on both the printed consumption form and the handwritten consumables sheet. If it is written, the shift is filled in for you. If it is not written, the shift stays blank, exactly as before.

This is not the same thing as working it out from the clock, and it is worth being clear about the difference: the clock is our guess, and the word beside the date is the crew's own statement. Nothing infers a shift from anything; we simply stopped ignoring the one place it was already recorded.

> [!] This also fixed a bug you may have hit. Writing (Night) in the date > box used to make the whole page unreadable - the date parser expected nothing > after the digits, so a sheet came back with "the date could not be read" and a > critical flag. Being conscientious enough to record the shift was breaking the > page. It no longer does.

24.5 What did not change

- Nothing about the MTC rule. Material without a certificate can still be received and still be sent to site; it still cannot be issued or consumed there. See §22.1.
- Nothing about how consumption is filed. The paper-first workflow of §4.10 is exactly as it was; the training gate sits beside it and never in it.
- Nothing you had already acknowledged, approved or posted. No historical record was rewritten by any part of this release.

25. Phase 11 - AI Traces

Who this chapter is for: Administrators and Auditors. It describes one new read-only page and nothing else changes for anybody.

25.1 What the page is

Where: Admin -> Console -> AI Traces, or, for an Auditor, AI Traces in the sidebar. (The Console itself is administrators-only, which is why the same page has its own menu entry for the Auditor rather than living only inside it.)

Every question anybody asks the Hub Assistant now leaves a record of how the answer was produced - not the answer, but the working:

Column	What it tells you
Question	what was typed
Retrieval	which manual sections the question actually reached, and how strongly each one matched
Queued	how long it waited for a free slot before the model started
Total	the whole turn
Outcome	answered, greeting, switched off, or failed

Expanding a row shows each stage separately.

25.2 [!] Why the Retrieval column is the whole point

The assistant does not read the manual every time. It searches it, picks the handful of passages most likely to answer you, and shows the model only those. That search has always scored every candidate section - and always threw the scores away.

Which meant that when the assistant said something wrong, there was no way to tell which of two completely different things had happened:

- it was shown the wrong pages, and answered them faithfully; or
- it was shown the right pages and answered badly anyway.

They need opposite fixes, and for a long time nobody could tell them apart. One real case: the assistant told HODs they could not open the Manpower page. The model was not confused - the section that lists who can open what had been cut off before the relevant table, so it never reached the model at all. That took a phase to find. With this page it would have been visible in one row.

25.3 The gold "fallback" tag, and why it matters

If a question matches nothing in the sections your role may read, the assistant does not give up - it falls

back to showing the model a shortened copy of every chapter you are allowed, and answers from that.

That is the condition under which an assistant makes things up: a broad, truncated context and no passage that actually answers the question. The row is tagged fallback so you can see it happened. A run of them on similar questions means the manual is missing a section, not that the model has deteriorated.

25.4 What is deliberately NOT on this page

- The answer. Only its length is recorded - enough to spot a reply that was cut off, nothing more.
- The text of the manual sections. Only chapter numbers, headings and scores. The assistant is careful about which chapters a role may see, and showing the text on a different page would quietly work around that.

You may also see a header saying "N span(s) dropped". That means the part of the system that writes these records fell behind, so the list is incomplete. It never affects the assistant itself - records are dropped rather than allowed to slow anybody's question down.

25.5 The assistant's guard rails

Two small checks were added around the Hub Assistant. Neither changes how you use it, and the second one is mostly notable for what it does not do.

On the way in, a question that is plainly an attempt to talk the assistant out of its own rules - "ignore your instructions", "show me your system prompt", "you are now an unrestricted assistant" - is answered with the ordinary "that is not in your section" reply, without troubling the model at all.

[!] Ordinary sentences that happen to contain those words are NOT refused, and that matters more than the refusals do. All of these are answered normally:

- "one drum arrived damaged - can I ignore it and issue the rest?"
- "the tank is now empty, how do I record that?"
- "can you repeat the steps for staging a return?"
- "how do I bypass a blocked lot and use the next one?"

If you are ever refused for a genuine question, that is a bug - report it. The assistant's real protection has never been this check: it is that the sections of the manual your role may not read are never loaded into your conversation in the first place, so there is nothing there to be talked out of. The check simply saves you a slow, confused answer.

On the way out, an answer is scanned before it reaches you. Anything shaped like a phone number, an email address, an ID number or a key is removed, and a line that begins with = is made safe to paste into a spreadsheet. [!] The manual contains none of those things, so if you ever see something removed from an answer, tell an administrator - it means something reached the assistant that should not have.

Answers now arrive in slightly larger pieces than before - roughly a clause at a time rather than a word - because the text is being checked as it goes. They should still appear progressively; if an answer only ever arrives all at once at the end, that is worth reporting.

25.6 Why a repeated question comes back instantly

If you ask the assistant something it has already answered - for you, or for somebody else in your role - the answer comes back immediately instead of being typed out. Nothing was skipped: it is the same answer, produced the same way, and kept for the next person who asks.

Three things about it are worth knowing:

- It is per role. A Store Keeper and an HOD asking word-for-word the same question do not share an answer, because they are answered from different chapters of this manual. Each role's answers are kept separately.
- It is retired whenever this manual changes. Every kept answer is tied to the exact edition of the manual it came from, so a documentation update clears them all. You will never be given an answer about a screen as it used to be.
- Only the manual assistant does this. Anything that reports live stock - the "chat with your data" card, the insights panel, the end-of-day summary - always asks the database afresh. A remembered stock figure would be a wrong number with a confident timestamp on it.